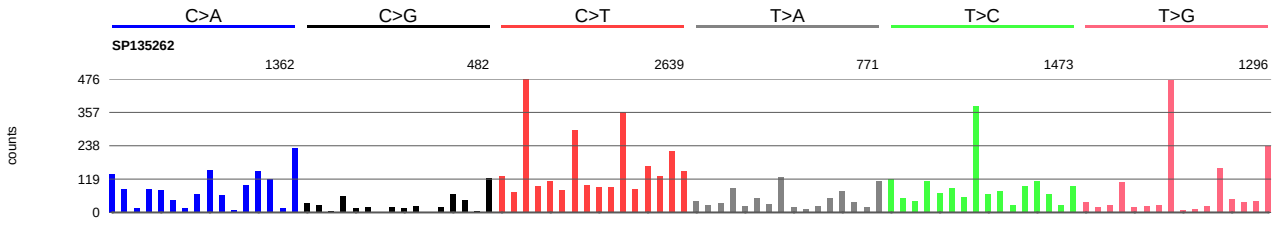
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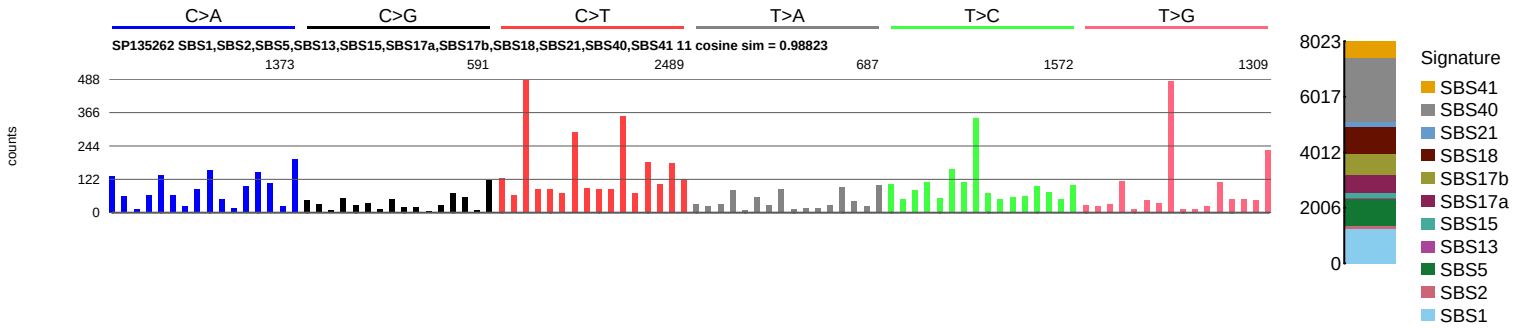
SP135262
1362
482
2639
771
1473
1296
476
357
238
119
0
counts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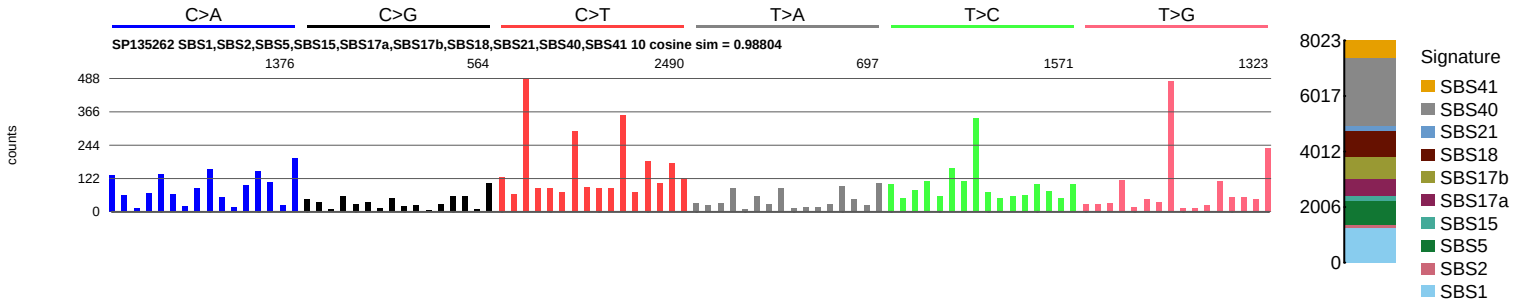
SP135262 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.98823
1373
591
2489
687
1572
1309
488
366
244
122
0
counts
8023
6017
4012
2006
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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SP135262 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98804
1376
564
2490
697
1571
1323
488
366
244
122
0
counts
8023
6017
4012
2006
0
Signature
SBS41
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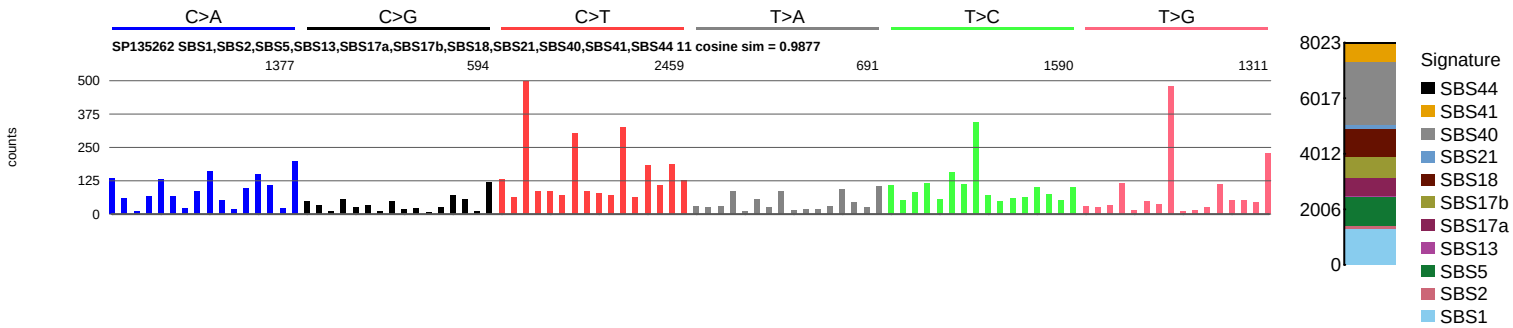
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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SP135262 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.9877
1377
594
2459
691
1590
1311
500
375
250
125
0
counts
8023
6017
4012
2006
0
Signature
SBS44
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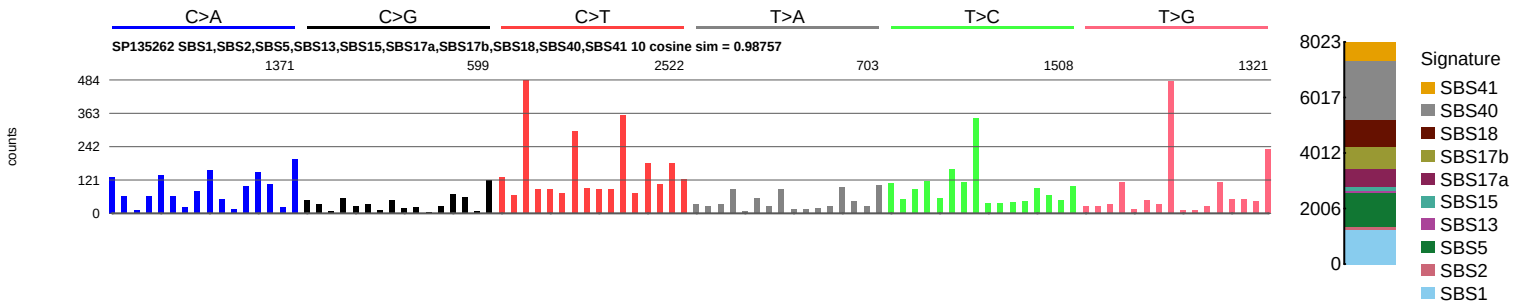
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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SP135262 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.98757
1371
599
2522
703
1508
1321
484
363
242
121
0
counts
8023
6017
4012
2006
0
Signature
SBS41
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SBS18
SBS17b
SBS17a
SBS15
SBS13
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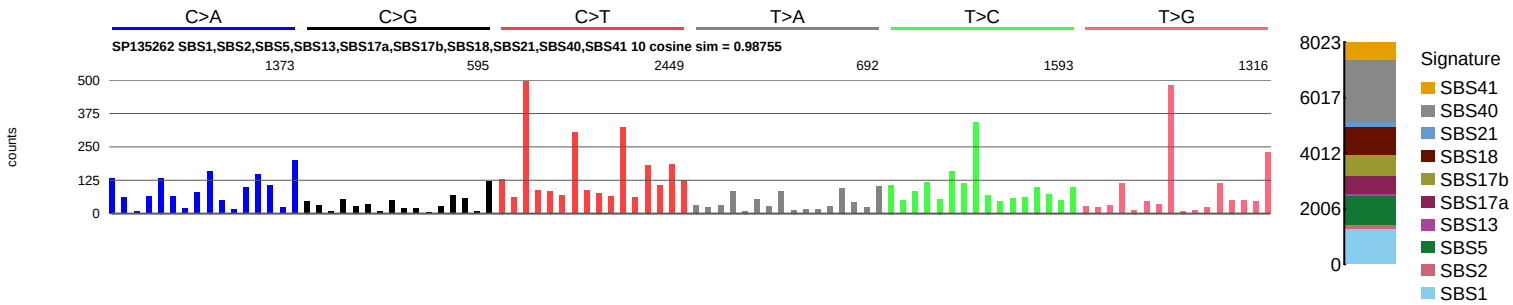
SBS2
SBS1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SP135262 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98755
1373
595
2449
692
1593
1316
500
375
250
125
0
counts
8023
6017
4012
2006
0
Signature
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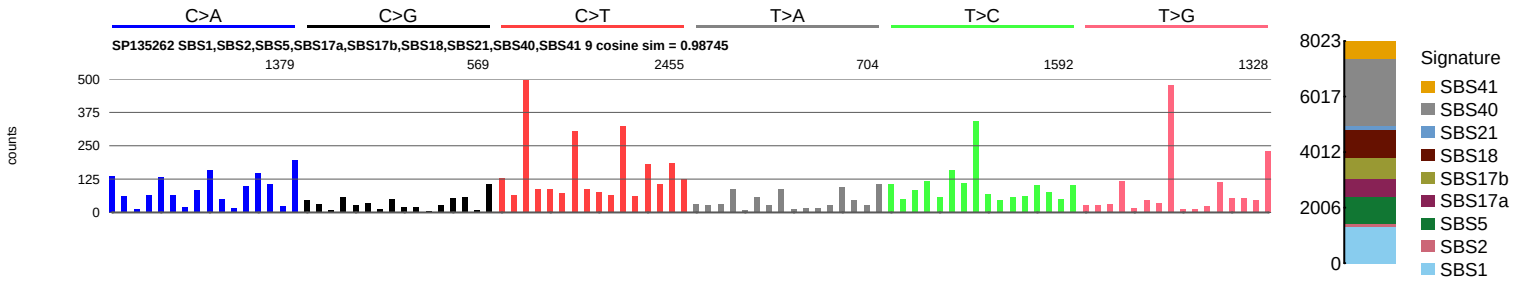
SBS2
SBS1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C>G
C>T
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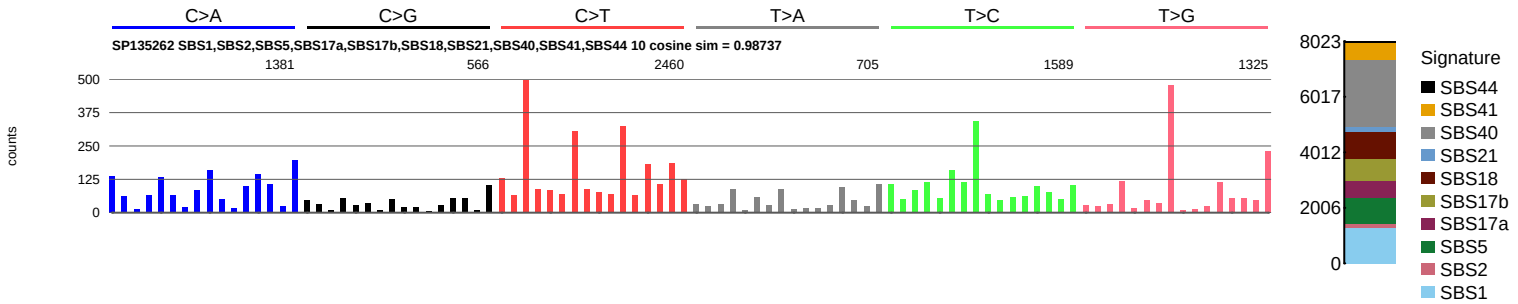
T>C
T>G
SP135262 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98745
1379
569
2455
704
1592
1328
counts
500
375
250
125
0
8023
6017
4012
2006
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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C>G
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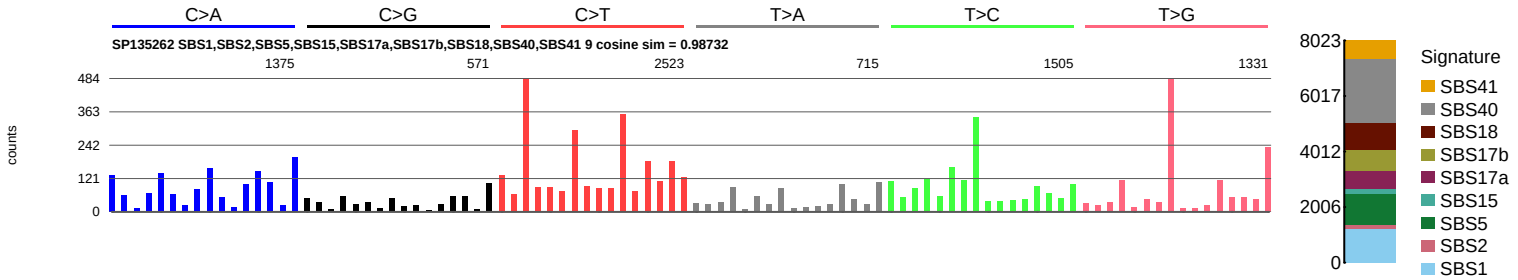
T>G
SP135262 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98737
1381
566
2460
705
1589
1325
counts
500
375
250
125
0
8023
6017
4012
2006
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98732
1375
571
2523
715
1505
1331
counts
484
363
242
121
0
8023
6017
4012
2006
0
Signature
SBS41
SBS40
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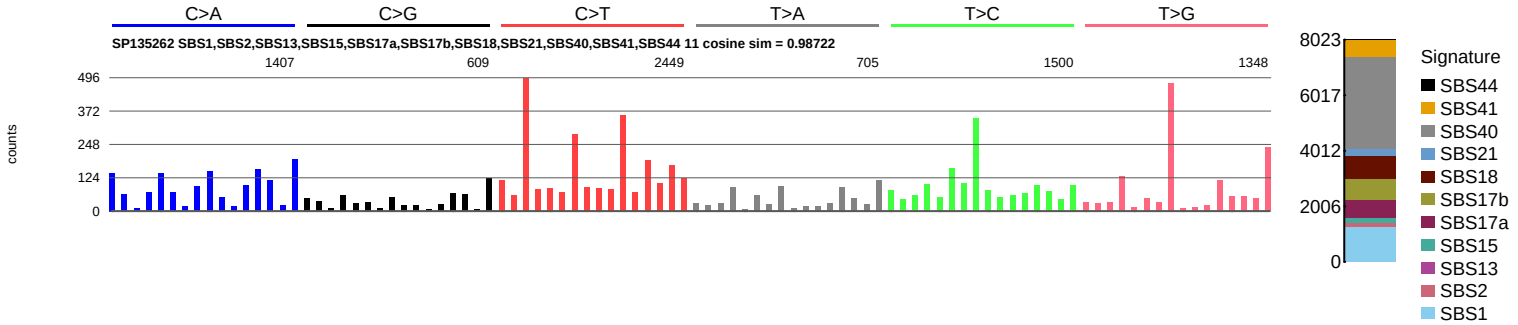
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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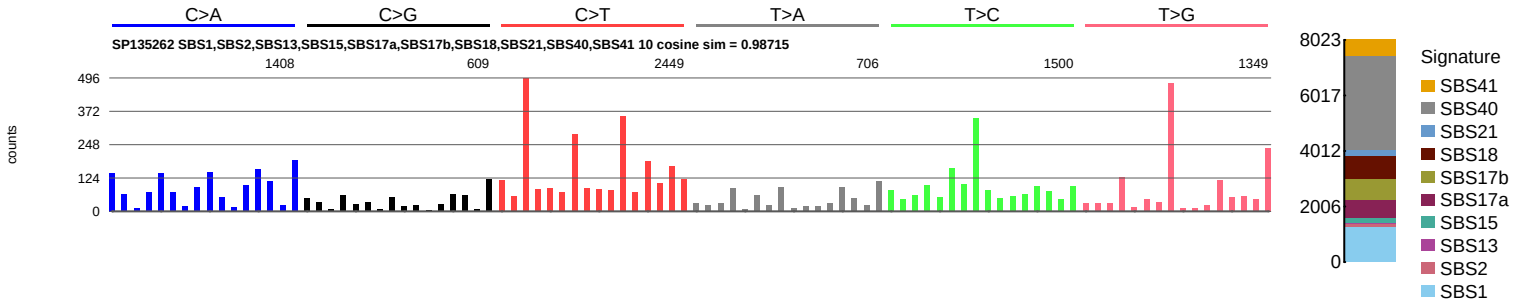
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98722
1407
609
2449
705
1500
1348
counts
496
372
248
124
0
8023
6017
4012
2006
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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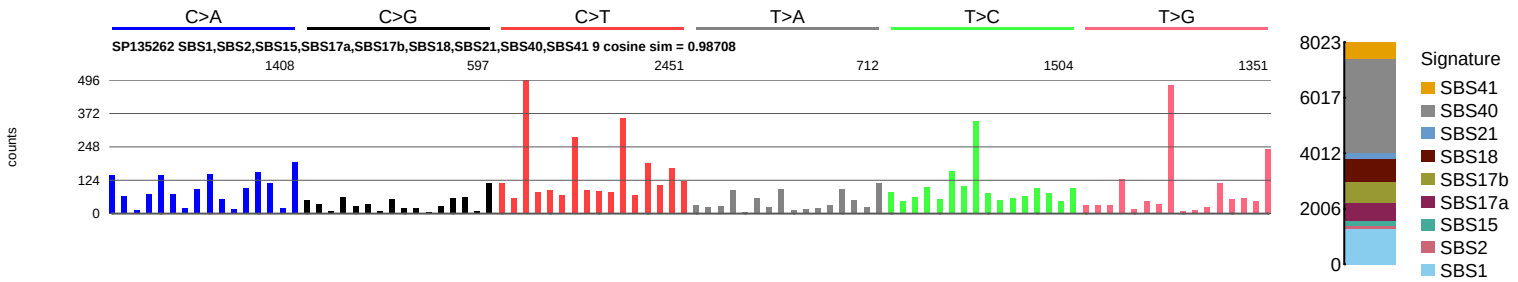
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98715
1408
609
2449
706
1500
1349
counts
496
372
248
124
0
8023
6017
4012
2006
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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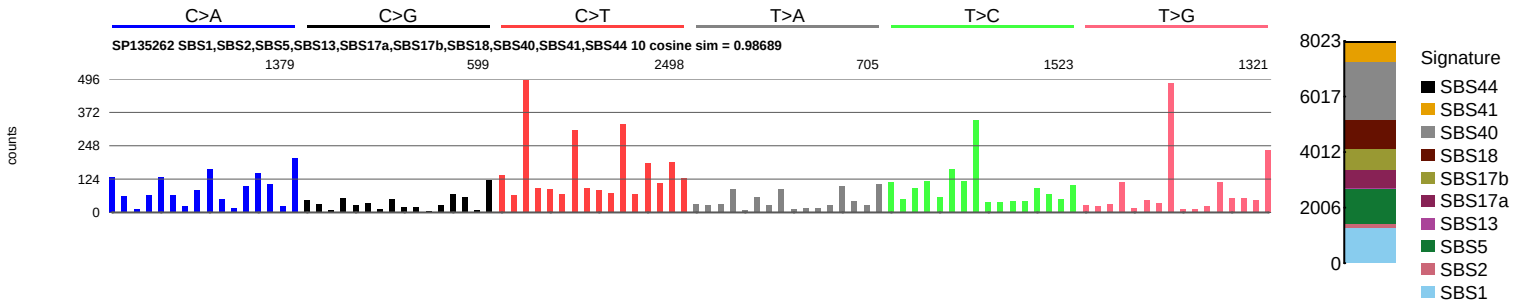
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98708
1408
597
2451
712
1504
1351
counts
496
372
248
124
0
8023
6017
4012
2006
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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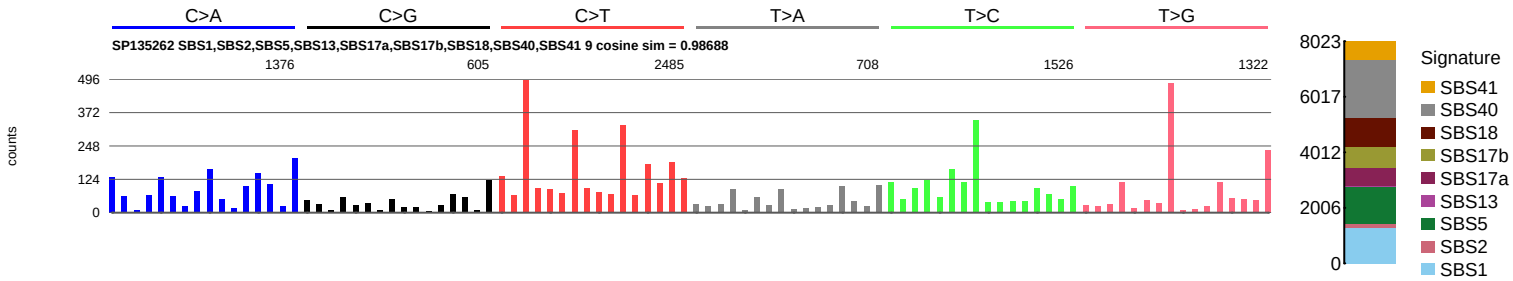
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98689
1379
599
2498
705
1523
1321
counts
496
372
248
124
0
8023
6017
4012
2006
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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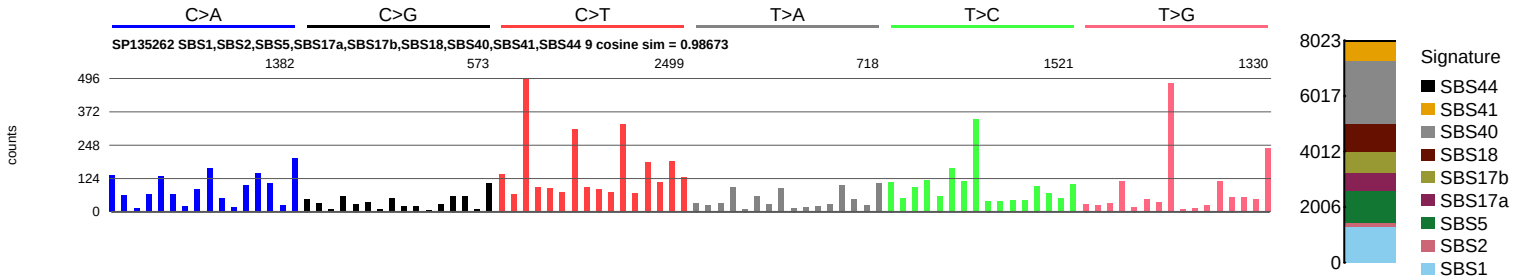
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98688
1376
605
2485
708
1526
1322
counts
496
372
248
124
0
8023
6017
4012
2006
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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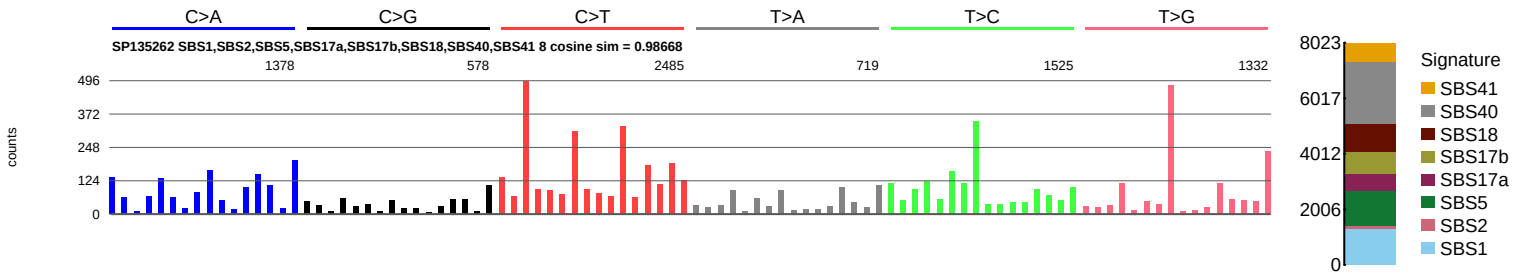
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98673
1382
573
2499
718
1521
1330
counts
496
372
248
124
0
8023
6017
4012
2006
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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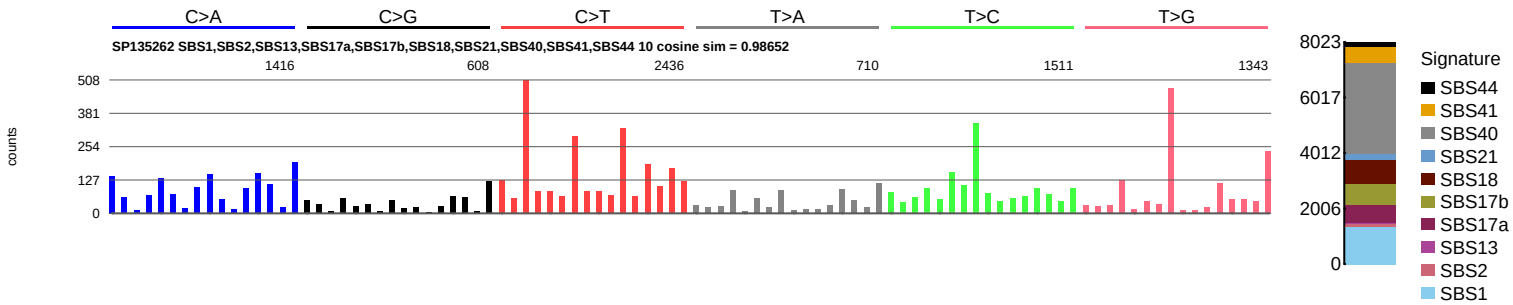
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98668
1378
578
2485
719
1525
1332
counts
496
372
248
124
0
8023
6017
4012
2006
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98652
1416
608
2436
710
1511
1343
counts
508
381
254
127
0
8023
6017
4012
2006
0
Signature
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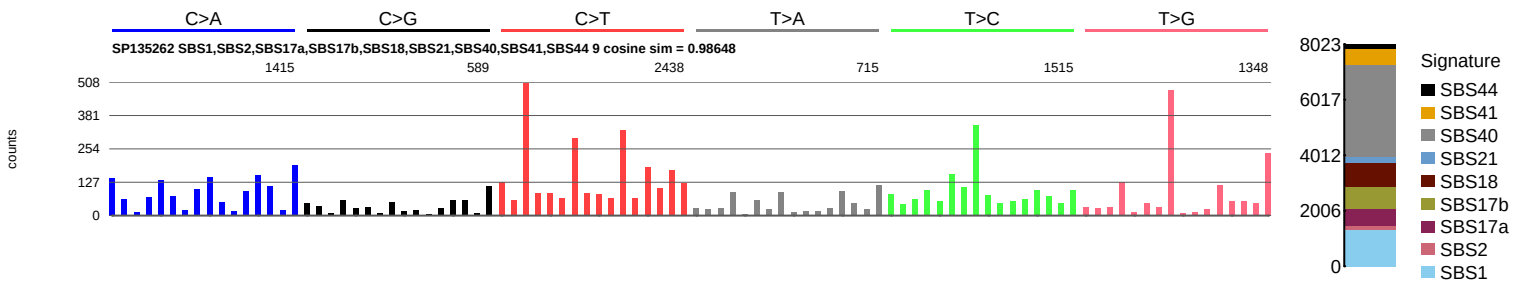
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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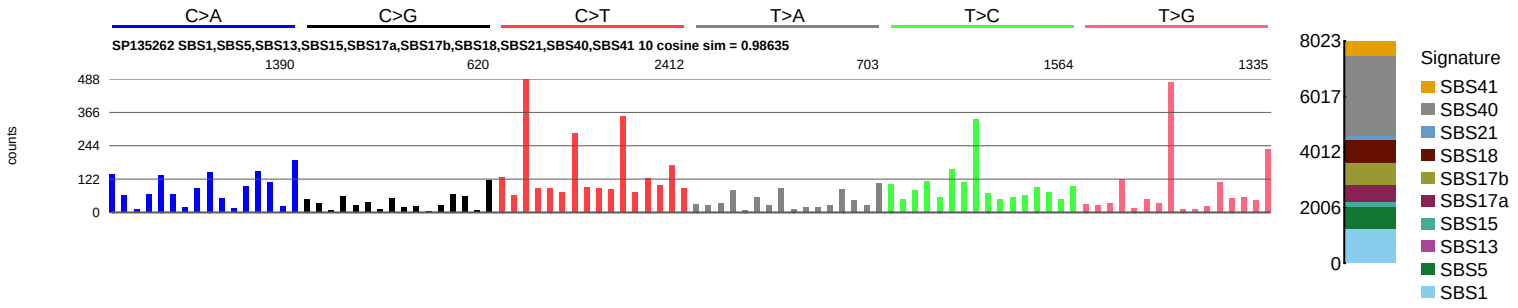
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98648
1415
589
2438
715
1515
1348
counts
508
381
254
127
0
8023
6017
4012
2006
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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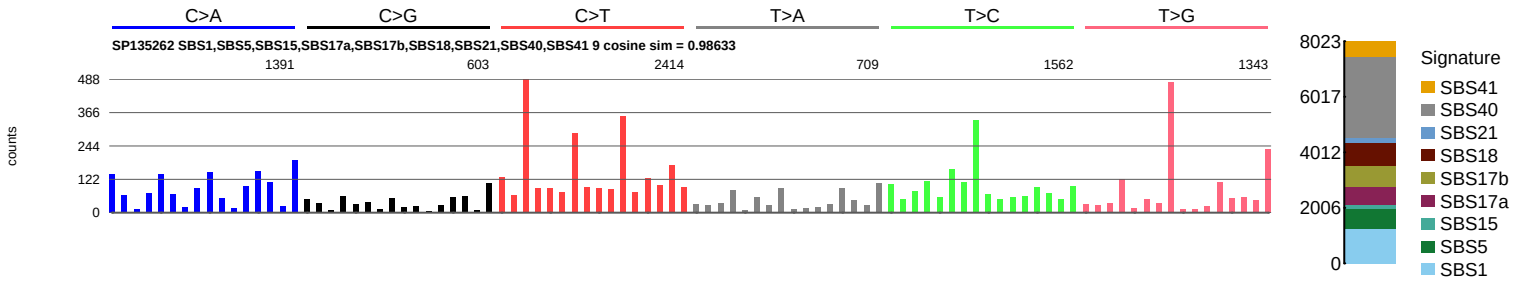
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98635
1390
620
2412
703
1564
1335
counts
488
366
244
122
0
8023
6017
4012
2006
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1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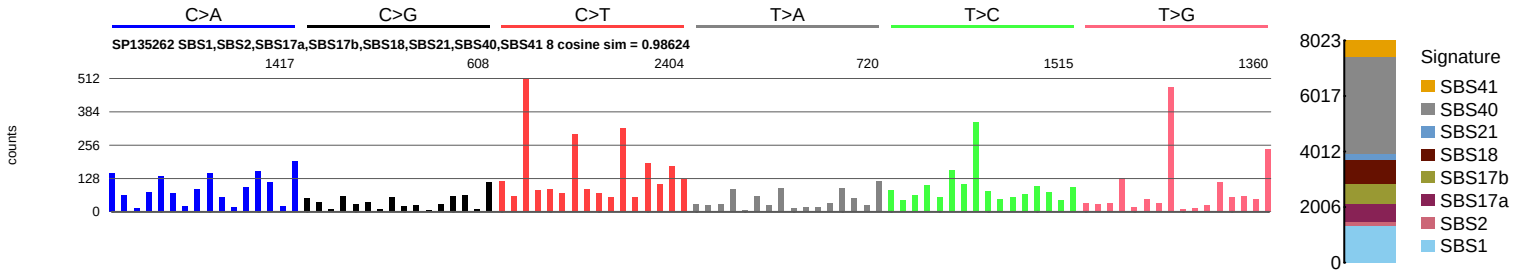
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98633
1391
603
2414
709
1562
1343
counts
488
366
244
122
0
8023
6017
4012
2006
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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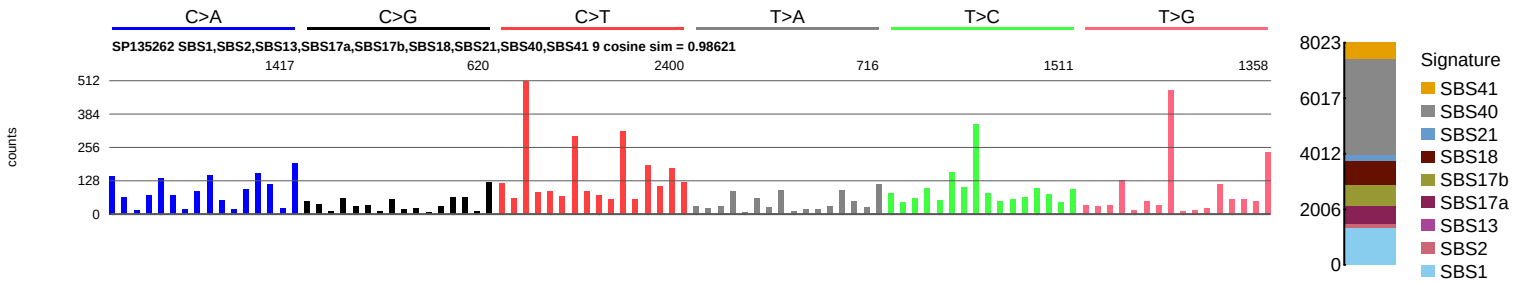
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98624
1417
608
2404
720
1515
1360
counts
512
384
256
128
0
8023
6017
4012
2006
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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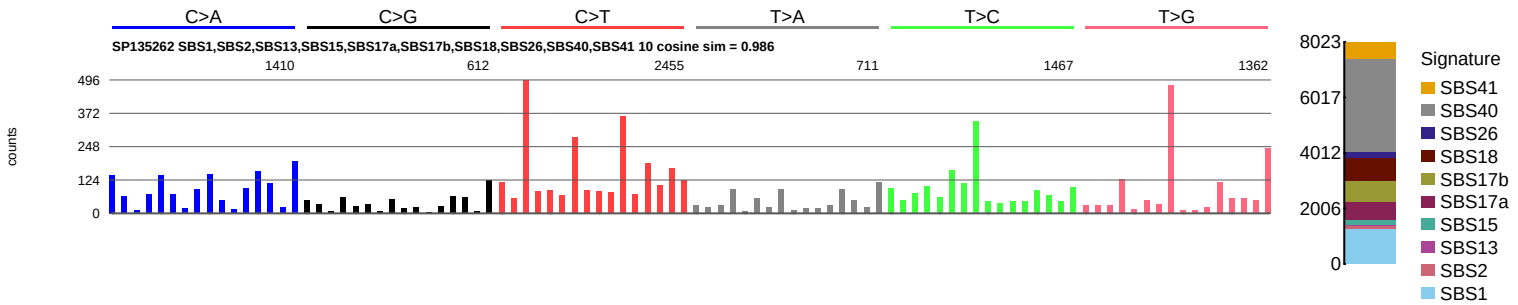
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98621
1417
620
2400
716
1511
1358
counts
512
384
256
128
0
8023
6017
4012
2006
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.986
1410
612
2455
711
1467
1362
counts
496
372
248
124
0
8023
6017
4012
2006
0
Signature
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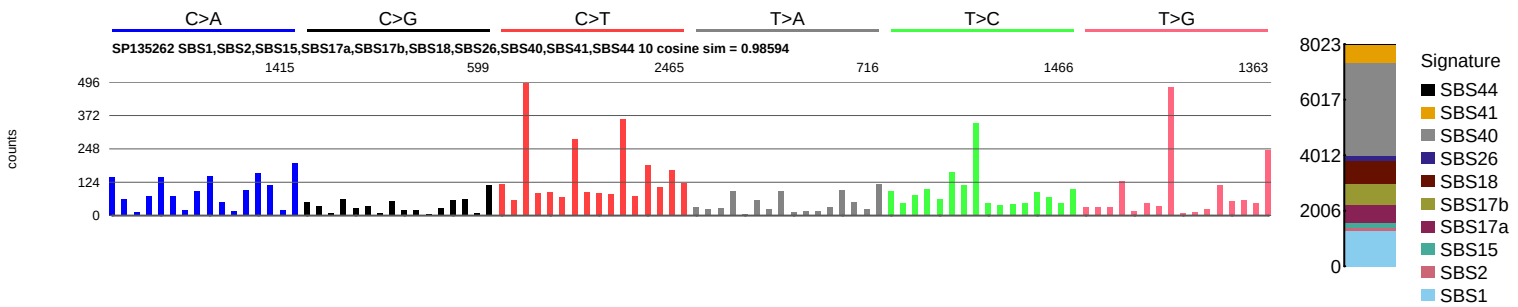
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98594
1415
599
2465
716
1466
1363
counts
496
372
248
124
0
8023
6017
4012
2006
0
Signature
SBS44
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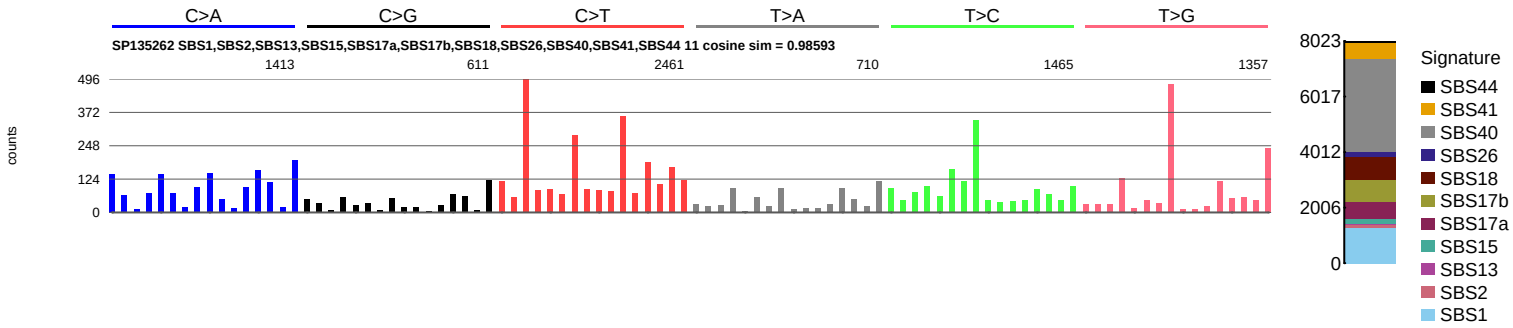
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
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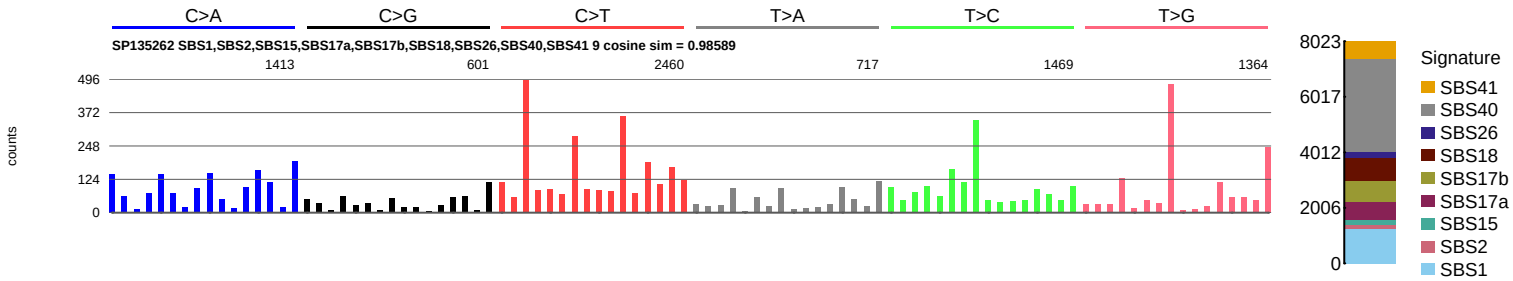
T>G
SP135262 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98593
1413
611
2461
710
1465
1357
counts
496
372
248
124
0
8023
6017
4012
2006
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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SP135262 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98589
1413
601
2460
717
1469
1364
counts
496
372
248
124
0
8023
6017
4012
2006
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Signature
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SBS40
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SBS18
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SBS17a
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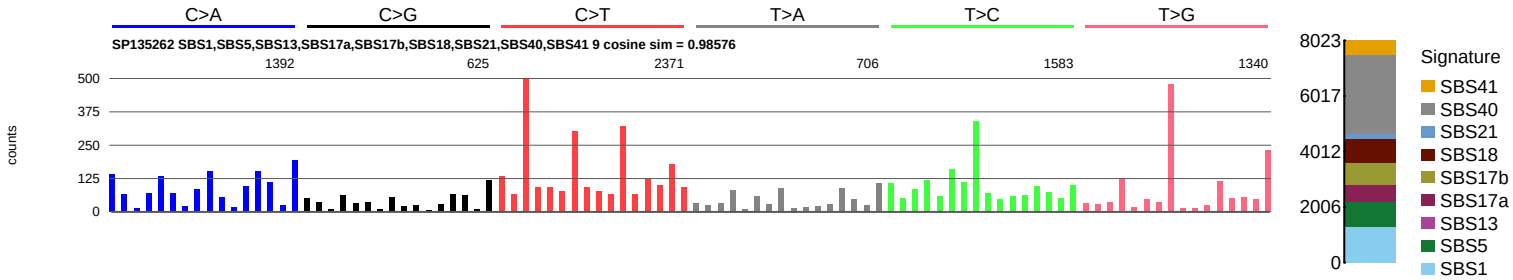
SBS2
SBS1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T>G
SP135262 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98576
1392
625
2371
706
1583
1340
counts
500
375
250
125
0
8023
6017
4012
2006
0
Signature
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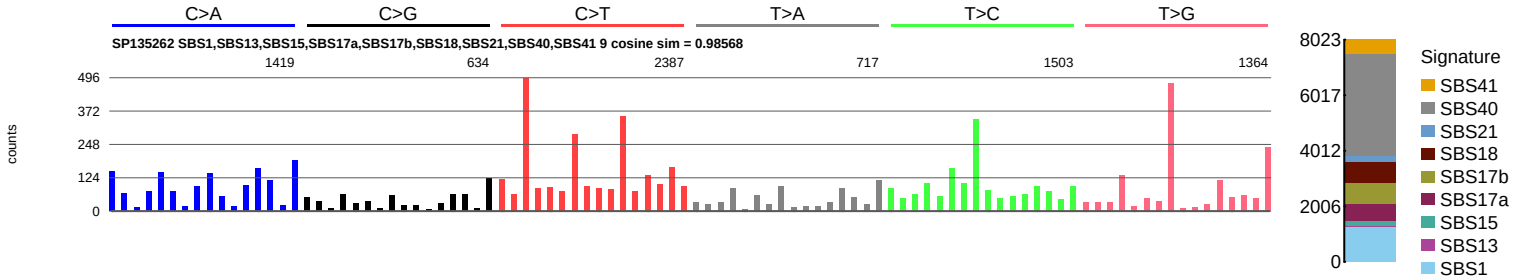
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1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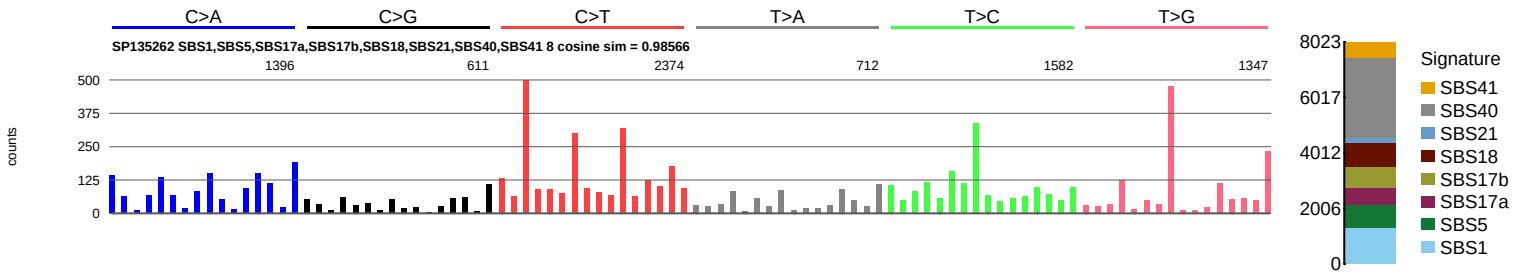
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98568
1419
634
2387
717
1503
1364
counts
496
372
248
124
0
8023
6017
4012
2006
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1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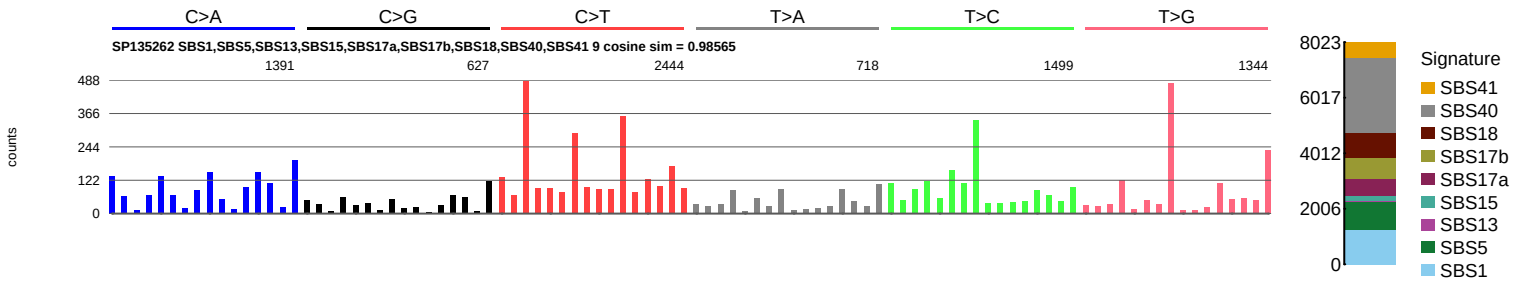
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98566
1396
611
2374
712
1582
1347
counts
500
375
250
125
0
8023
6017
4012
2006
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1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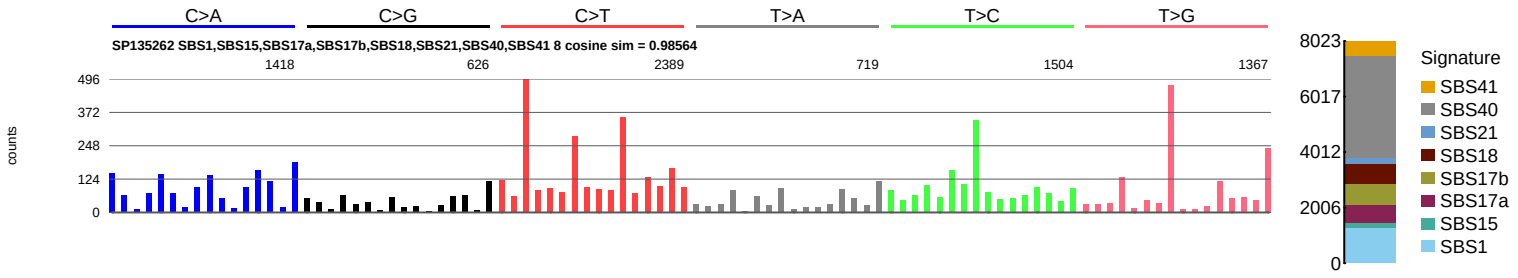
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98565
1391
627
2444
718
1499
1344
counts
488
366
244
122
0
8023
6017
4012
2006
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1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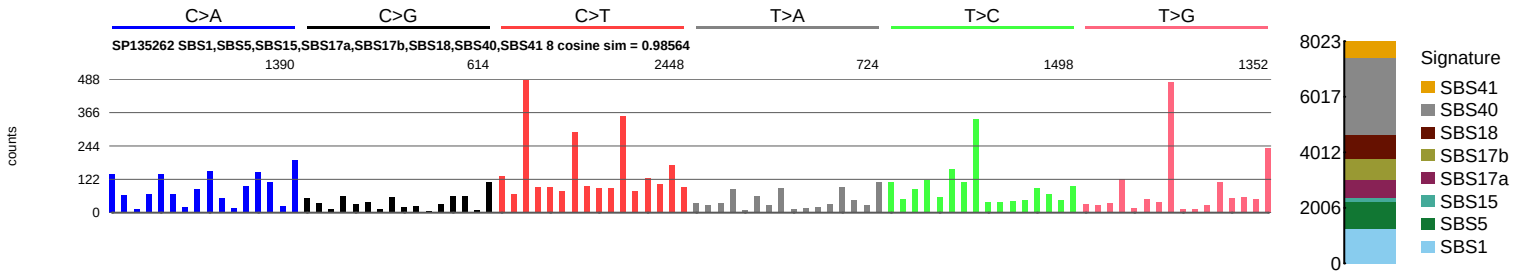
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98564
1418
626
2389
719
1504
1367
counts
496
372
248
124
0
8023
6017
4012
2006
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1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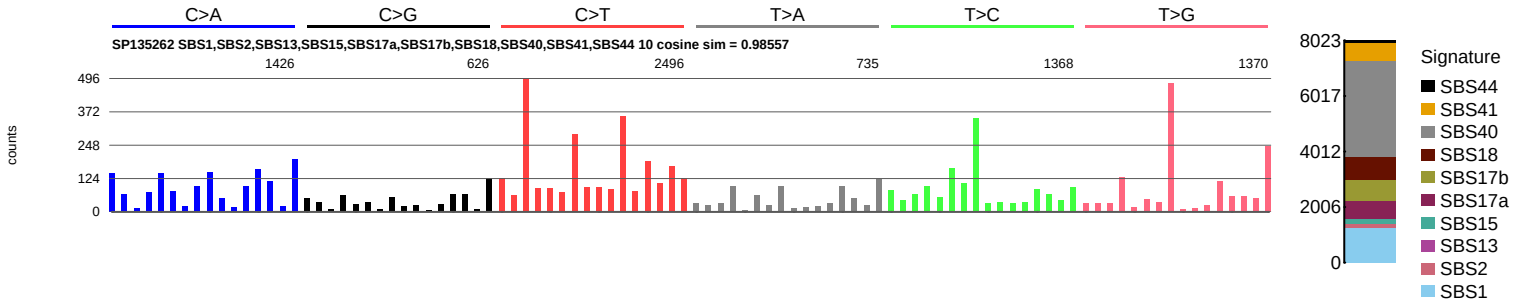
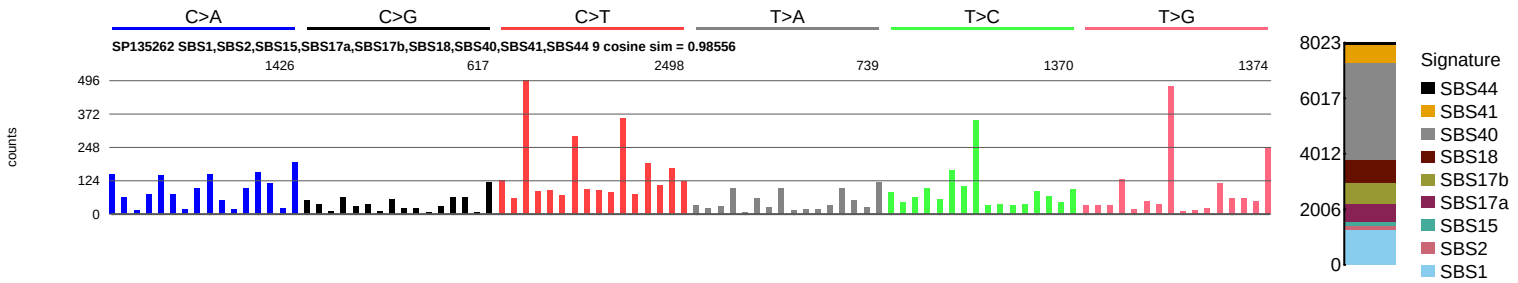
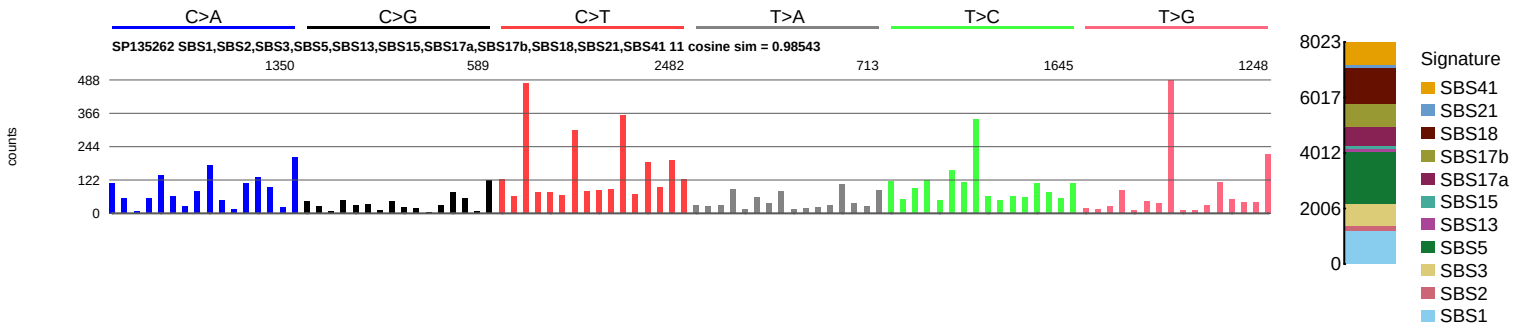
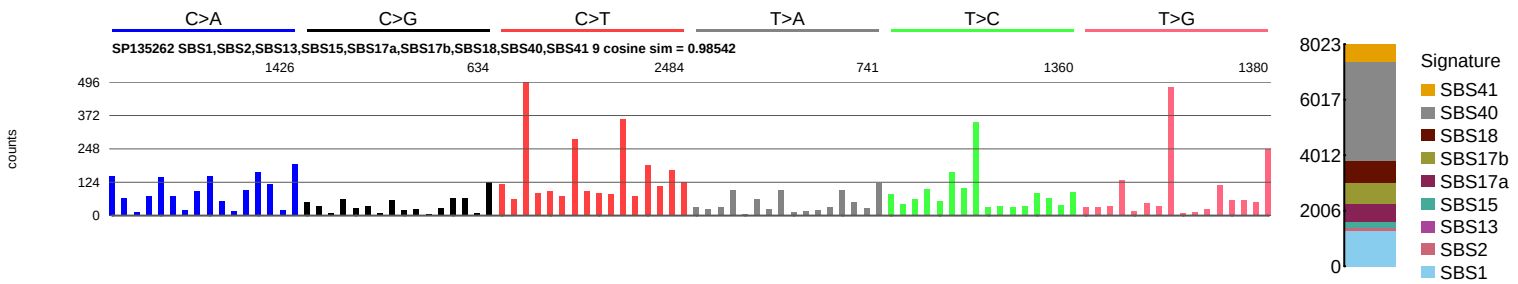
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98564
1390
614
2448
724
1498
1352
counts
488
366
244
122
0
8023
6017
4012
2006
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98557
1426
626
2496
735
1368
1370
counts
496
372
248
124
0
8023
6017
4012
2006
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98556
1426
617
2498
739
1370
1374
counts
496
372
248
124
0
8023
6017
4012
2006
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 11 cosine sim = 0.98543
1350
589
2482
713
1645
1248
counts
488
366
244
122
0
8023
6017
4012
2006
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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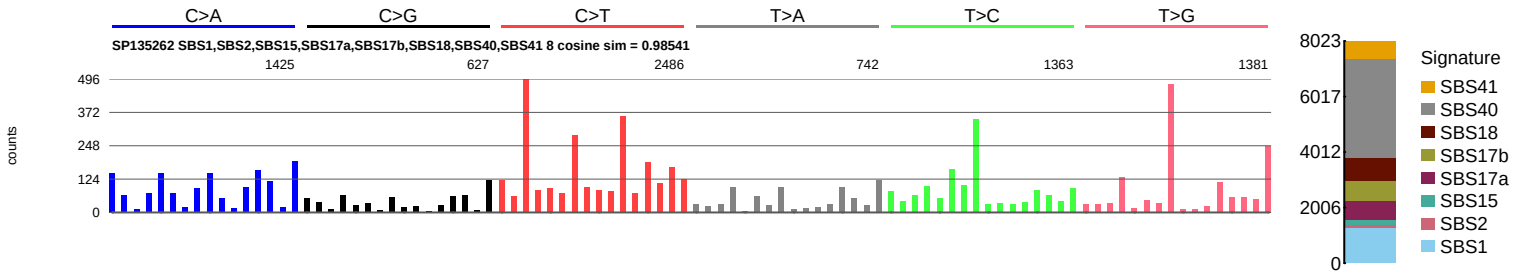
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98542
1426
634
2484
741
1360
1380
counts
496
372
248
124
0
8023
6017
4012
2006
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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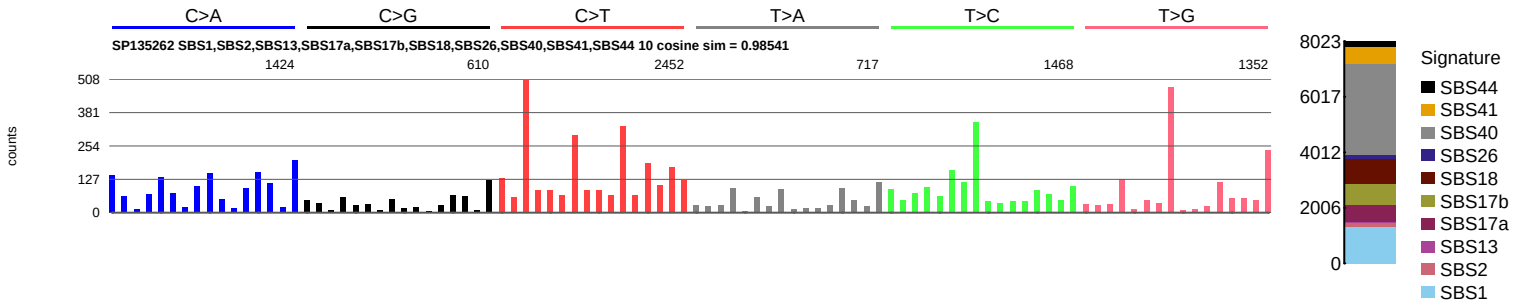
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98541
1425
627
2486
742
1363
1381
counts
496
372
248
124
0
8023
6017
4012
2006
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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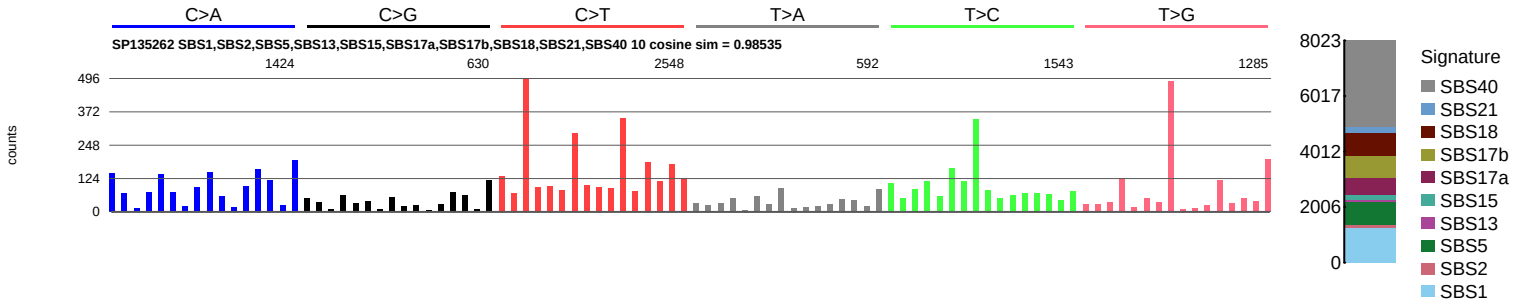
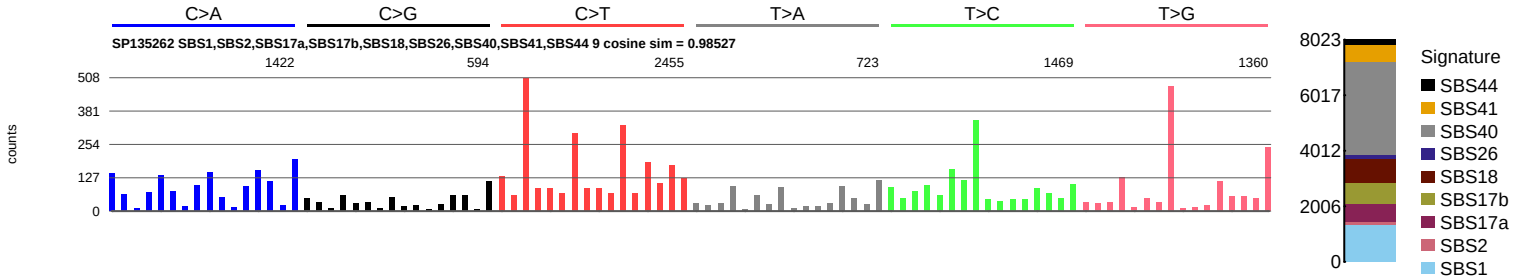
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98541
1424
610
2452
717
1468
1352
counts
508
381
254
127
0
8023
6017
4012
2006
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 10 cosine sim = 0.98535
1424
630
2548
592
1543
1285
counts
496
372
248
124
0
8023
6017
4012
2006
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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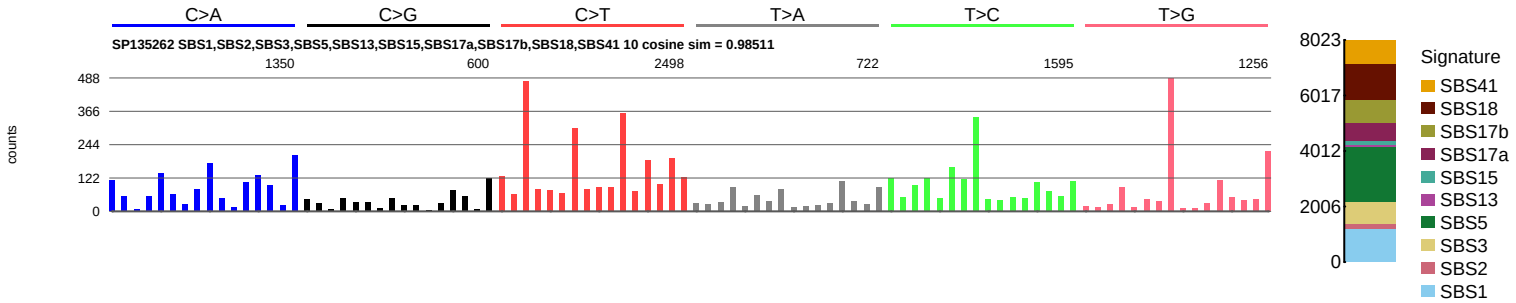
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98527
1422
594
2455
723
1469
1360
counts
508
381
254
127
0
8023
6017
4012
2006
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1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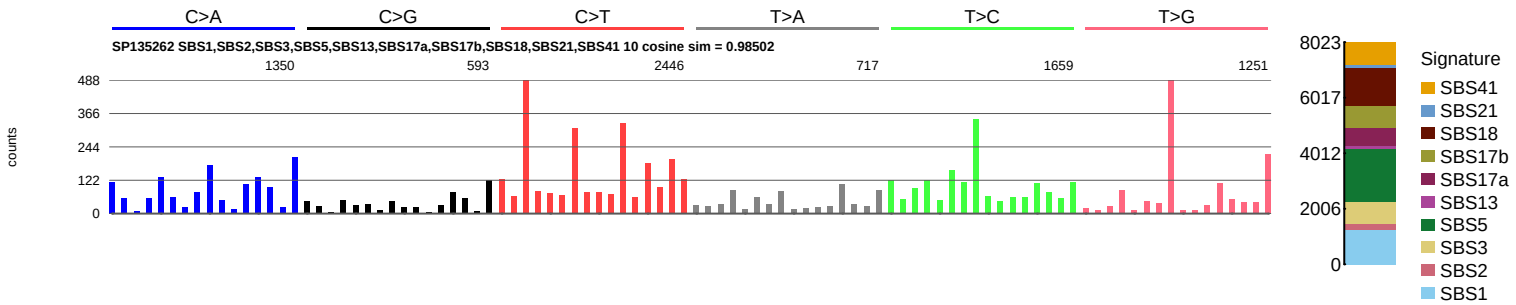
T>C
T>G
SP135262 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 10 cosine sim = 0.98511
1350
600
2498
722
1595
1256
counts
488
366
244
122
0
8023
6017
4012
2006
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
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T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98502
1350
593
2446
717
1659
1251
counts
488
366
244
122
0
8023
6017
4012
2006
0
Signature
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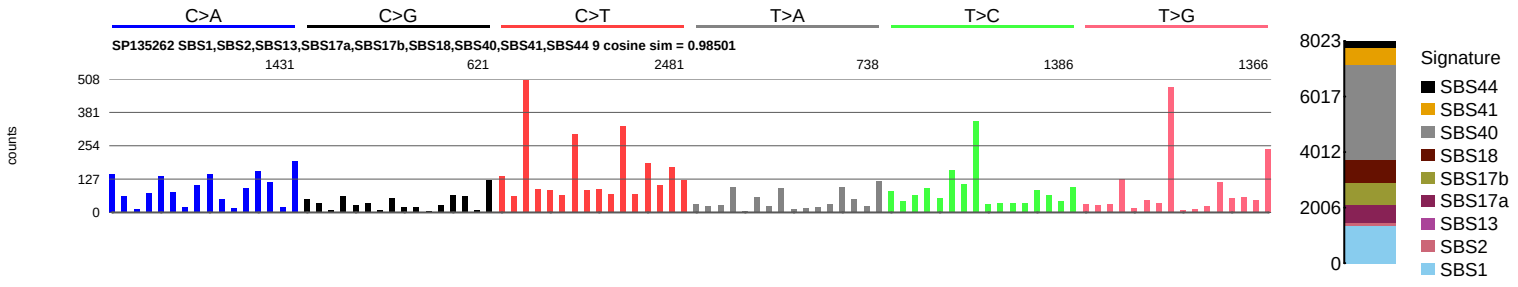
SBS2
SBS1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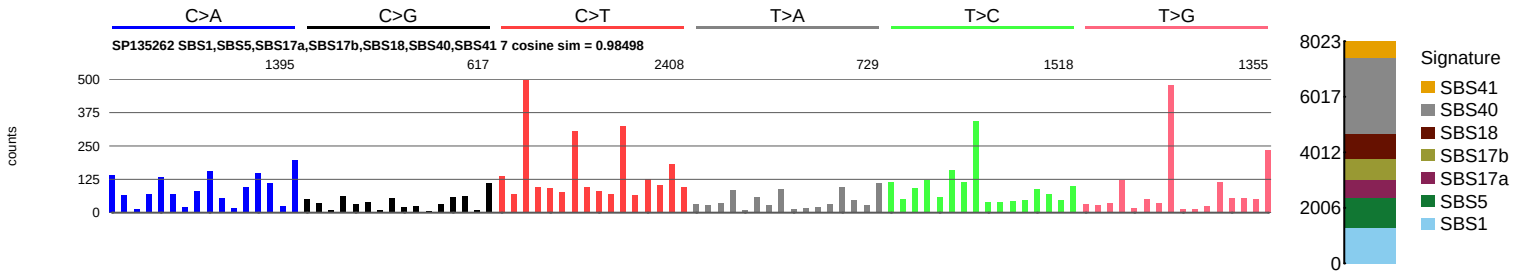
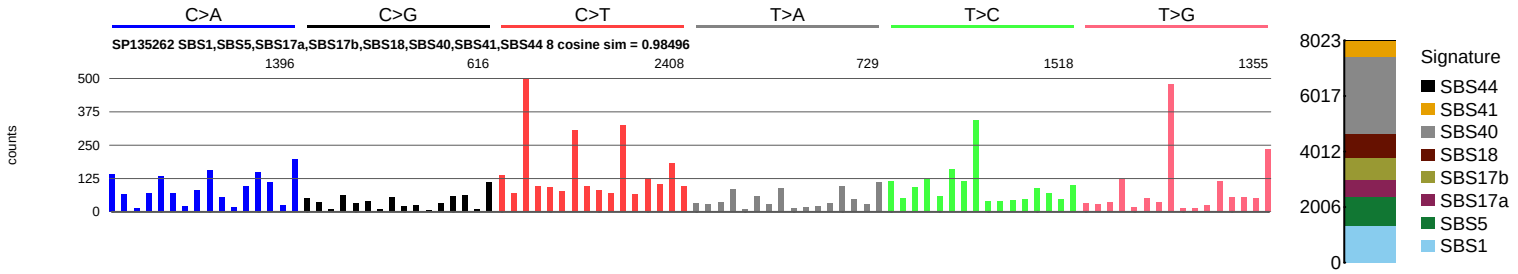
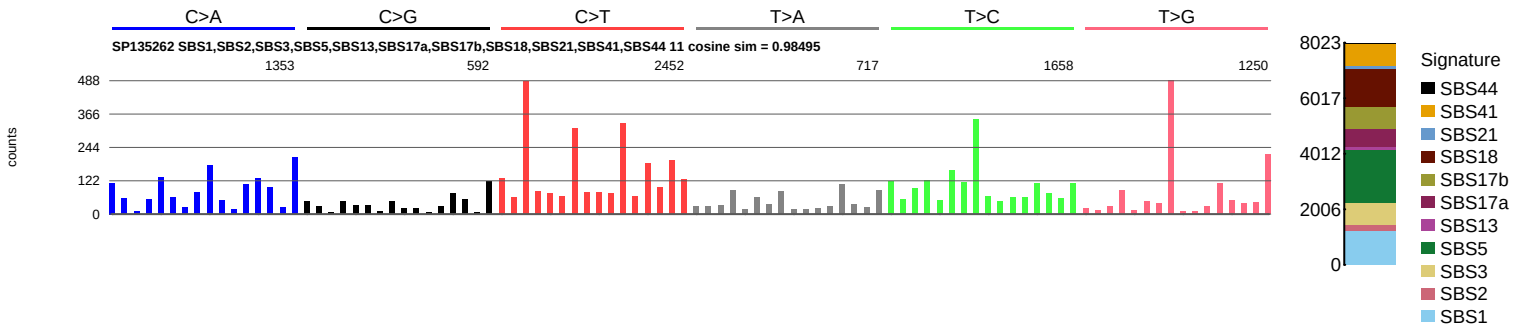
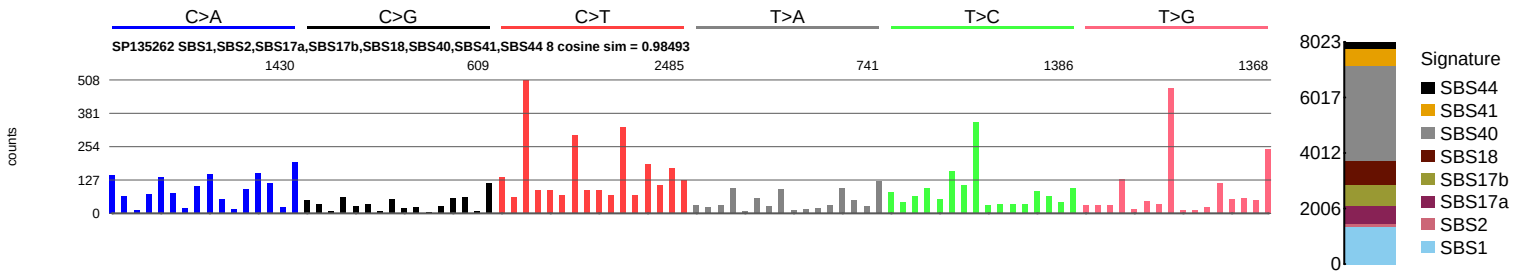
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98501
1431
621
2481
738
1386
1366
counts
508
381
254
127
0
8023
6017
4012
2006
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98498
1395
617
2408
729
1518
1355
counts
500
375
250
125
0
8023
6017
4012
2006
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98496
1396
616
2408
729
1518
1355
counts
500
375
250
125
0
8023
6017
4012
2006
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.98495
1353
592
2452
717
1658
1250
counts
488
366
244
122
0
8023
6017
4012
2006
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98493
1430
609
2485
741
1386
1368
counts
508
381
254
127
0
8023
6017
4012
2006
0
Signature
SBS44
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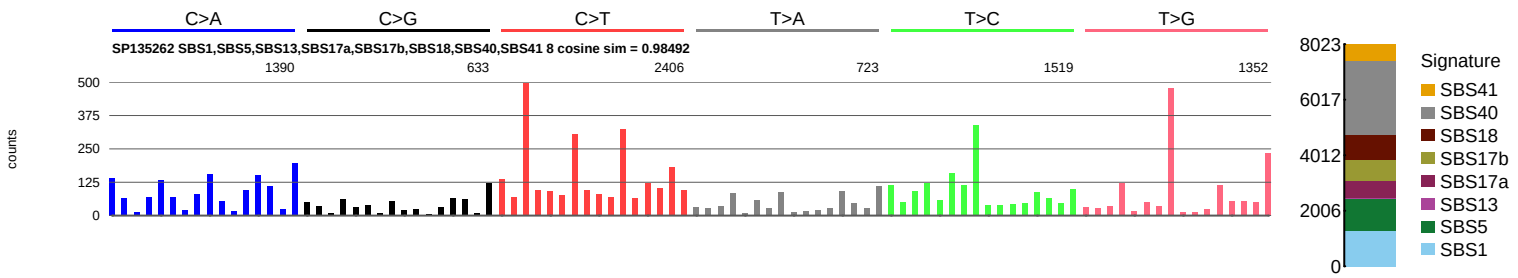
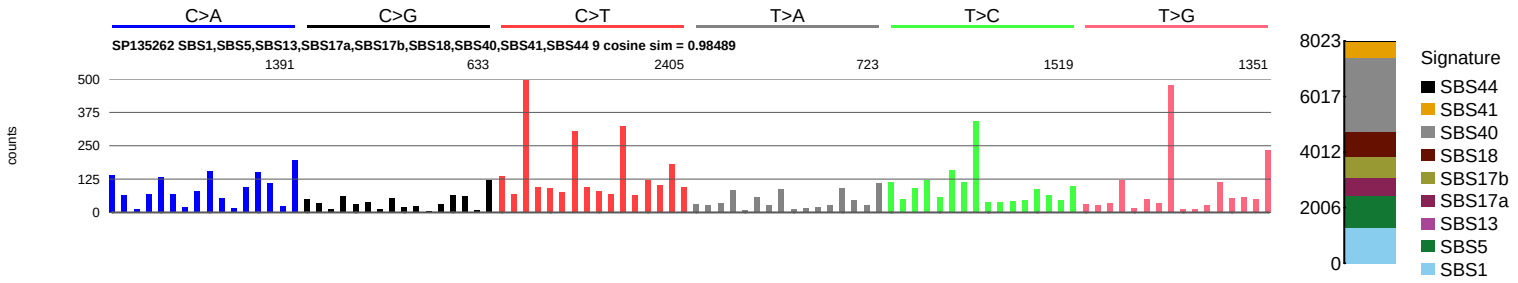
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98492
1390
633
2406
723
1519
1352
counts
500
375
250
125
0
8023
6017
4012
2006
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1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C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98489
1391
633
2405
723
1519
1351
counts
500
375
250
125
0
8023
6017
4012
2006
0
Signature
SBS44
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SBS40
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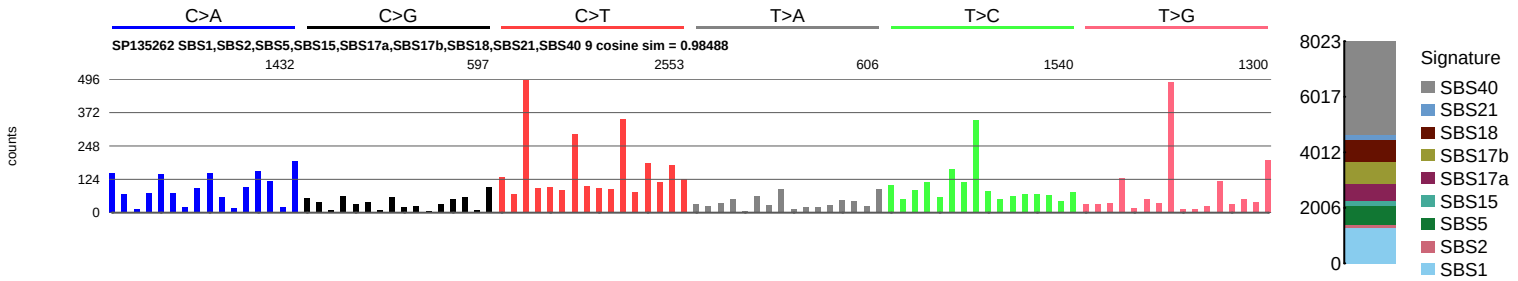
SBS1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SP135262 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98488
1432
597
2553
606
1540
1300
counts
496
372
248
124
0
8023
6017
4012
2006
0
Signature
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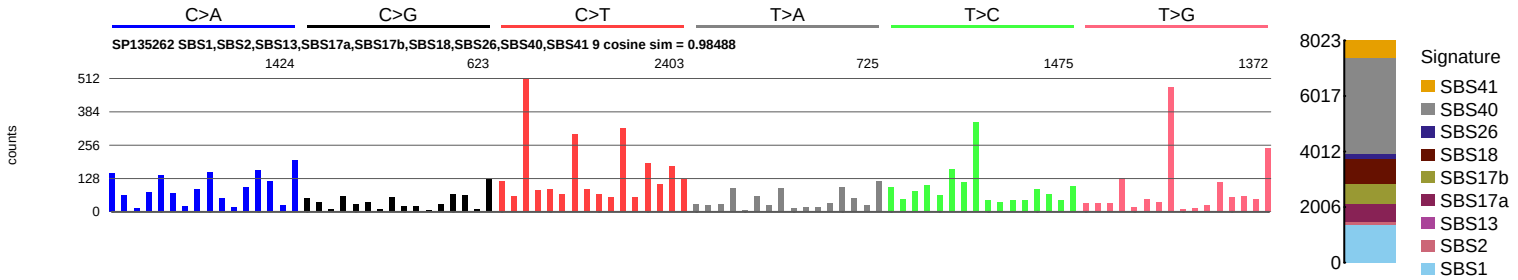
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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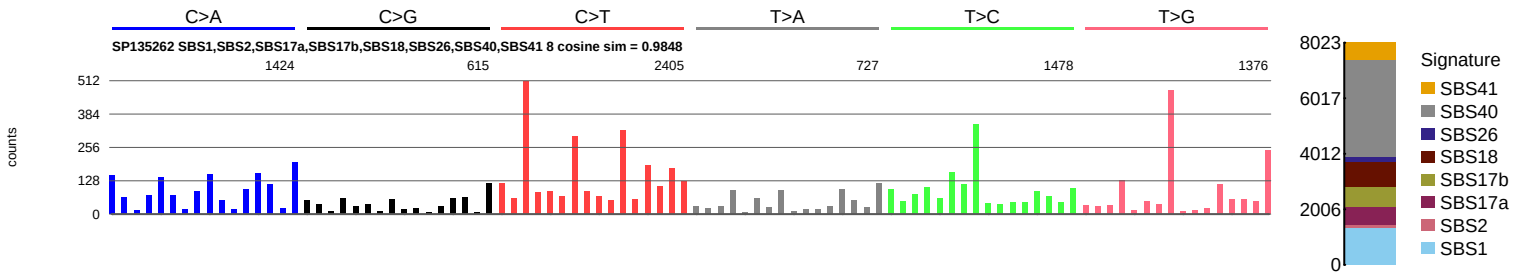
SP135262 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98488
1424
623
2403
725
1475
1372
counts
512
384
256
128
0
8023
6017
4012
2006
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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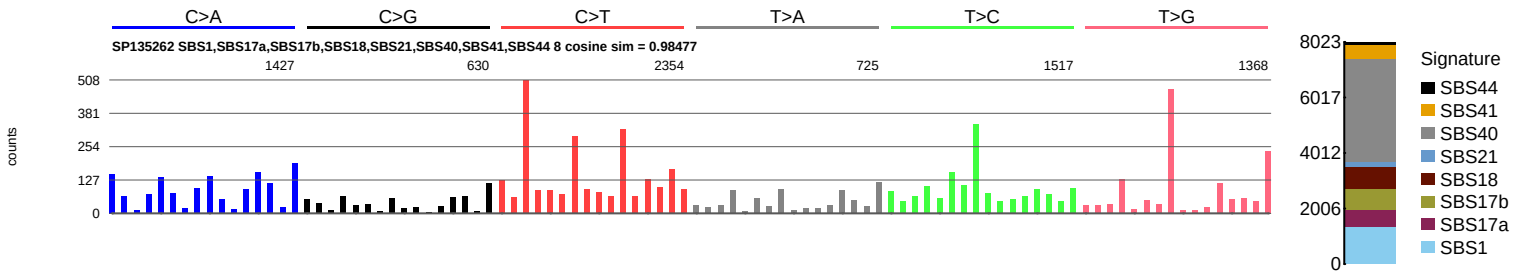
SP135262 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.9848
1424
615
2405
727
1478
1376
counts
512
384
256
128
0
8023
6017
4012
2006
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1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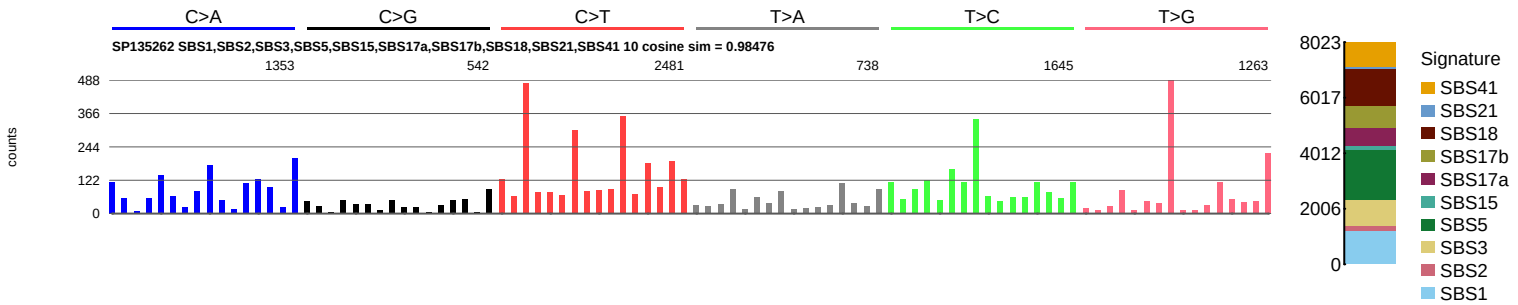
SP135262 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98477
1427
630
2354
725
1517
1368
counts
508
381
254
127
0
8023
6017
4012
2006
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS1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C>G
C>T
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T>C
T>G
SP135262 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98476
1353
542
2481
738
1645
1263
counts
488
366
244
122
0
8023
6017
4012
2006
0
Signature
SBS41
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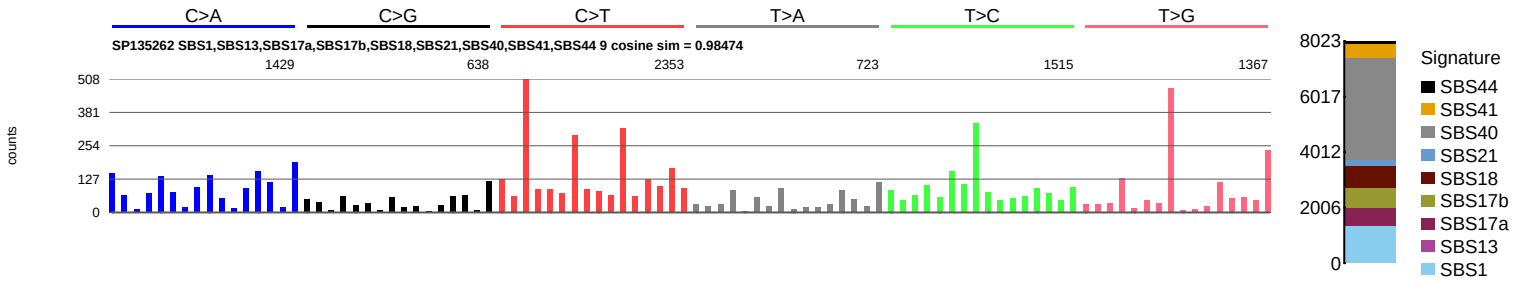
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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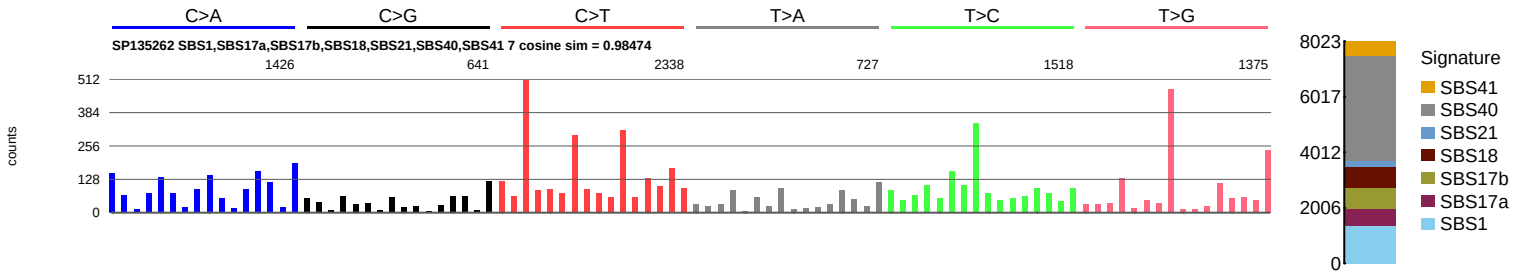
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98474
1429
638
2353
723
1515
1367
counts
508
381
254
127
0
8023
6017
4012
2006
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1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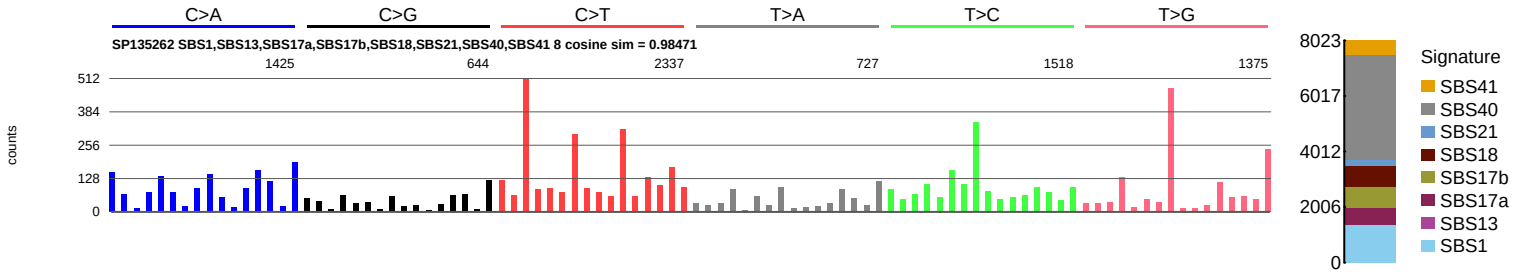
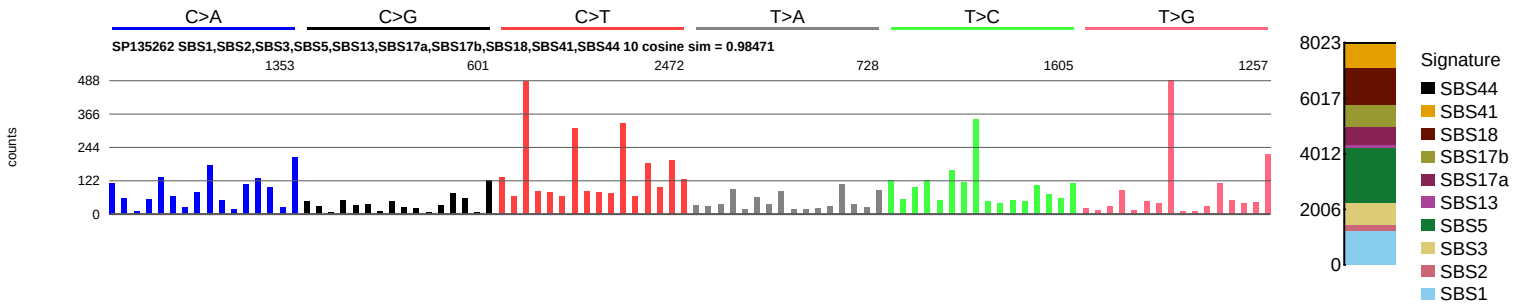
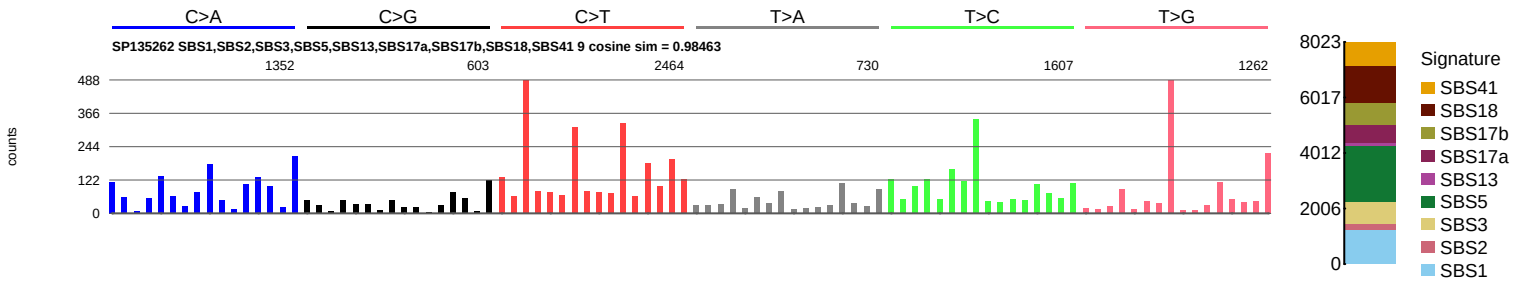
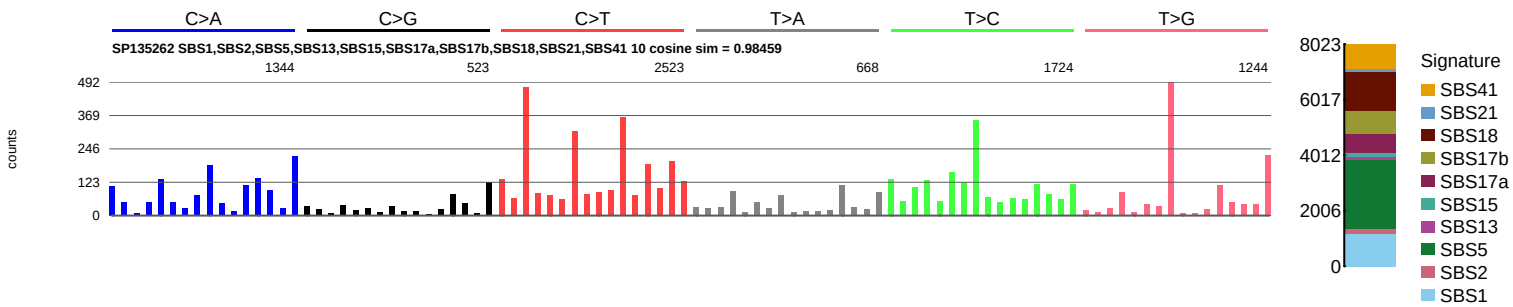
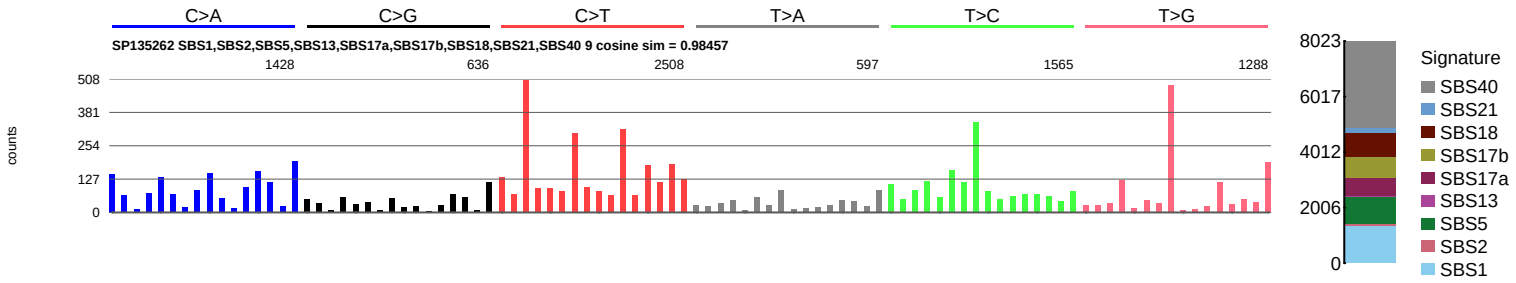
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98474
1426
641
2338
727
1518
1375
counts
512
384
256
128
0
8023
6017
4012
2006
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98471
1425
644
2337
727
1518
1375
counts
512
384
256
128
0
8023
6017
4012
2006
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.98471
1353
601
2472
728
1605
1257
counts
488
366
244
122
0
8023
6017
4012
2006
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98463
1352
603
2464
730
1607
1262
counts
488
366
244
122
0
8023
6017
4012
2006
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98459
1344
523
2523
668
1724
1244
counts
492
369
246
123
0
8023
6017
4012
2006
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98457
1428
636
2508
597
1565
1288
counts
508
381
254
127
0
8023
6017
4012
2006
0
Signature
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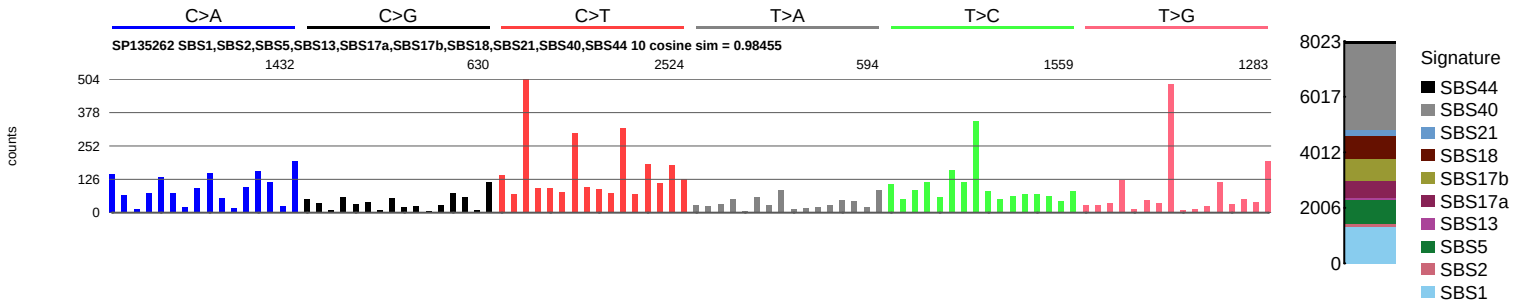
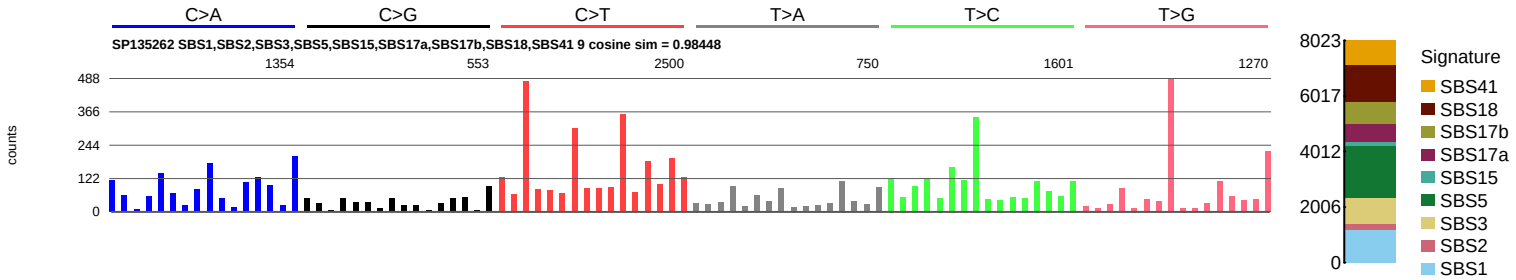
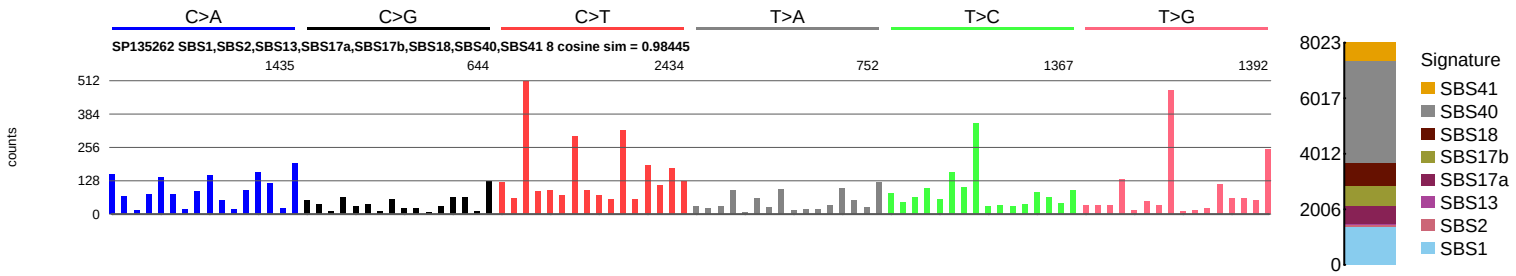
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98455
1432
630
2524
594
1559
1283
counts
504
378
252
126
0
8023
6017
4012
2006
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98448
1354
553
2500
750
1601
1270
counts
488
366
244
122
0
8023
6017
4012
2006
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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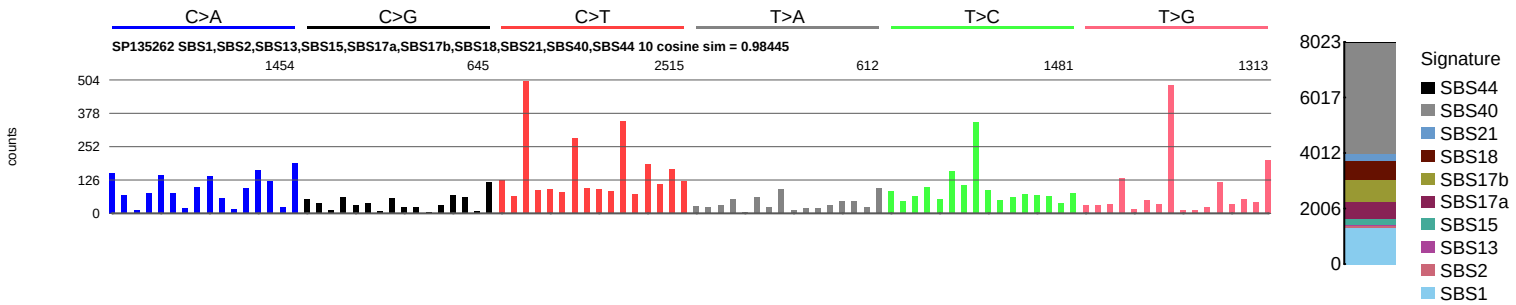
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98445
1435
644
2434
752
1367
1392
counts
512
384
256
128
0
8023
6017
4012
2006
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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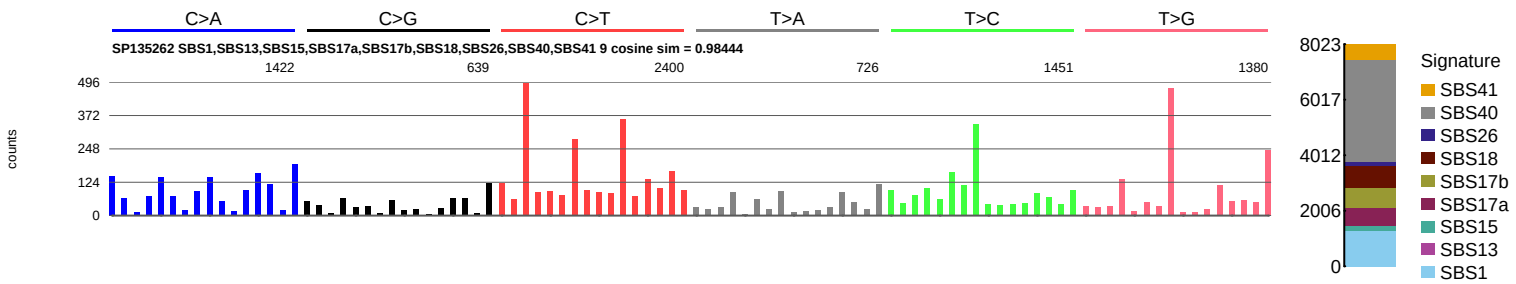
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98445
1454
645
2515
612
1481
1313
counts
504
378
252
126
0
8023
6017
4012
2006
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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C>G
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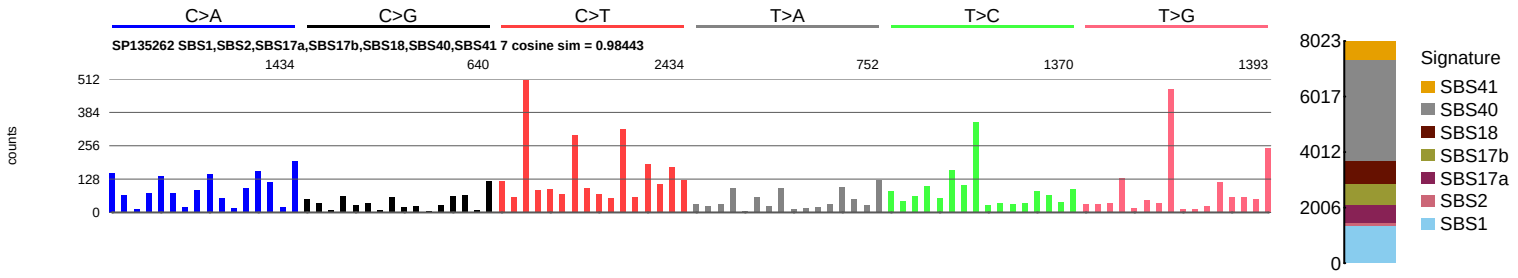
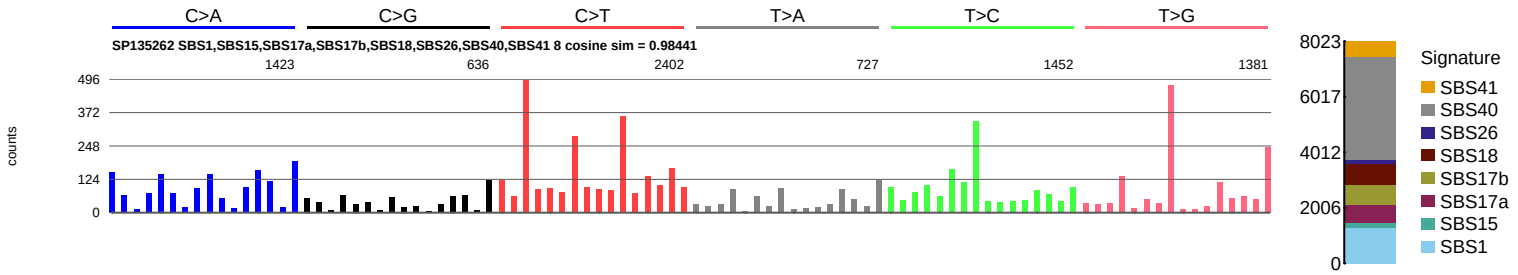
T>A
T>C
T>G
SP135262 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98444
1422
639
2400
726
1451
1380
counts
496
372
248
124
0
8023
6017
4012
2006
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98443
1434
640
2434
752
1370
1393
counts
512
384
256
128
0
8023
6017
4012
2006
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1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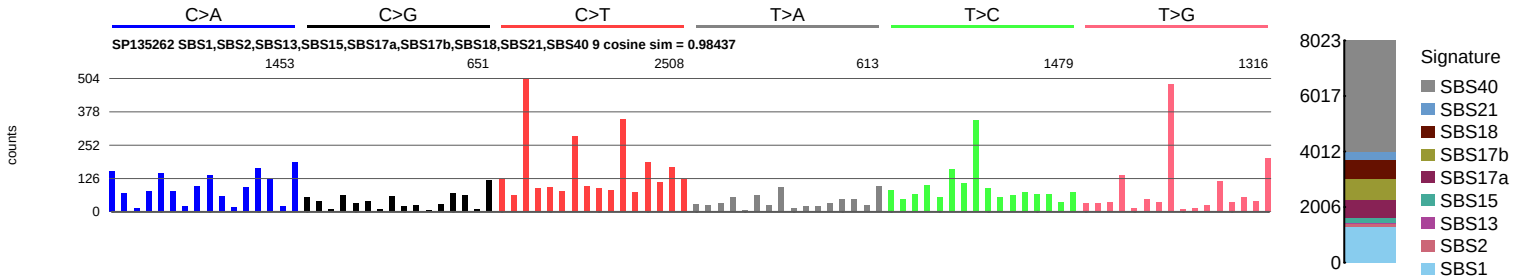
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98441
1423
636
2402
727
1452
1381
counts
496
372
248
124
0
8023
6017
4012
2006
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1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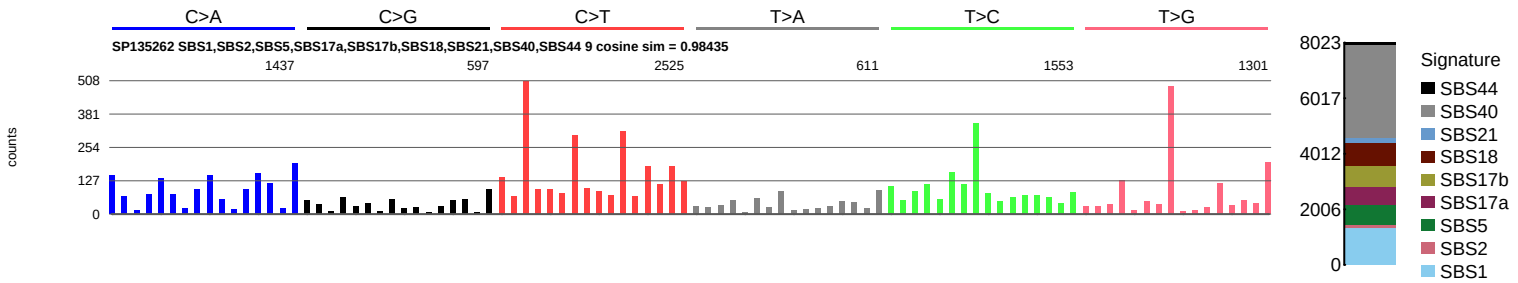
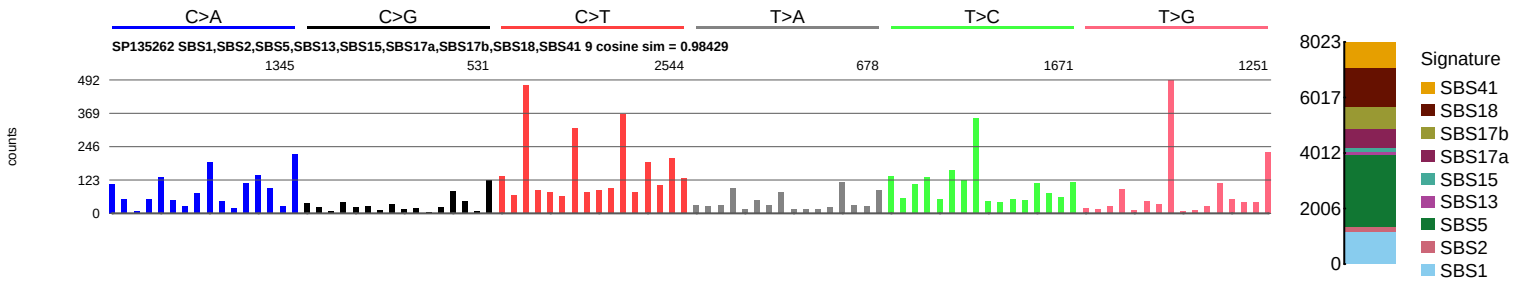
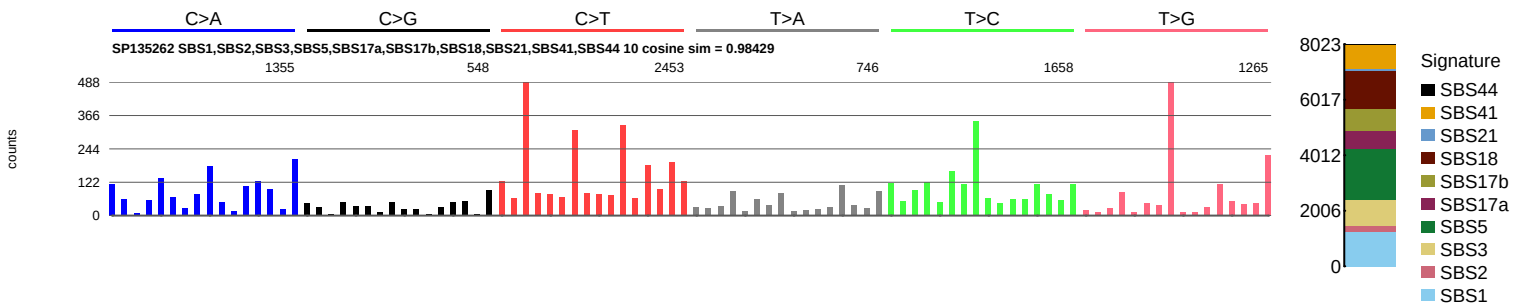
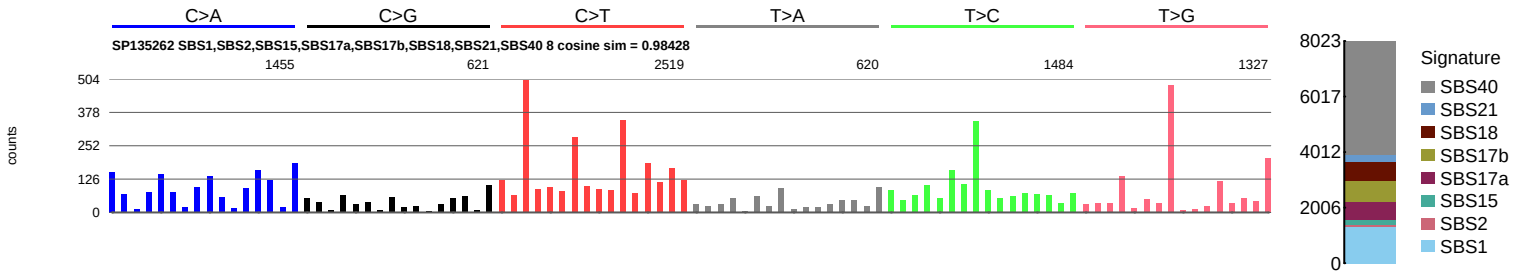
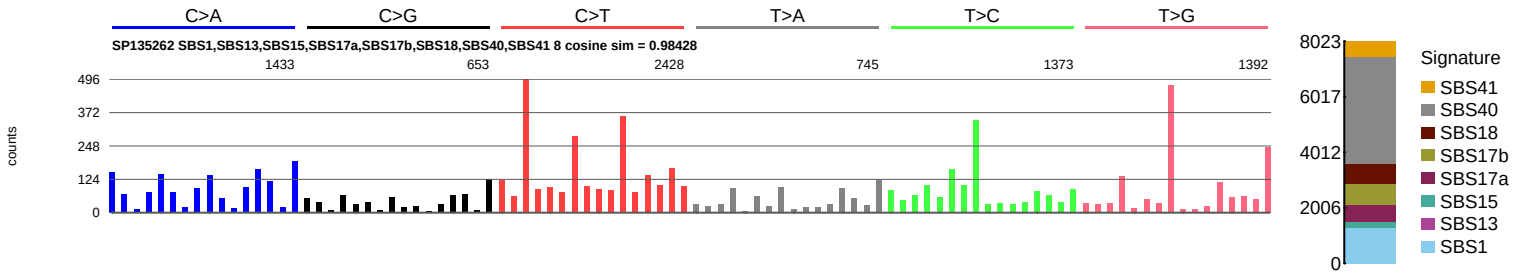
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98437
1453
651
2508
613
1479
1316
counts
504
378
252
126
0
8023
6017
4012
2006
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98435
1437
597
2525
611
1553
1301
counts
508
381
254
127
0
8023
6017
4012
2006
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98429
1345
531
2544
678
1671
1251
counts
492
369
246
123
0
8023
6017
4012
2006
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98429
1355
548
2453
746
1658
1265
counts
488
366
244
122
0
8023
6017
4012
2006
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98428
1455
621
2519
620
1484
1327
counts
504
378
252
126
0
8023
6017
4012
2006
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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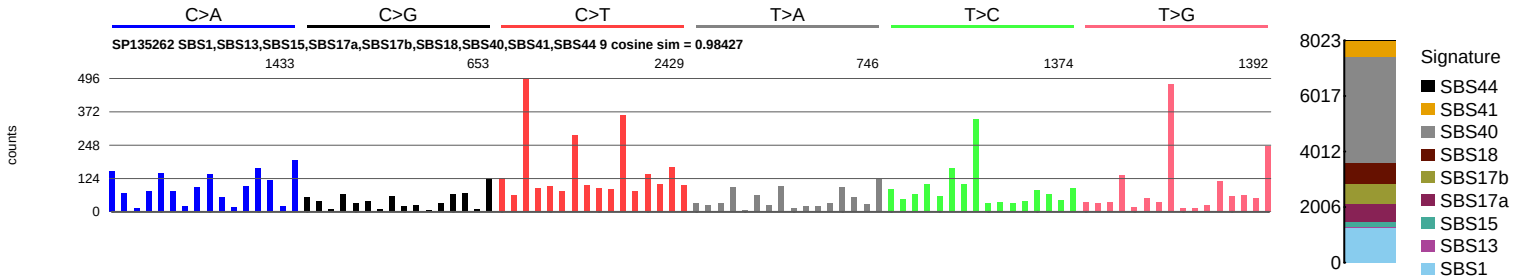
SP135262 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98428
1433
653
2428
745
1373
1392
counts
496
372
248
124
0
8023
6017
4012
2006
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1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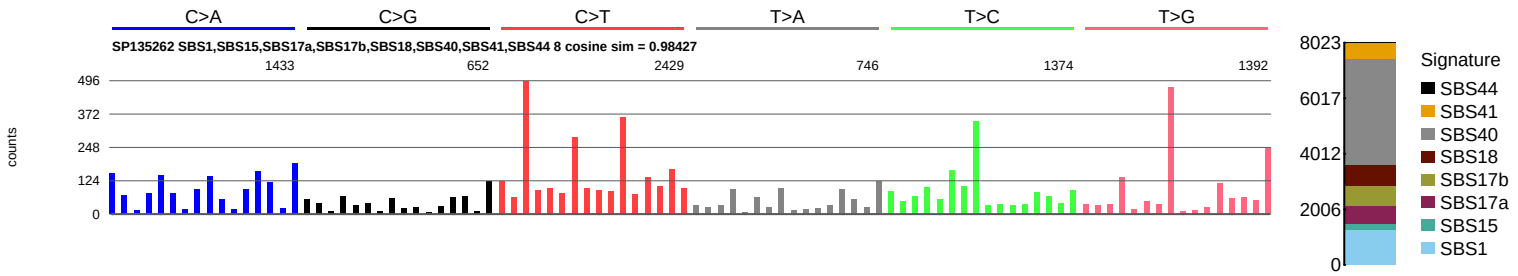
SP135262 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98427
1433
653
2429
746
1374
1392
counts
496
372
248
124
0
8023
6017
4012
2006
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1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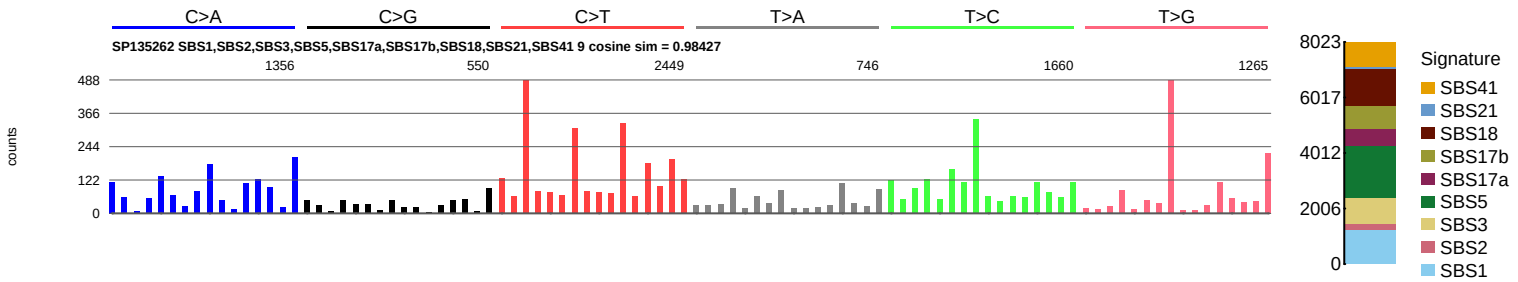
SP135262 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98427
1433
652
2429
746
1374
1392
counts
496
372
248
124
0
8023
6017
4012
2006
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1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T>A
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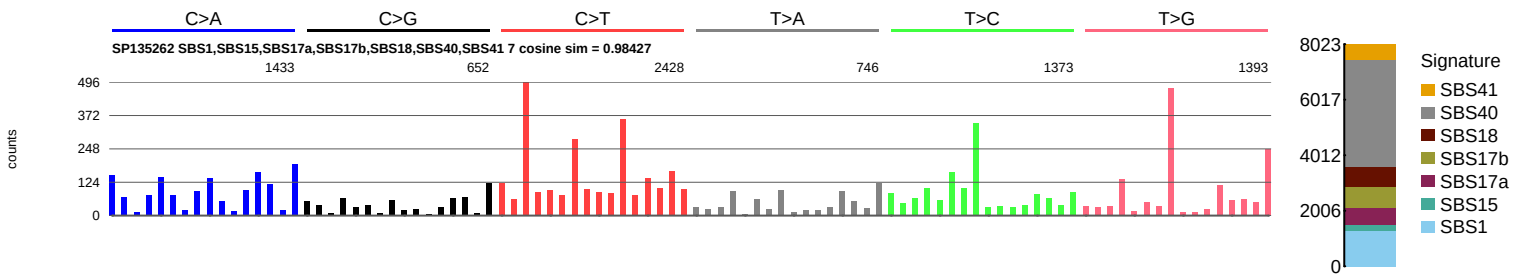
SP135262 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98427
1356
550
2449
746
1660
1265
counts
488
366
244
122
0
8023
6017
4012
2006
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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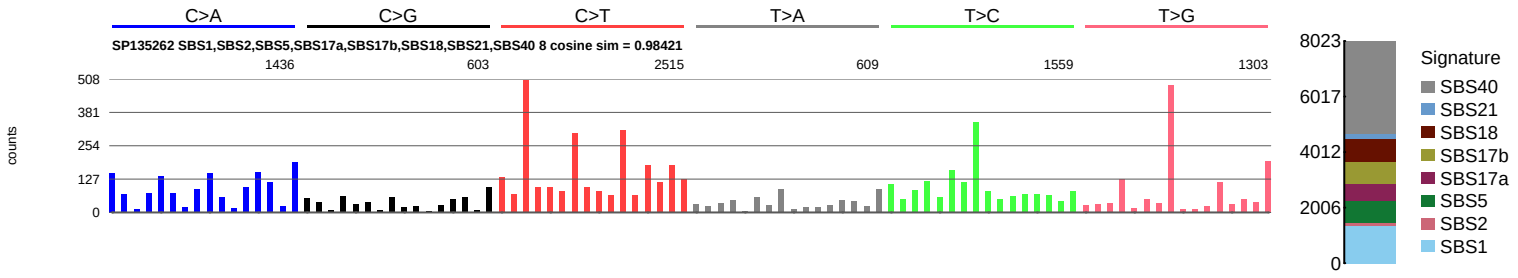
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98427
1433
652
2428
746
1373
1393
counts
496
372
248
124
0
8023
6017
4012
2006
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1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C>G
C>T
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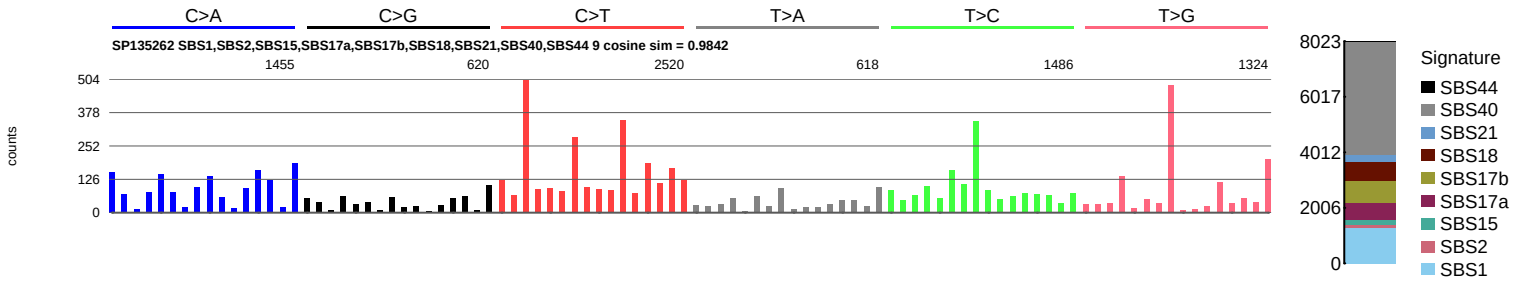
T>C
T>G
SP135262 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98421
1436
603
2515
609
1559
1303
counts
508
381
254
127
0
8023
6017
4012
2006
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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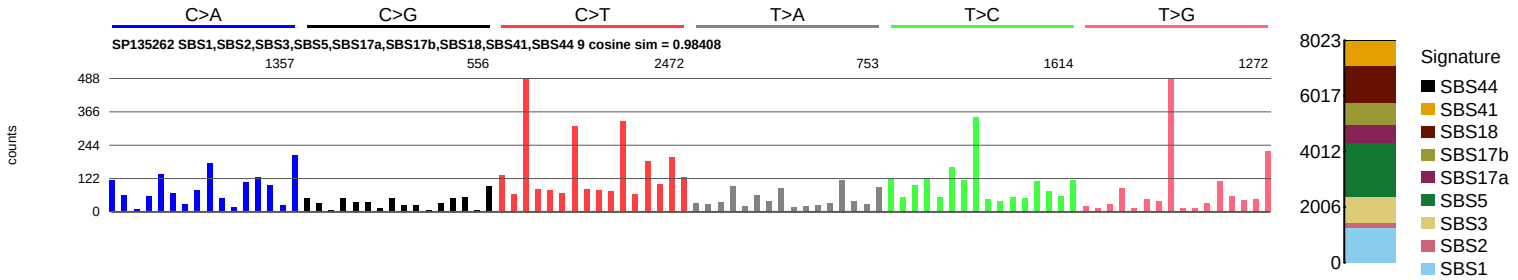
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.9842
1455
620
2520
618
1486
1324
counts
504
378
252
126
0
8023
6017
4012
2006
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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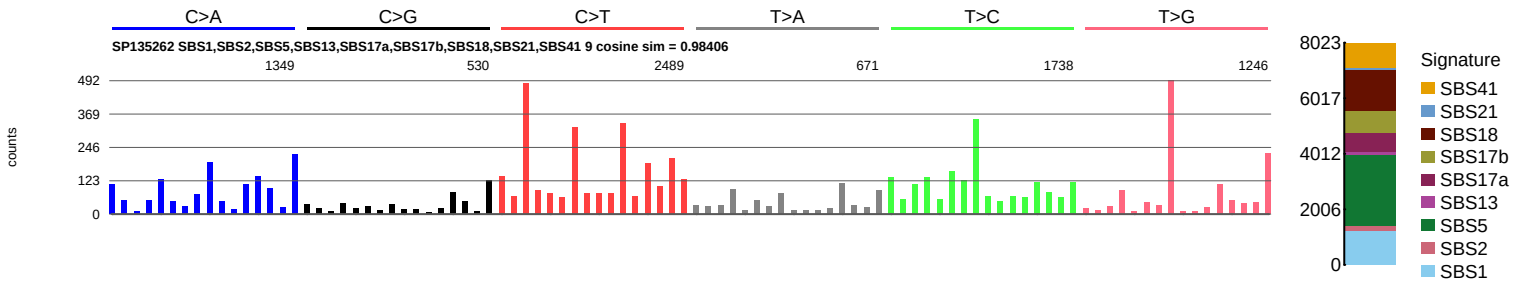
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98408
1357
556
2472
753
1614
1272
counts
488
366
244
122
0
8023
6017
4012
2006
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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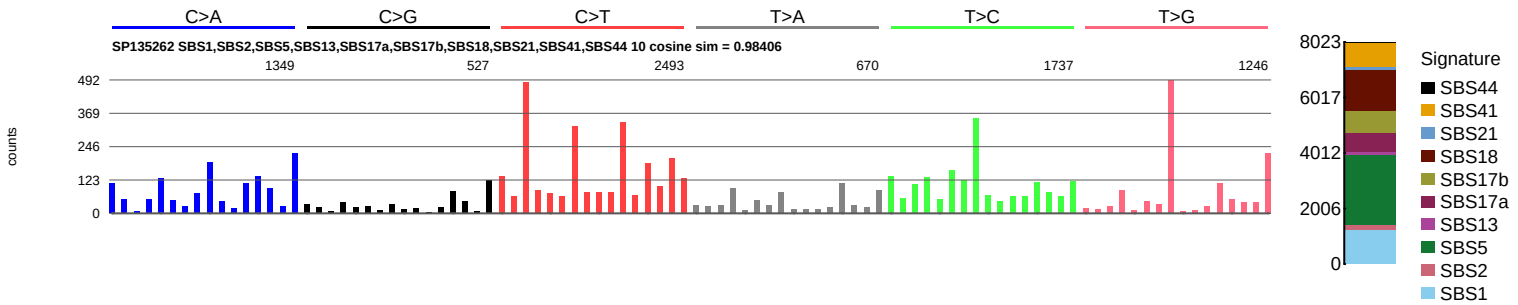
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98406
1349
530
2489
671
1738
1246
counts
492
369
246
123
0
8023
6017
4012
2006
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98406
1349
527
2493
670
1737
1246
counts
492
369
246
123
0
8023
6017
4012
2006
0
Signature
SBS44
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SBS21
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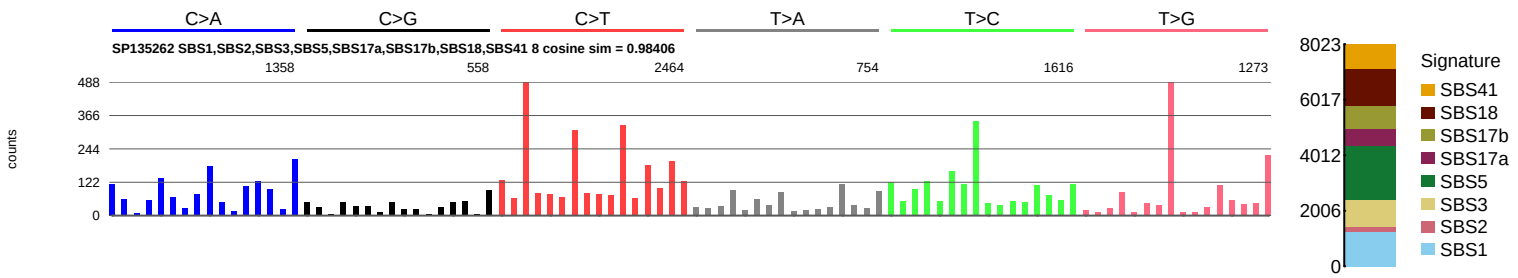
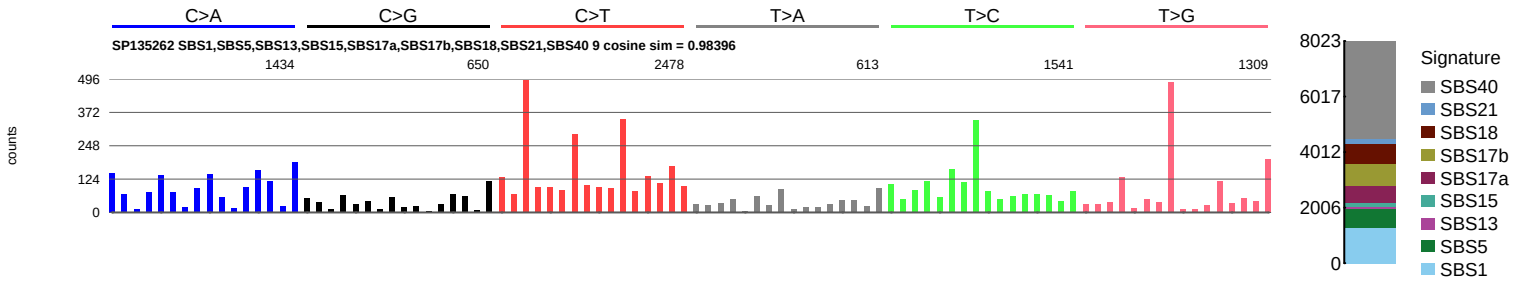
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98406
1358
558
2464
754
1616
1273
counts
488
366
244
122
0
8023
6017
4012
2006
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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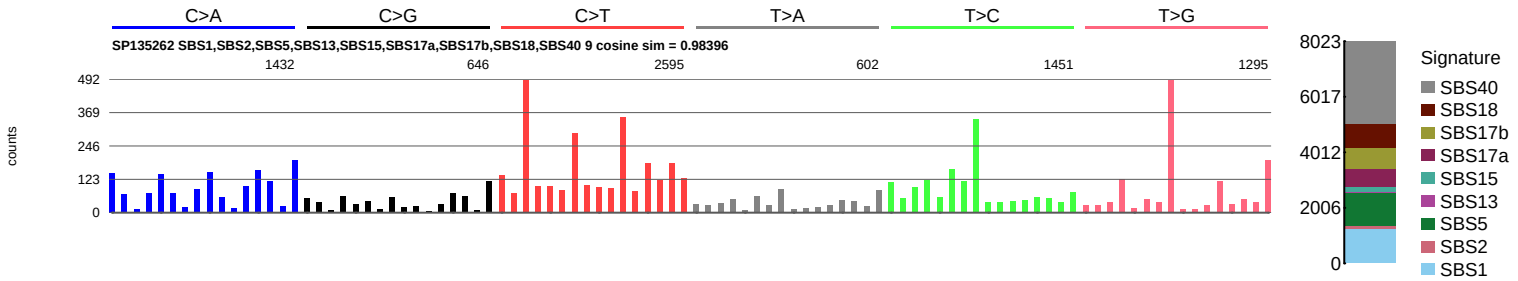
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98396
1434
650
2478
613
1541
1309
counts
496
372
248
124
0
8023
6017
4012
2006
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1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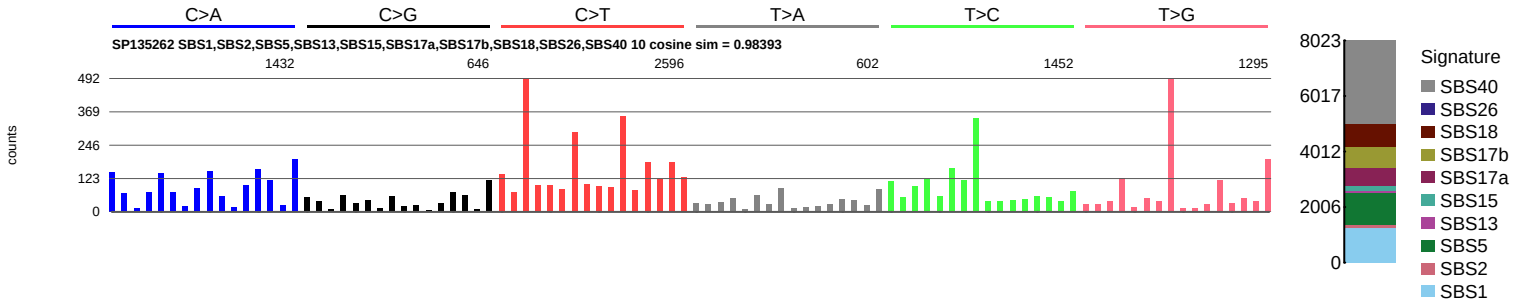
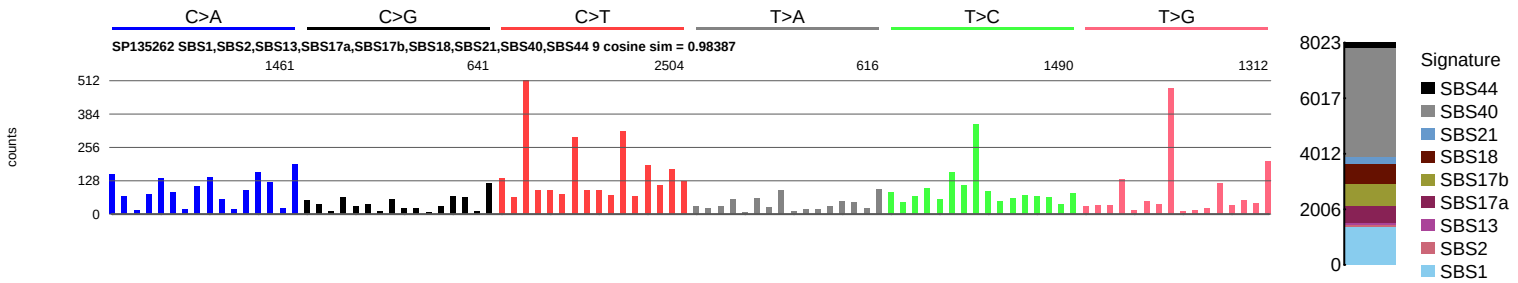
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 9 cosine sim = 0.98396
1432
646
2595
602
1451
1295
counts
492
369
246
123
0
8023
6017
4012
2006
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 10 cosine sim = 0.98393
1432
646
2596
602
1452
1295
counts
492
369
246
123
0
8023
6017
4012
2006
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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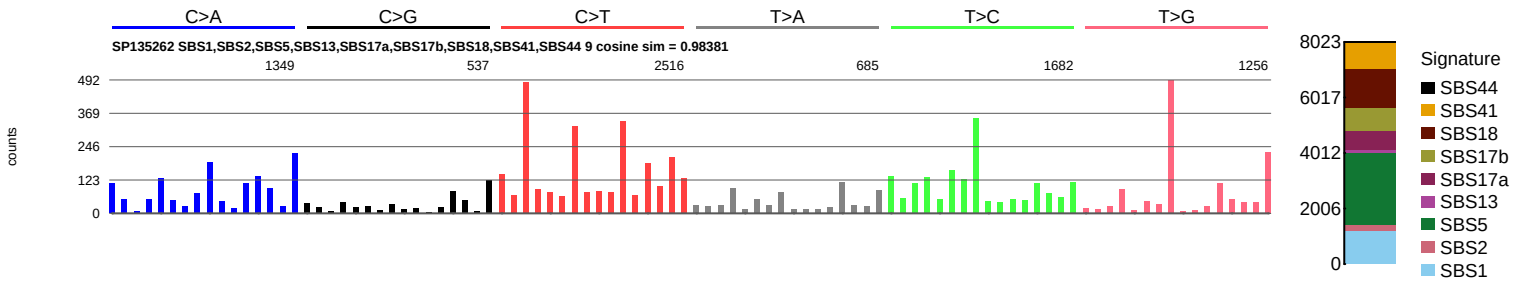
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98387
1461
641
2504
616
1490
1312
counts
512
384
256
128
0
8023
6017
4012
2006
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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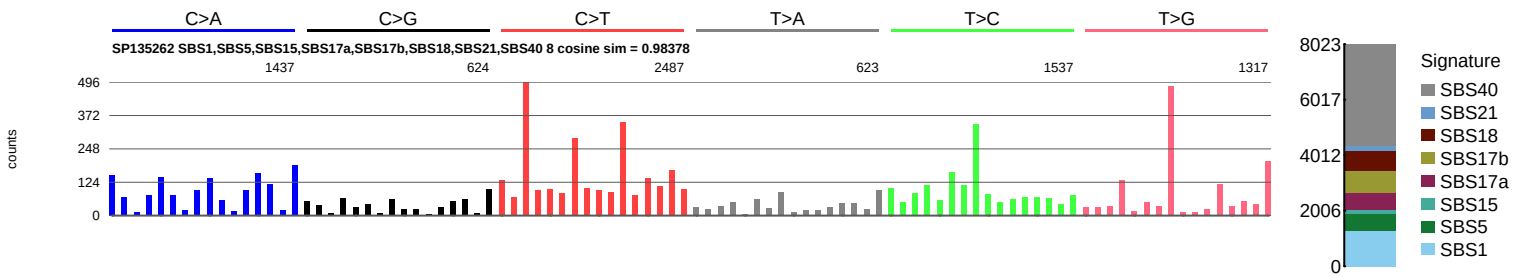
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98381
1349
537
2516
685
1682
1256
counts
492
369
246
123
0
8023
6017
4012
2006
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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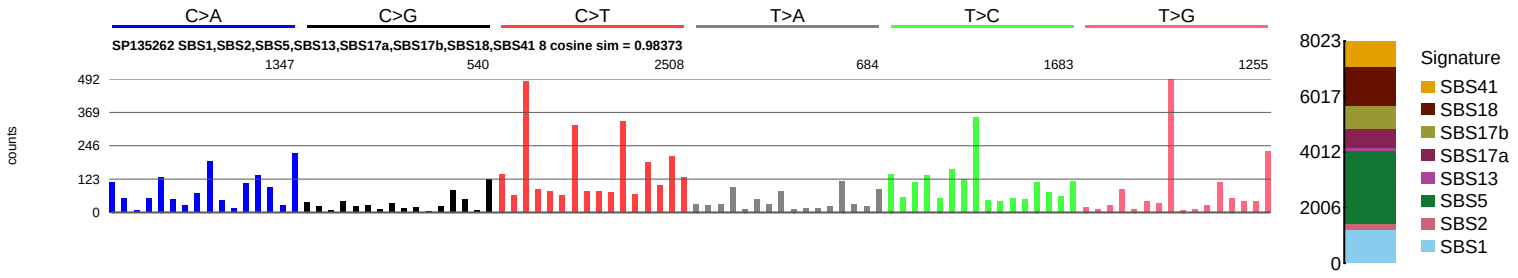
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98378
1437
624
2487
623
1537
1317
counts
496
372
248
124
0
8023
6017
4012
2006
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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C>T
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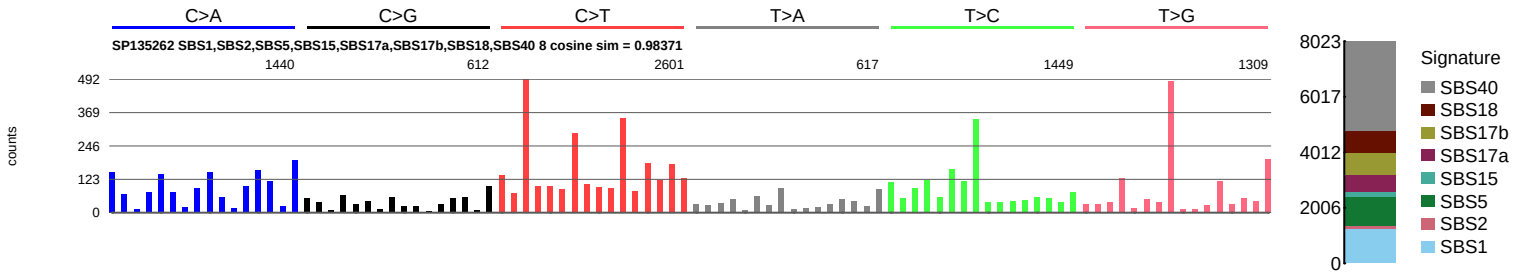
T>C
T>G
SP135262 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98373
1347
540
2508
684
1683
1255
counts
492
369
246
123
0
8023
6017
4012
2006
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98371
1440
612
2601
617
1449
1309
counts
492
369
246
123
0
8023
6017
4012
2006
0
Signature
SBS40
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SBS17a
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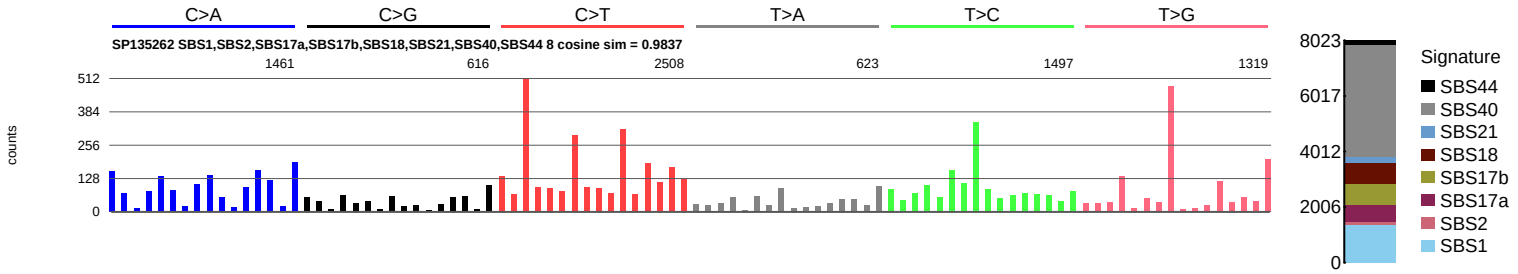
SBS5
SBS2
SBS1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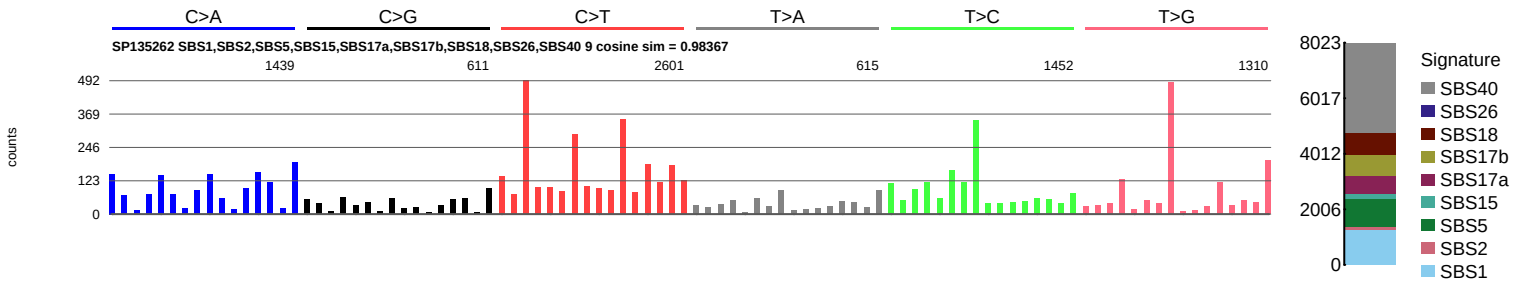
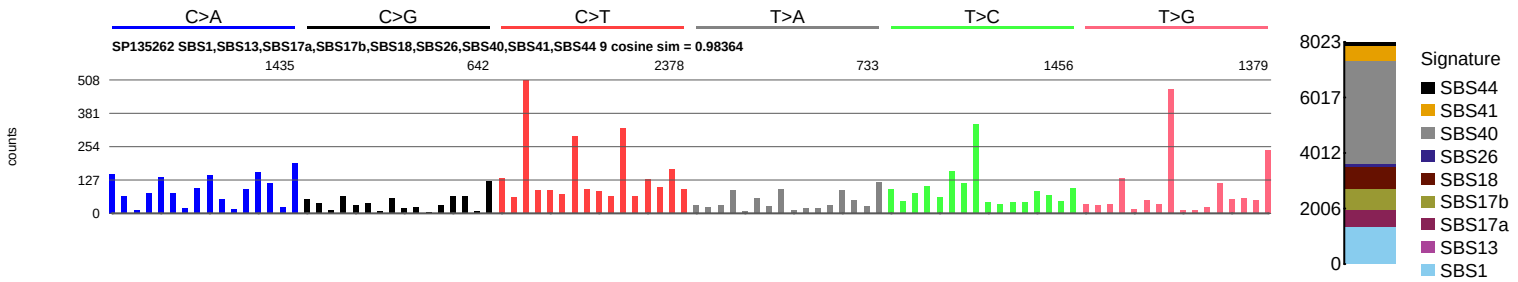
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.9837
1461
616
2508
623
1497
1319
counts
512
384
256
128
0
8023
6017
4012
2006
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98367
1439
611
2601
615
1452
1310
counts
492
369
246
123
0
8023
6017
4012
2006
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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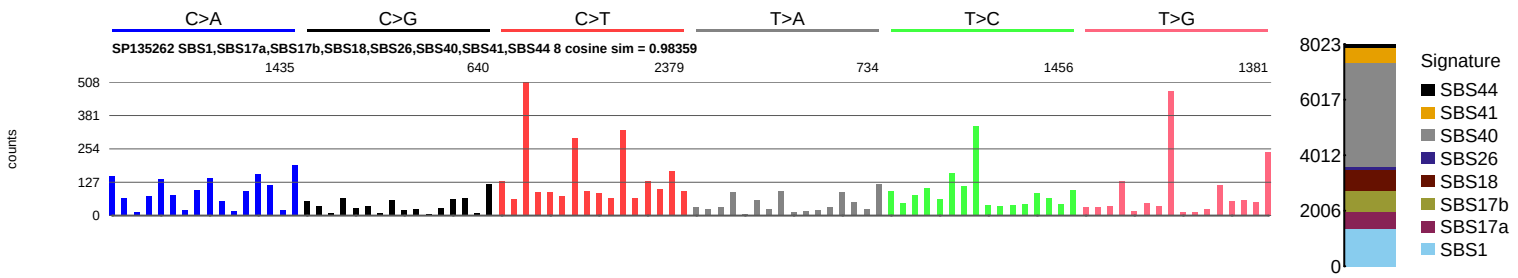
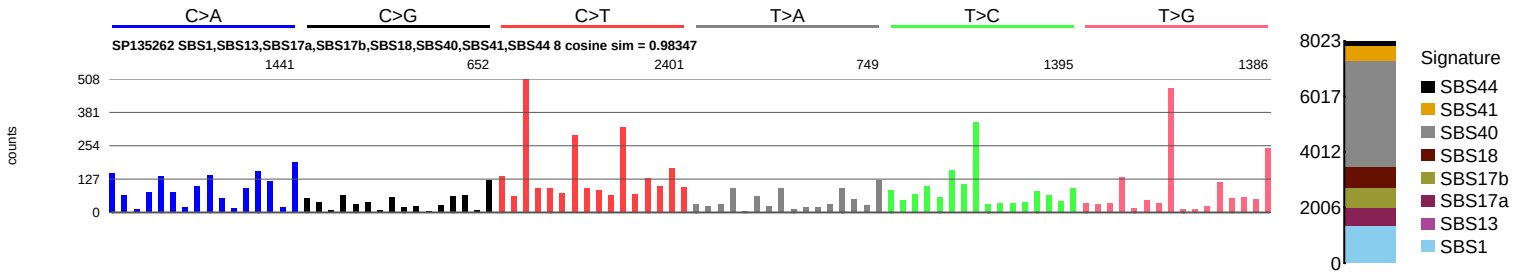
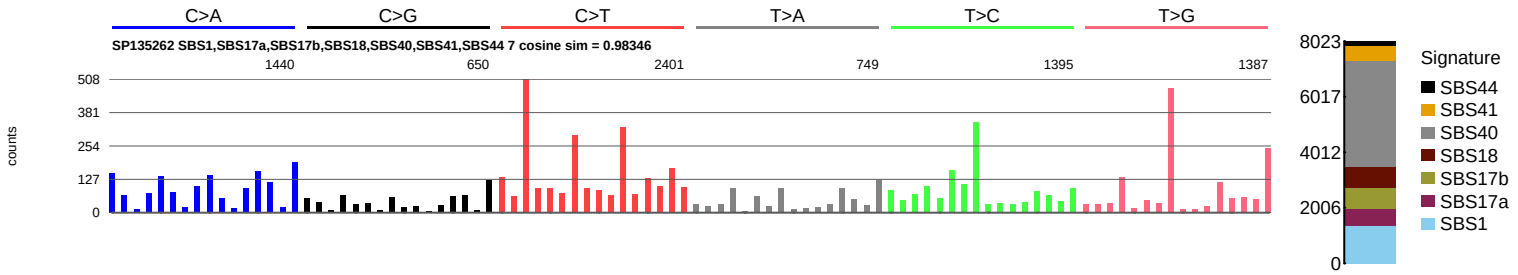
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98364
1435
642
2378
733
1456
1379
counts
508
381
254
127
0
8023
6017
4012
2006
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98359
1435
640
2379
734
1456
1381
counts
508
381
254
127
0
8023
6017
4012
2006
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98347
1441
652
2401
749
1395
1386
counts
508
381
254
127
0
8023
6017
4012
2006
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.98346
1440
650
2401
749
1395
1387
counts
508
381
254
127
0
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4012
2006
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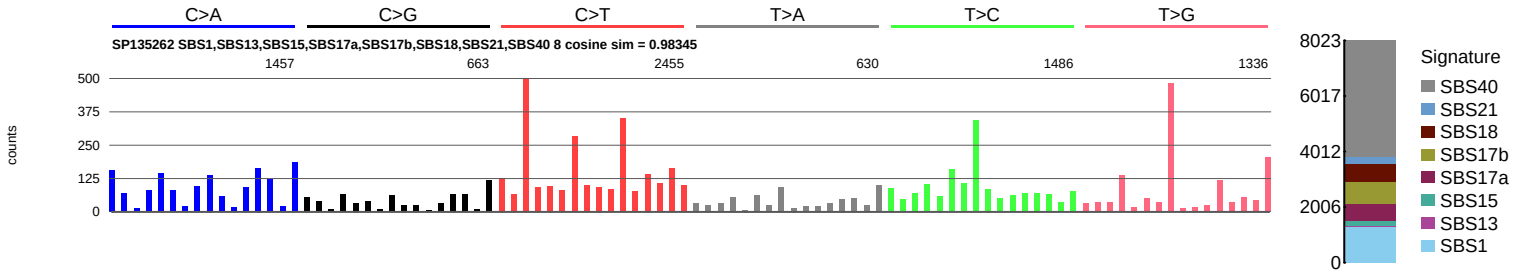
SBS1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SP135262 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98345
1457
663
2455
630
1486
1336
counts
500
375
250
125
0
8023
6017
4012
2006
0
Signature
SBS40
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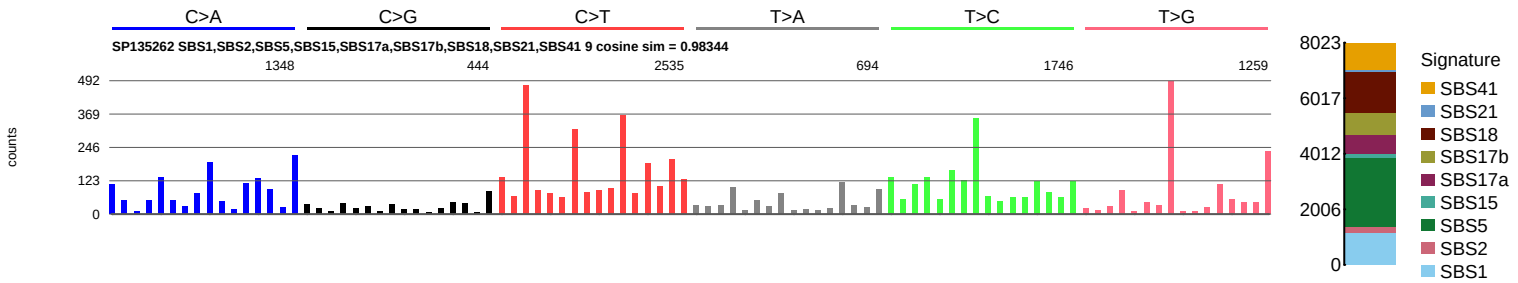
SBS13
SBS1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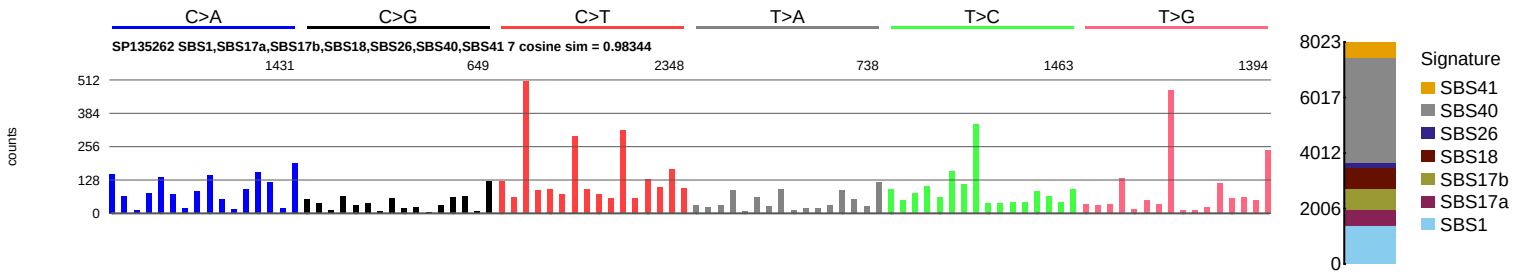
SP135262 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98344
1348
444
2535
694
1746
1259
counts
492
369
246
123
0
8023
6017
4012
2006
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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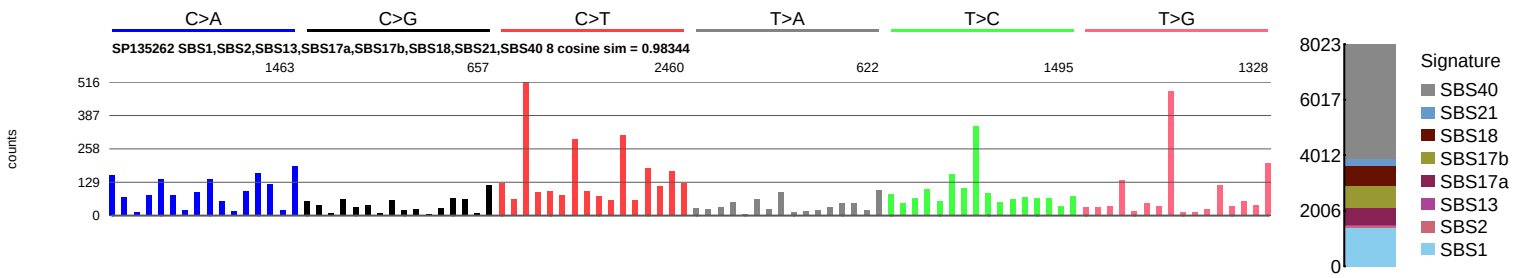
SP135262 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98344
1431
649
2348
738
1463
1394
counts
512
384
256
128
0
8023
6017
4012
2006
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1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SP135262 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98344
1463
657
2460
622
1495
1328
counts
516
387
258
129
0
8023
6017
4012
2006
0
Signature
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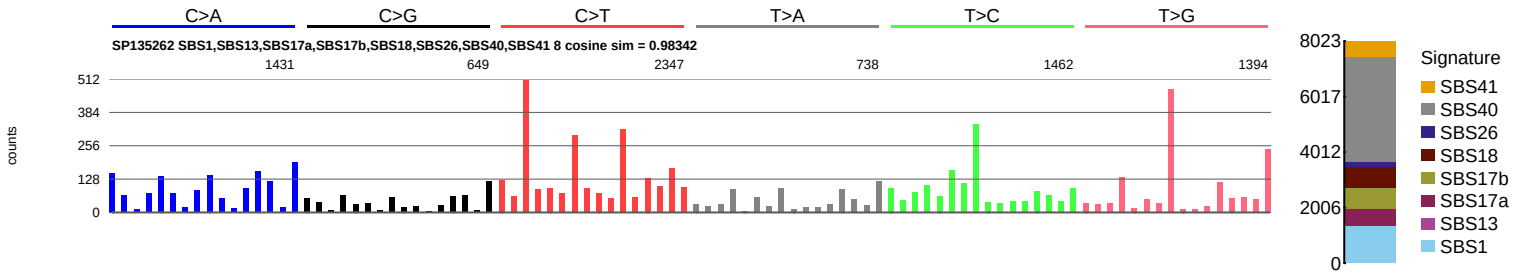
SBS2
SBS1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T>C
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SP135262 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98342
1431
649
2347
738
1462
1394
counts
512
384
256
128
0
8023
6017
4012
2006
0
Signature
SBS41
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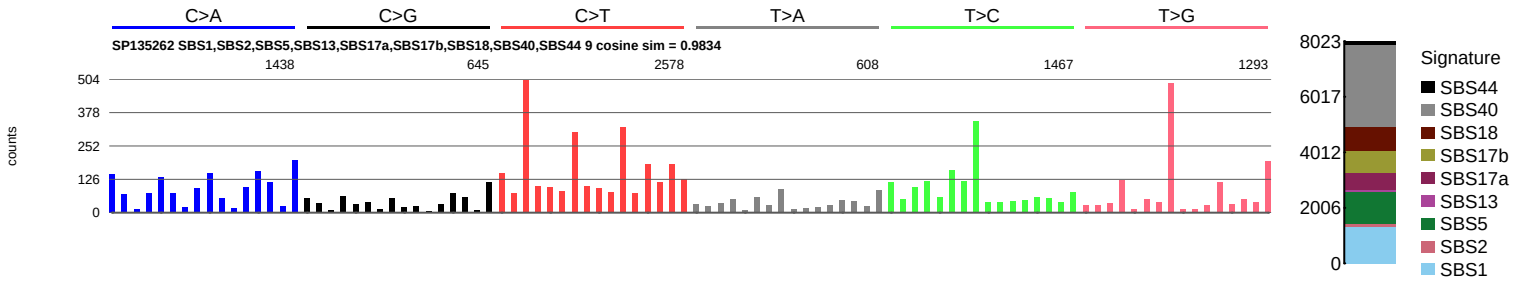
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS1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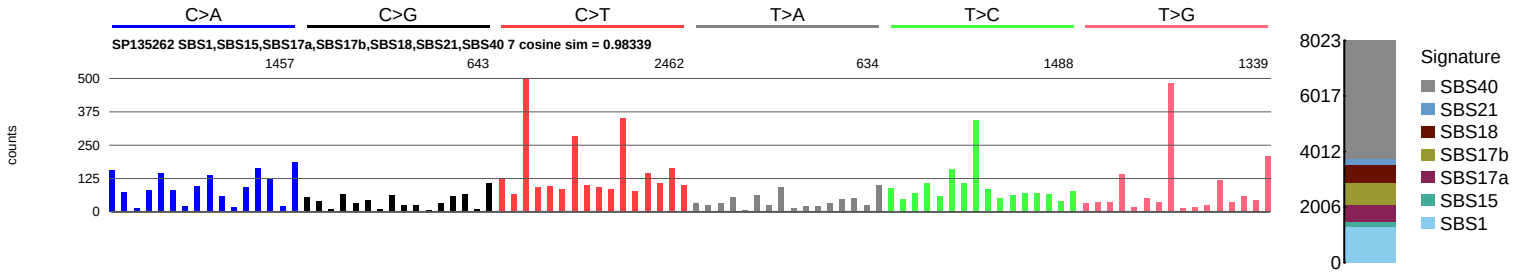
T>C
T>G
SP135262 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.9834
1438
645
2578
608
1467
1293
counts
504
378
252
126
0
8023
6017
4012
2006
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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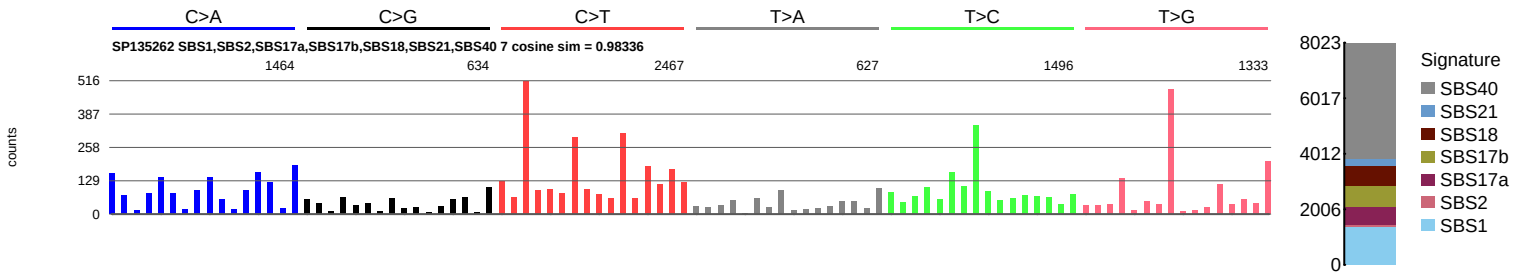
SP135262 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98339
1457
643
2462
634
1488
1339
counts
500
375
250
125
0
8023
6017
4012
2006
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1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C>G
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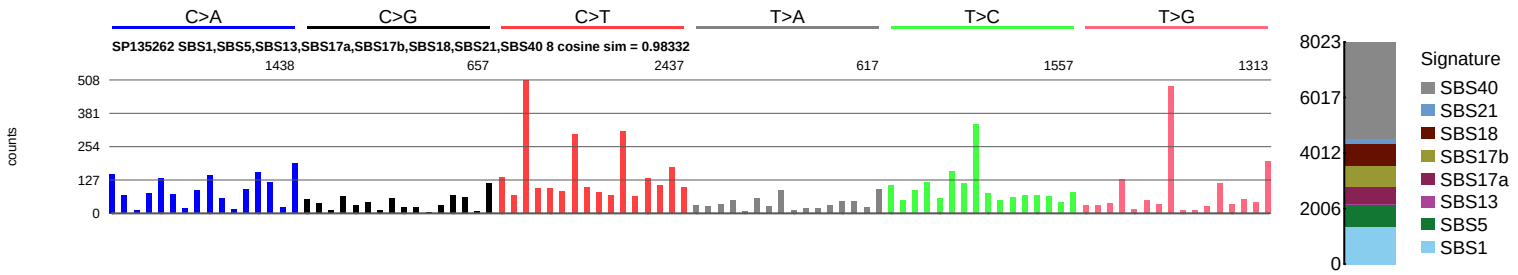
T>C
T>G
SP135262 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98336
1464
634
2467
627
1496
1333
counts
516
387
258
129
0
8023
6017
4012
2006
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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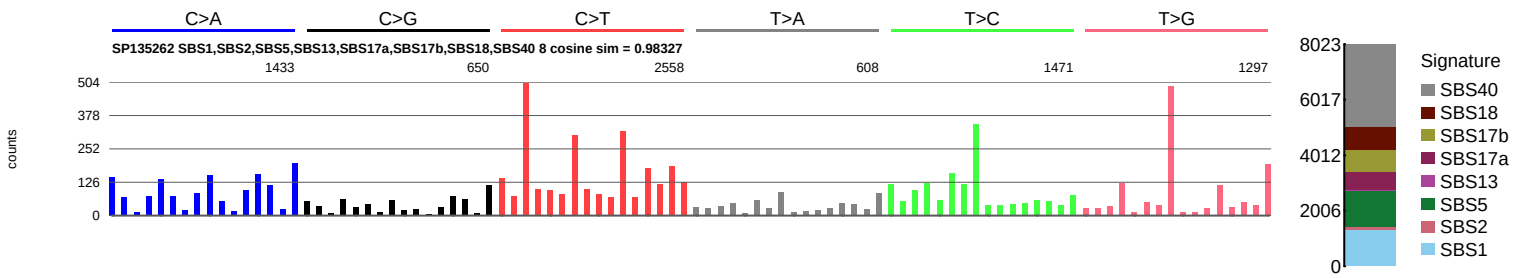
SP135262 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98332
1438
657
2437
617
1557
1313
counts
508
381
254
127
0
8023
6017
4012
2006
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1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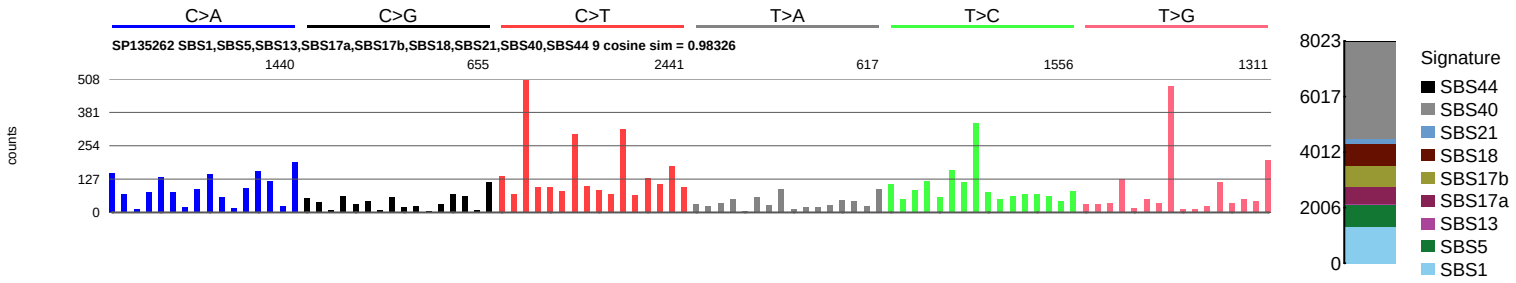
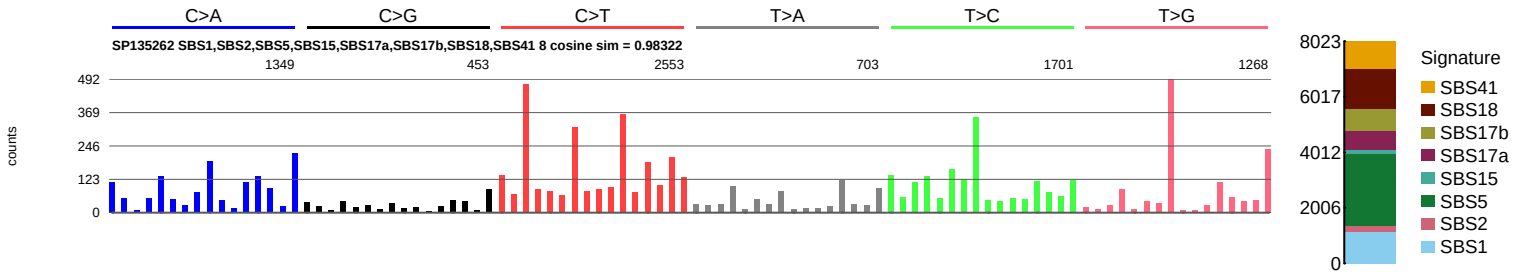
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98327
1433
650
2558
608
1471
1297
counts
504
378
252
126
0
8023
6017
4012
2006
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98326
1440
655
2441
617
1556
1311
counts
508
381
254
127
0
8023
6017
4012
2006
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1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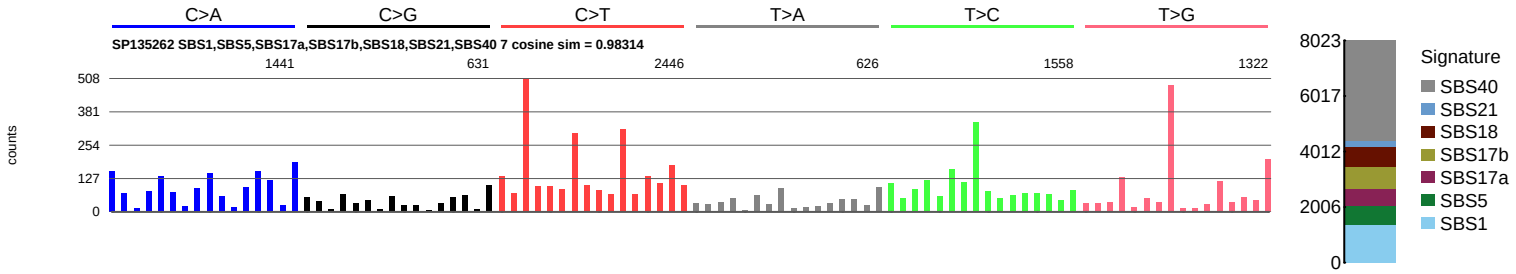
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98322
1349
453
2553
703
1701
1268
counts
492
369
246
123
0
8023
6017
4012
2006
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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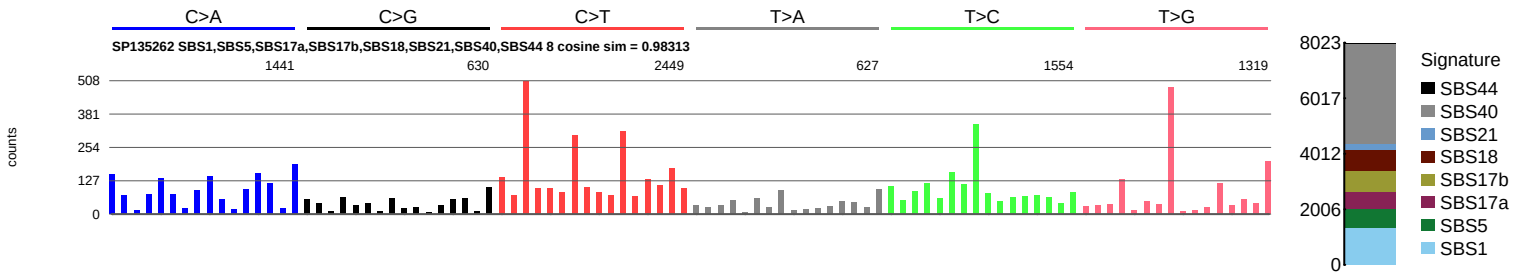
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98314
1441
631
2446
626
1558
1322
counts
508
381
254
127
0
8023
6017
4012
2006
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1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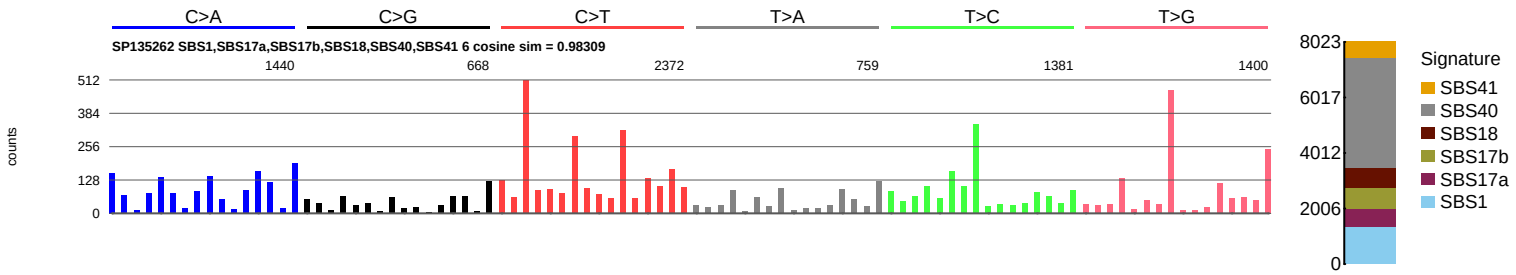
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98313
1441
630
2449
627
1554
1319
counts
508
381
254
127
0
8023
6017
4012
2006
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1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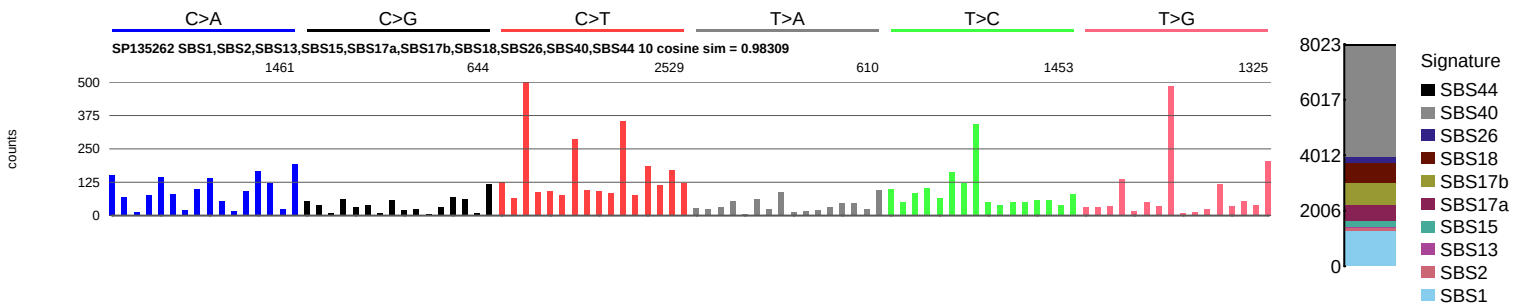
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS17a,SBS17b,SBS18,SBS40,SBS41 6 cosine sim = 0.98309
1440
668
2372
759
1381
1400
counts
512
384
256
128
0
8023
6017
4012
2006
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS1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C>G
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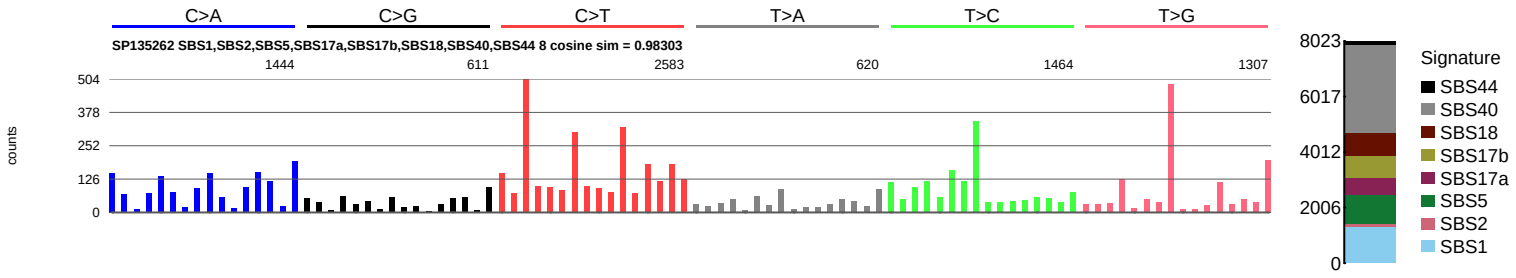
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98309
1461
644
2529
610
1453
1325
counts
500
375
250
125
0
8023
6017
4012
2006
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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T>C
T>G
SP135262 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98303
1444
611
2583
620
1464
1307
counts
504
378
252
126
0
8023
6017
4012
2006
0
Signature
SBS44
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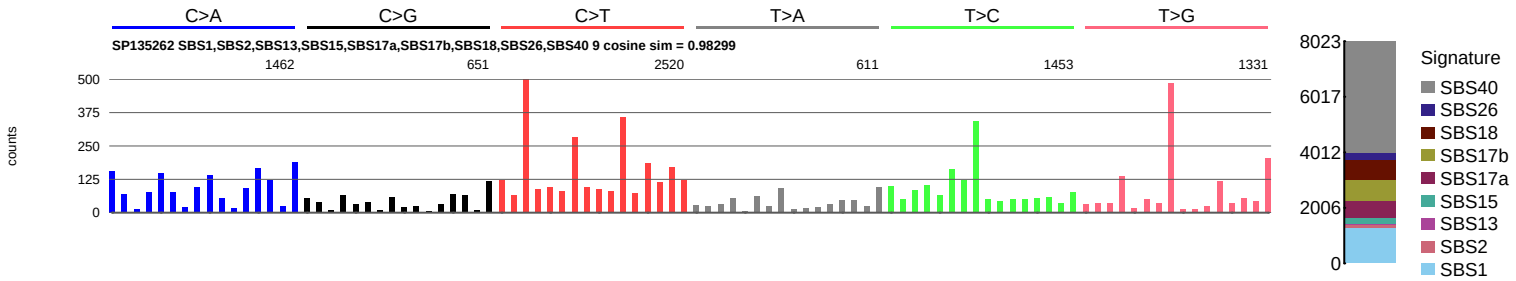
SBS17b
SBS17a
SBS5
SBS2
SBS1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C>T
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T>C
T>G
SP135262 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98299
1462
651
2520
611
1453
1331
counts
500
375
250
125
0
8023
6017
4012
2006
0
Signature
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SBS26
SBS18
SBS17b
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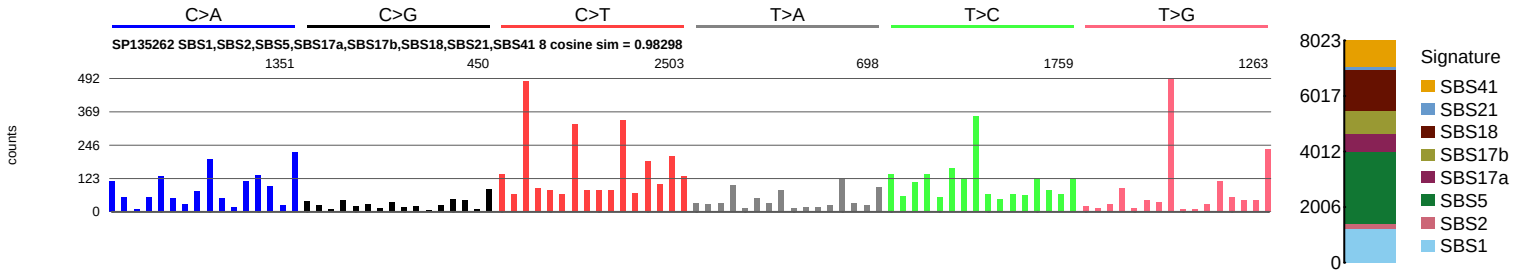
SBS15
SBS13
SBS2
SBS1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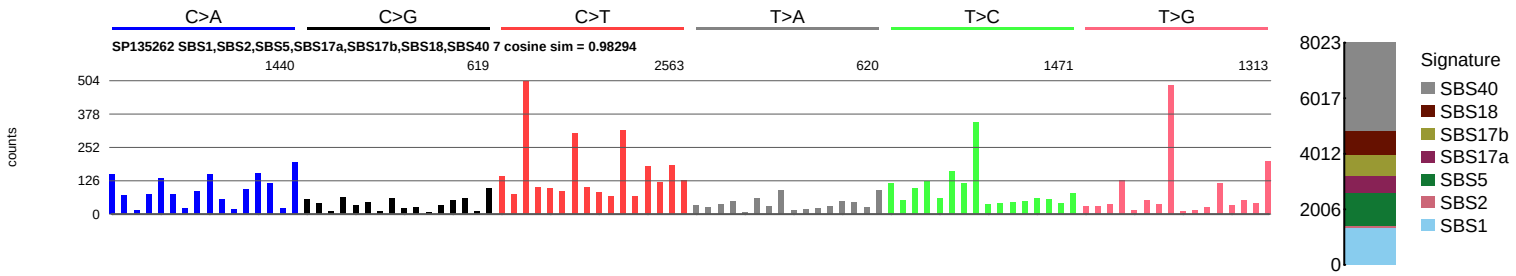
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98298
1351
450
2503
698
1759
1263
counts
492
369
246
123
0
8023
6017
4012
2006
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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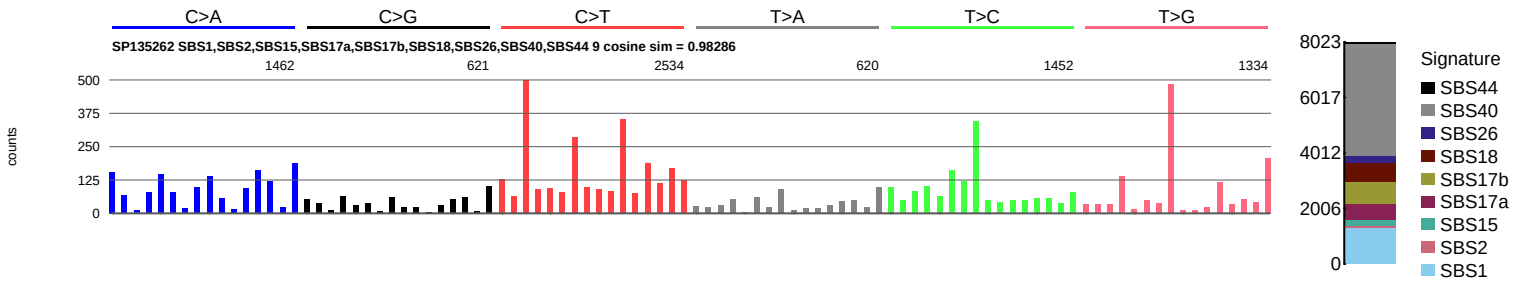
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98294
1440
619
2563
620
1471
1313
counts
504
378
252
126
0
8023
6017
4012
2006
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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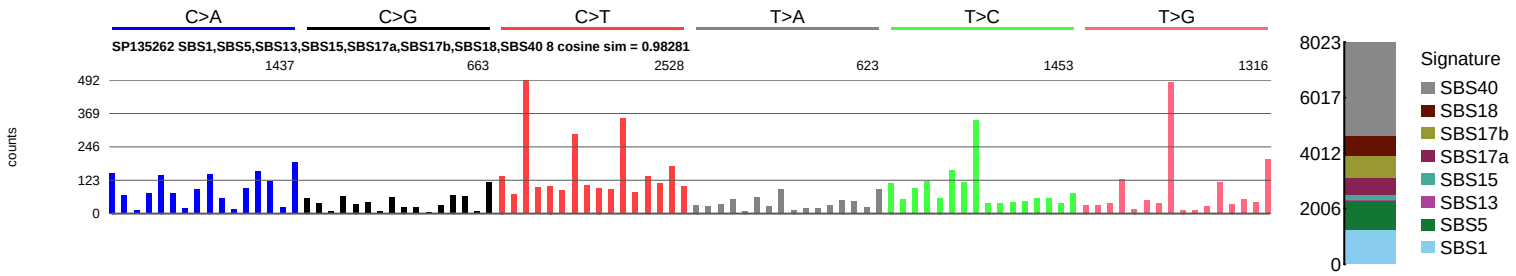
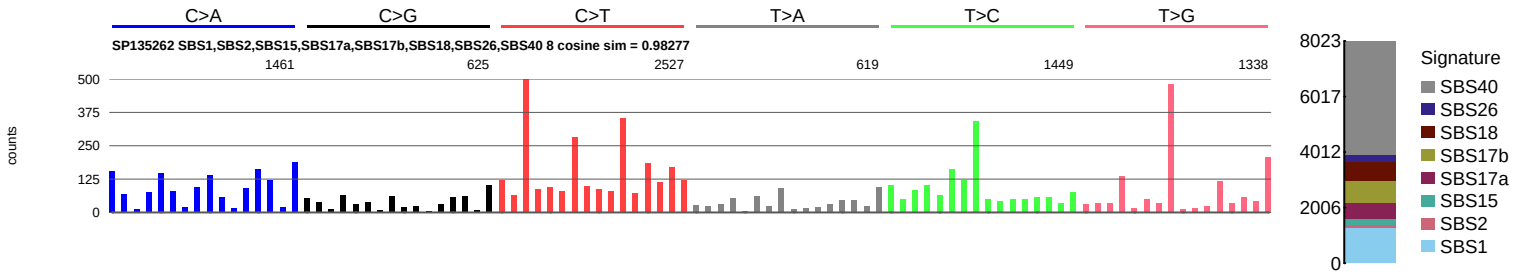
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98286
1462
621
2534
620
1452
1334
counts
500
375
250
125
0
8023
6017
4012
2006
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98281
1437
663
2528
623
1453
1316
counts
492
369
246
123
0
8023
6017
4012
2006
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98277
1461
625
2527
619
1449
1338
counts
500
375
250
125
0
8023
6017
4012
2006
0
Signature
SBS40
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SBS2
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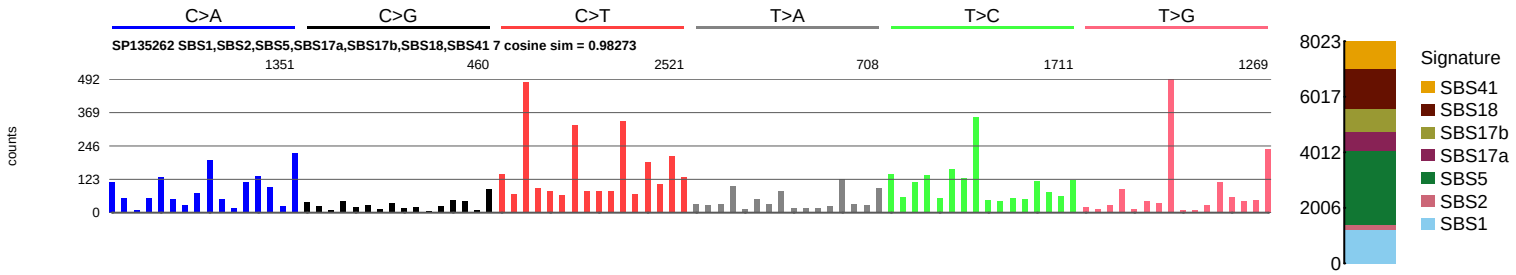
C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98273
1351
460
2521
708
1711
1269
counts
492
369
246
123
0
8023
6017
4012
2006
0
Signature
SBS41
SBS18
SBS17b
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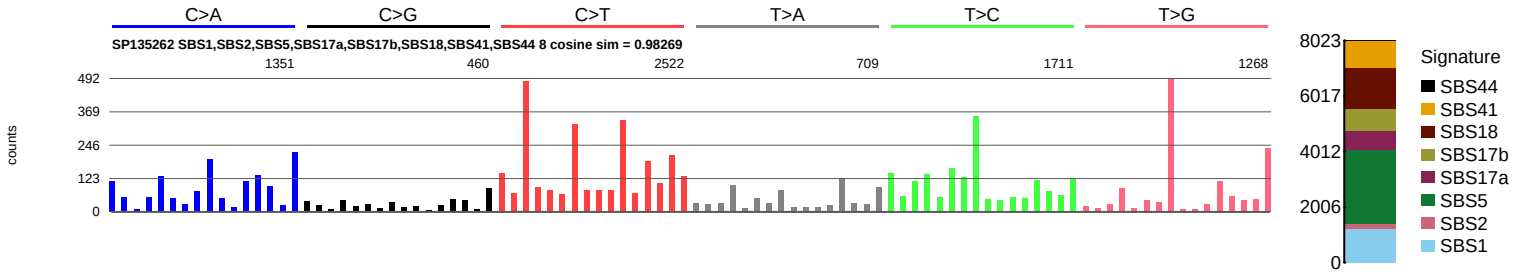
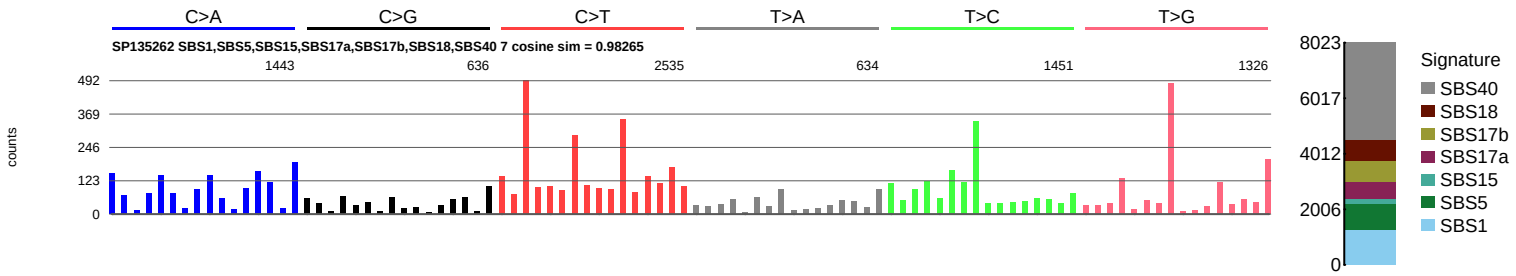
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98269
1351
460
2522
709
1711
1268
counts
492
369
246
123
0
8023
6017
4012
2006
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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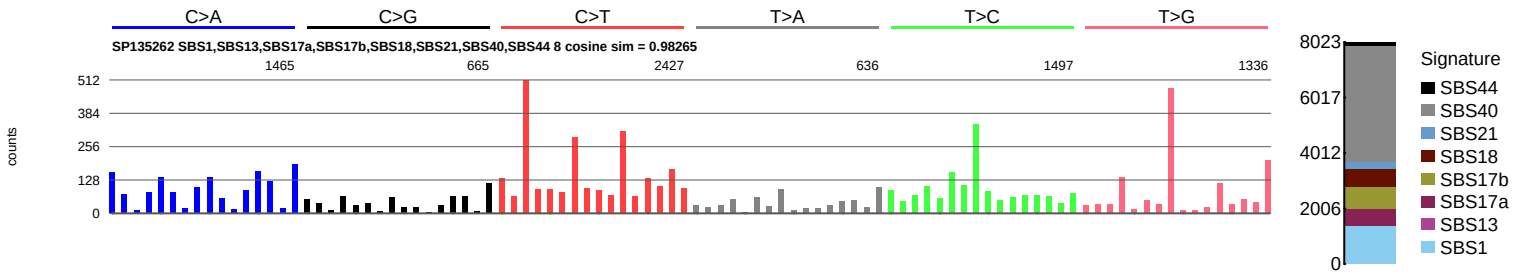
SP135262 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98265
1443
636
2535
634
1451
1326
counts
492
369
246
123
0
8023
6017
4012
2006
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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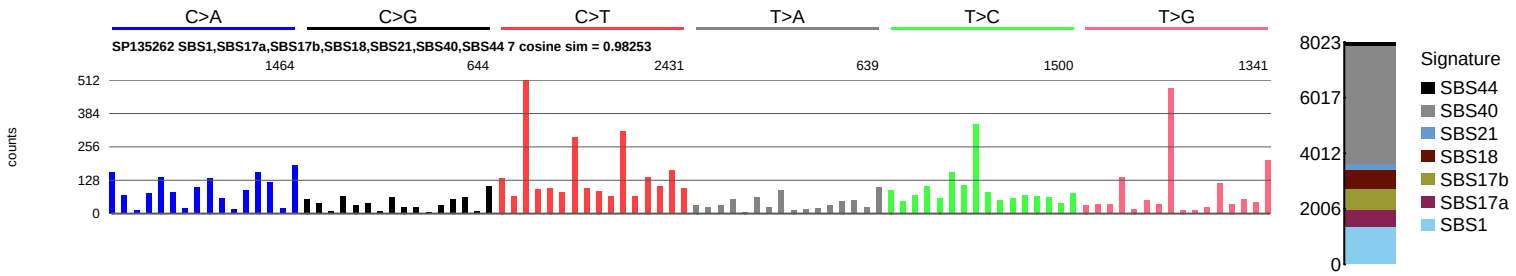
SP135262 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98265
1465
665
2427
636
1497
1336
counts
512
384
256
128
0
8023
6017
4012
2006
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1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1464
644
2431
639
1500
1341
counts
512
384
256
128
0
8023
6017
4012
2006
0
Signature
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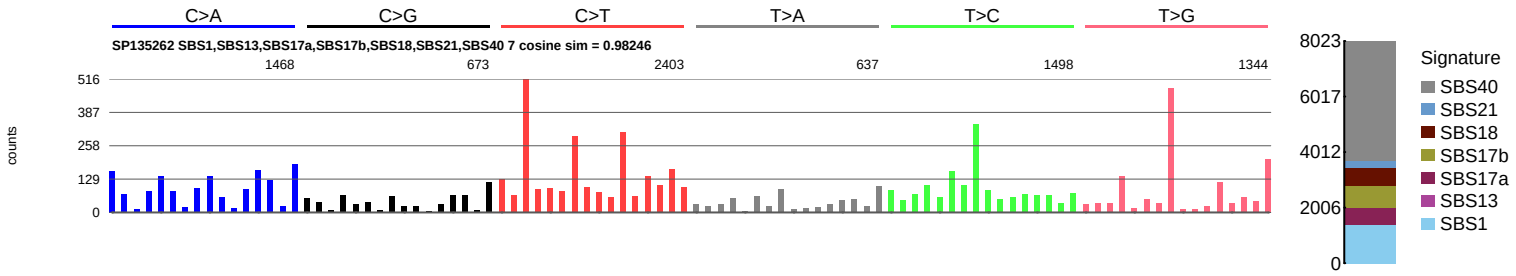
SBS21
SBS18
SBS17b
SBS17a
SBS1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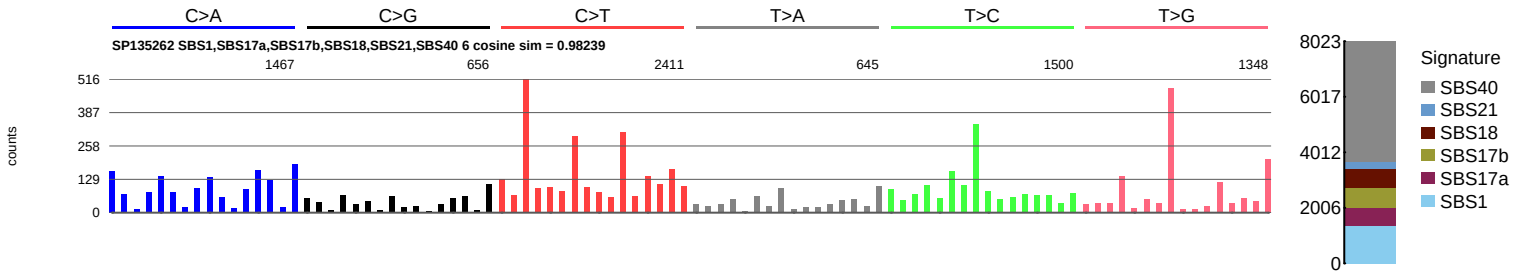
SP135262 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98246
1468
673
2403
637
1498
1344
counts
516
387
258
129
0
8023
6017
4012
2006
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1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SP135262 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40 6 cosine sim = 0.98239
1467
656
2411
645
1500
1348
counts
516
387
258
129
0
8023
6017
4012
2006
0
Signature
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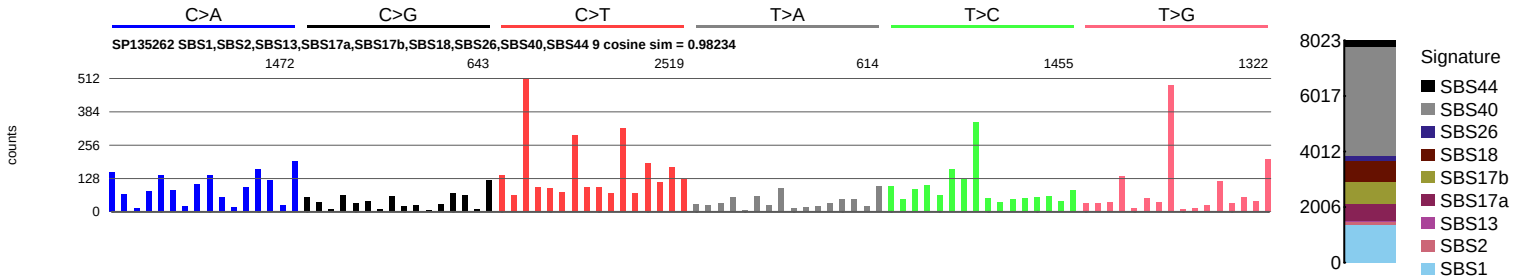
SBS17a
SBS1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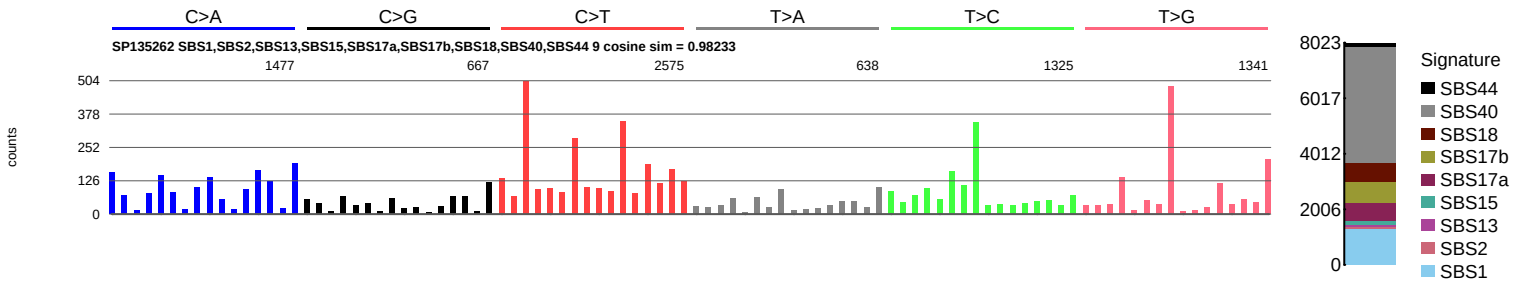
SP135262 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98234
1472
643
2519
614
1455
1322
counts
512
384
256
128
0
8023
6017
4012
2006
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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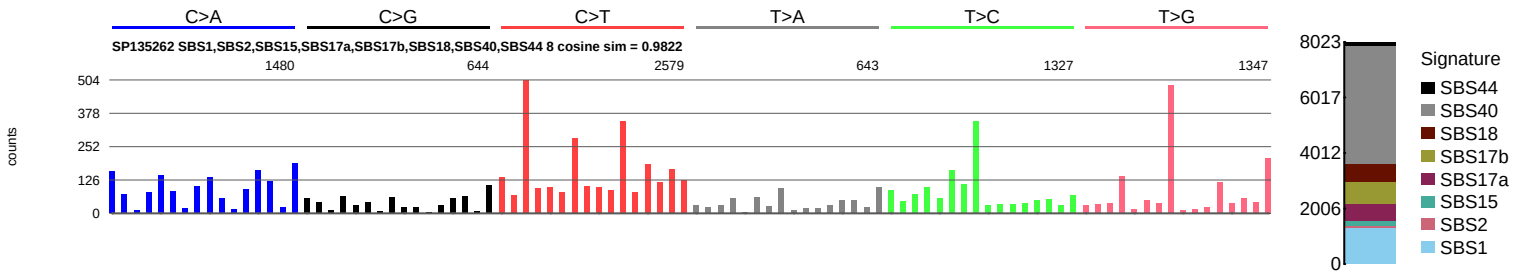
SP135262 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98233
1477
667
2575
638
1325
1341
counts
504
378
252
126
0
8023
6017
4012
2006
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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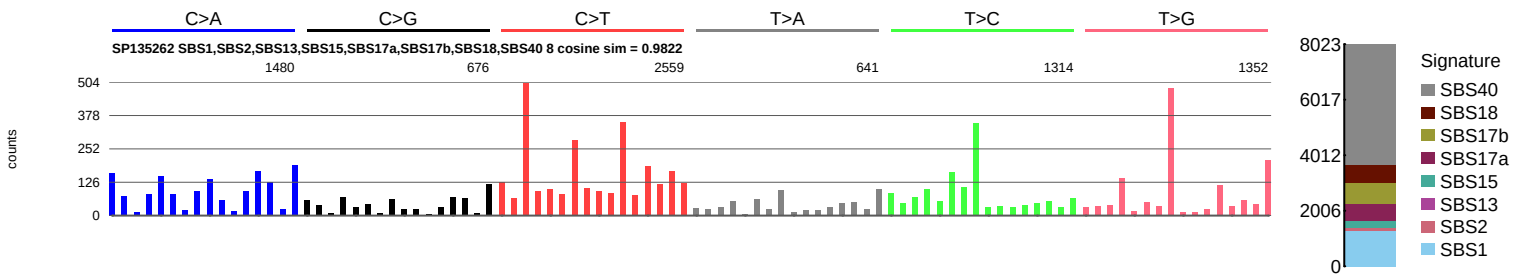
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.9822
1480
644
2579
643
1327
1347
counts
504
378
252
126
0
8023
6017
4012
2006
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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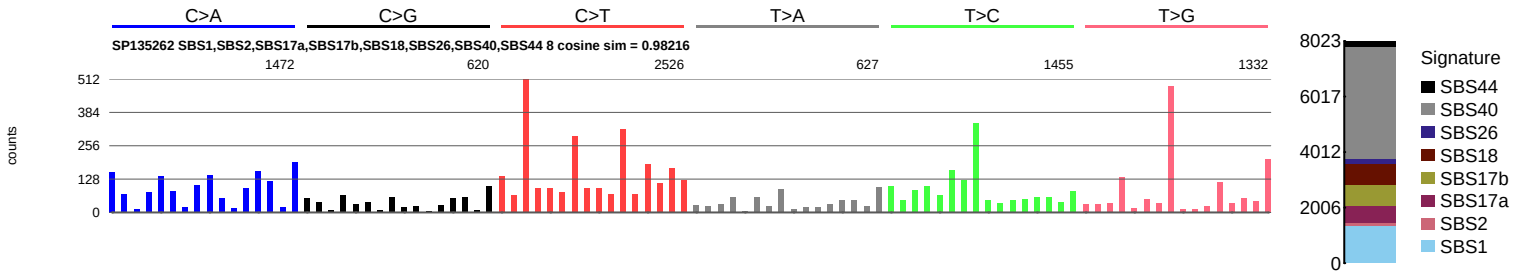
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.9822
1480
676
2559
641
1314
1352
counts
504
378
252
126
0
8023
6017
4012
2006
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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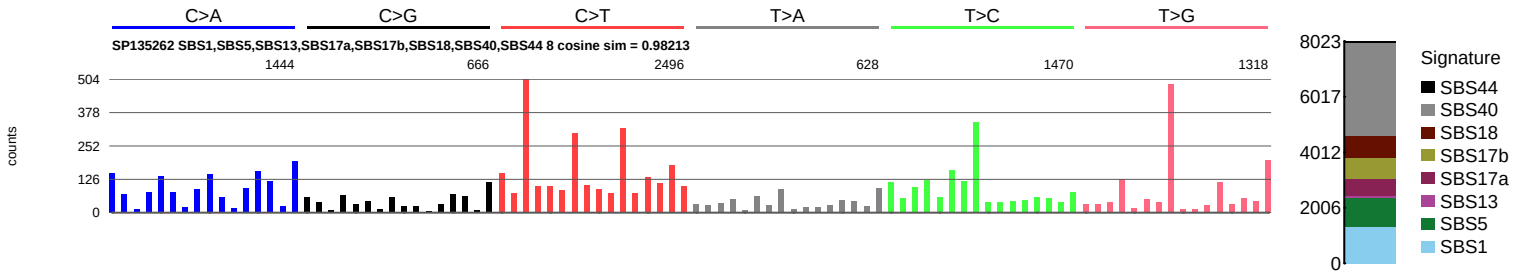
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98216
1472
620
2526
627
1455
1332
counts
512
384
256
128
0
8023
6017
4012
2006
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1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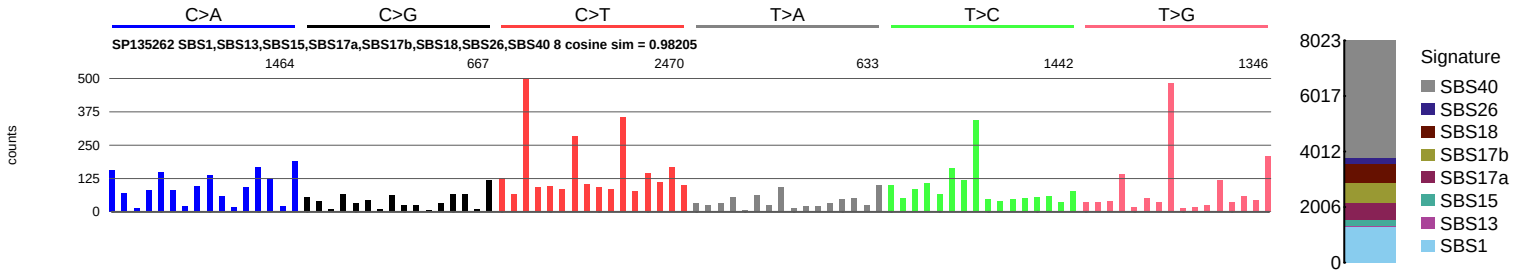
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98213
1444
666
2496
628
1470
1318
counts
504
378
252
126
0
8023
6017
4012
2006
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1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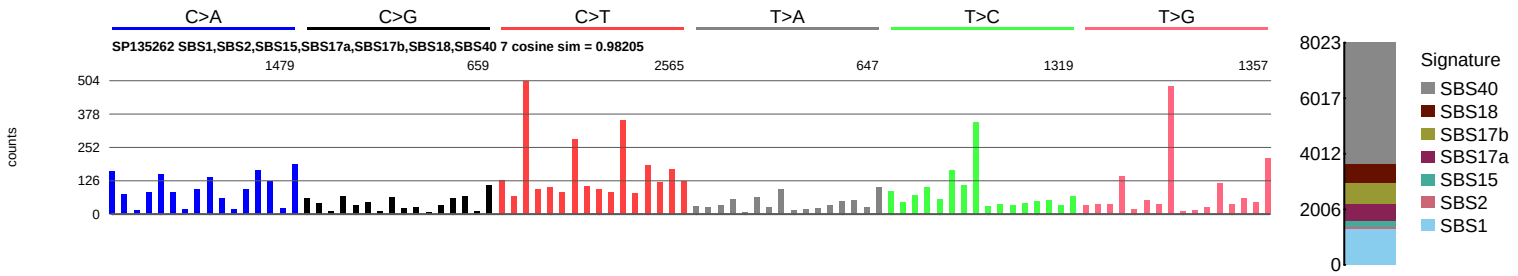
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98205
1464
667
2470
633
1442
1346
counts
500
375
250
125
0
8023
6017
4012
2006
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1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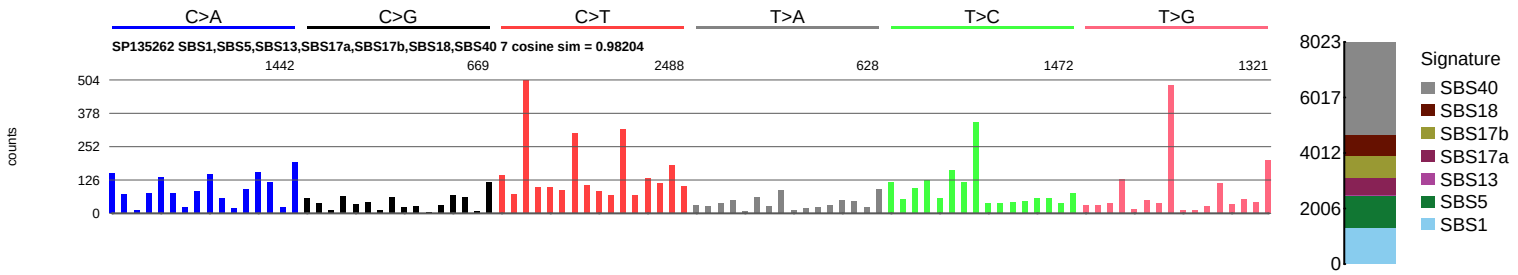
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98205
1479
659
2565
647
1319
1357
counts
504
378
252
126
0
8023
6017
4012
2006
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>G
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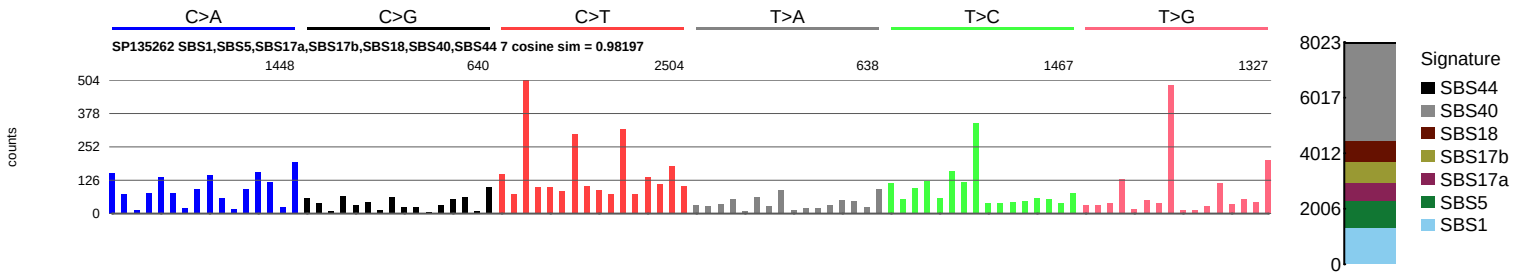
T>C
T>G
SP135262 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98204
1442
669
2488
628
1472
1321
counts
504
378
252
126
0
8023
6017
4012
2006
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1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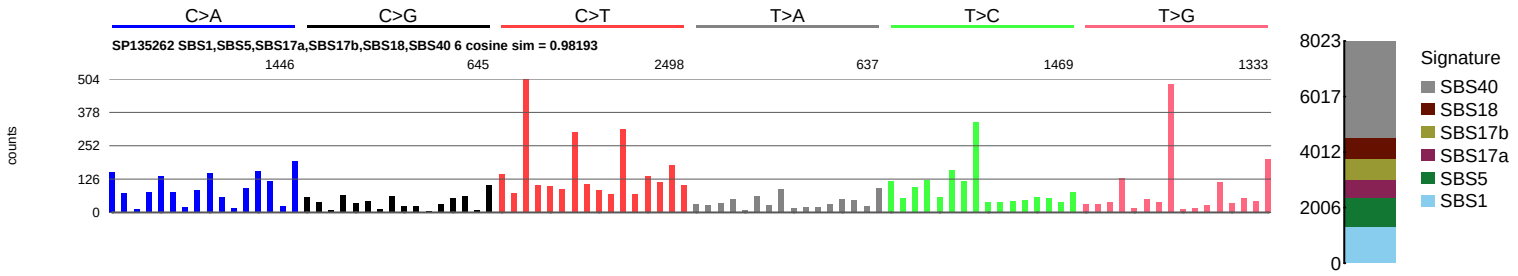
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.98197
1448
640
2504
638
1467
1327
counts
504
378
252
126
0
8023
6017
4012
2006
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1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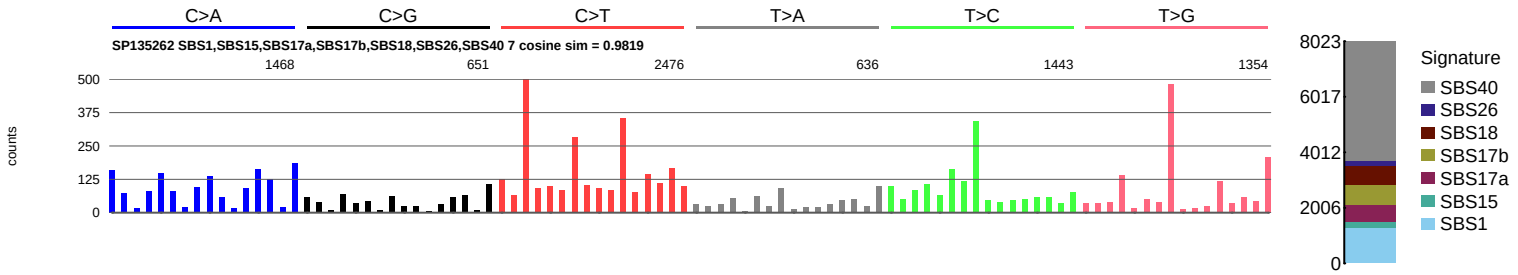
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.98193
1446
645
2498
637
1469
1333
counts
504
378
252
126
0
8023
6017
4012
2006
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1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C>G
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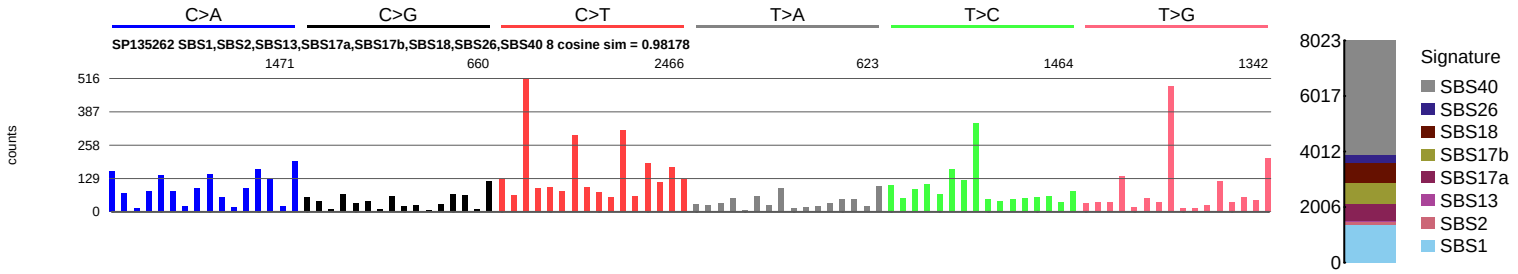
T>A
T>C
T>G
SP135262 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.9819
1468
651
2476
636
1443
1354
counts
500
375
250
125
0
8023
6017
4012
2006
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1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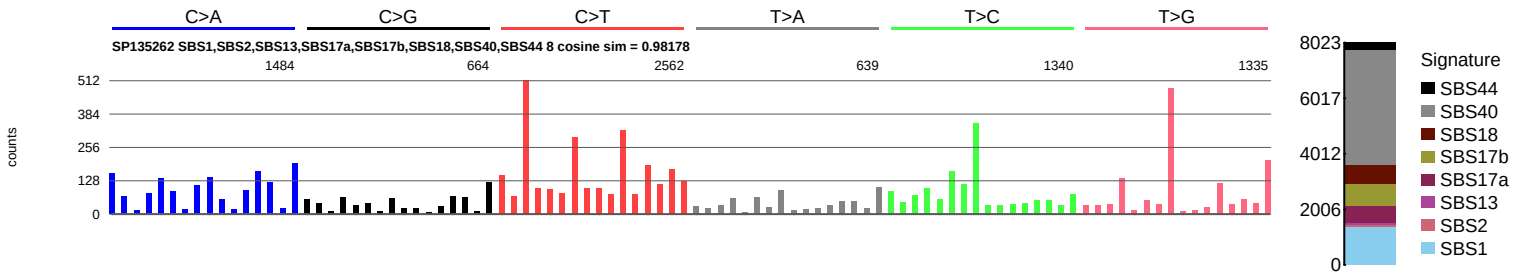
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98178
1471
660
2466
623
1464
1342
counts
516
387
258
129
0
8023
6017
4012
2006
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98178
1484
664
2562
639
1340
1335
counts
512
384
256
128
0
8023
6017
4012
2006
0
Signature
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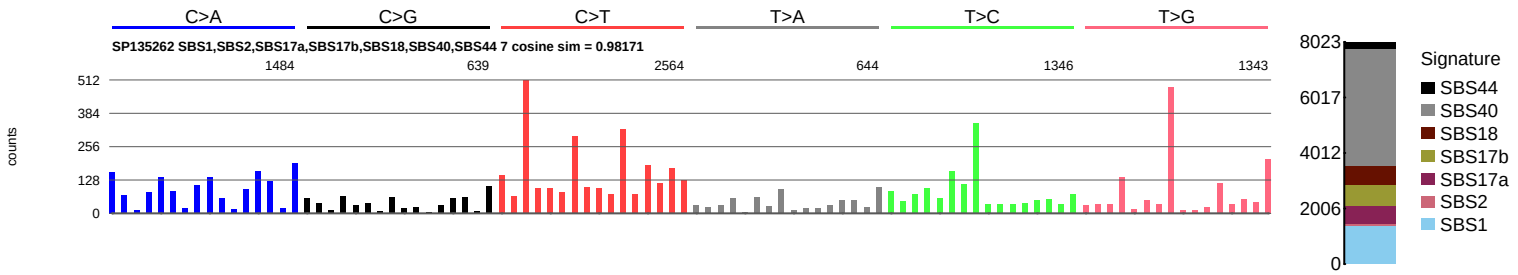
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.98171
1484
639
2564
644
1346
1343
counts
512
384
256
128
0
8023
6017
4012
2006
0
Signature
SBS44
SBS40
SBS18
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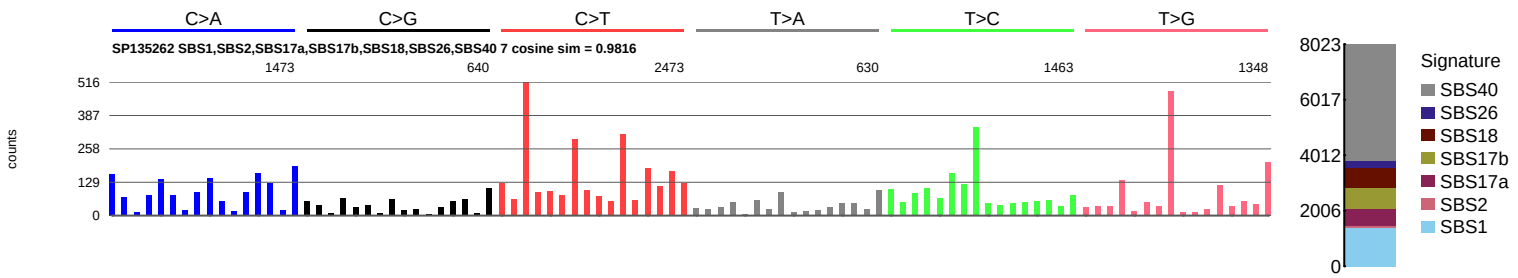
SBS2
SBS1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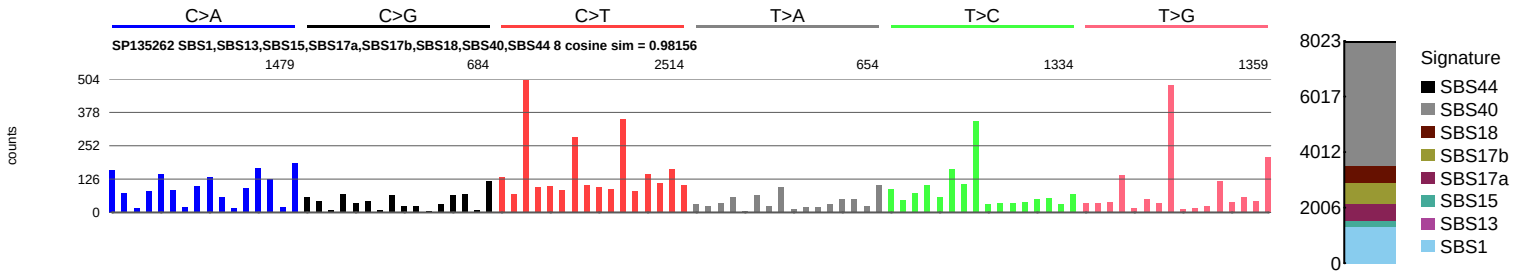
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.9816
1473
640
2473
630
1463
1348
counts
516
387
258
129
0
8023
6017
4012
2006
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1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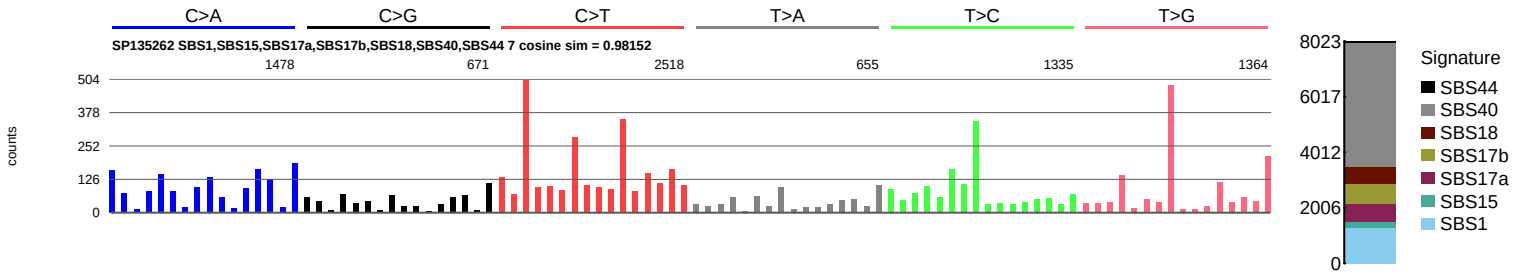
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98156
1479
684
2514
654
1334
1359
counts
504
378
252
126
0
8023
6017
4012
2006
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.98152
1478
671
2518
655
1335
1364
counts
504
378
252
126
0
8023
6017
4012
2006
0
Signature
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SBS18
SBS17b
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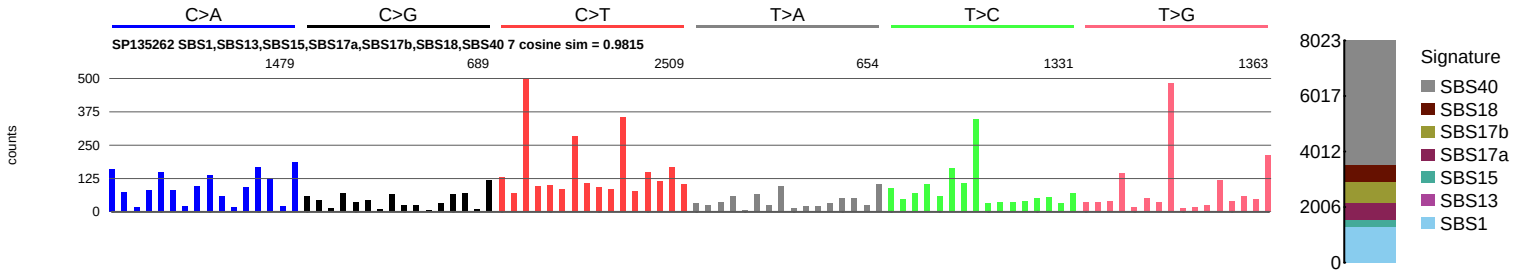
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.9815
1479
689
2509
654
1331
1363
counts
500
375
250
125
0
8023
6017
4012
2006
0
Signature
SBS40
SBS18
SBS17b
SBS17a
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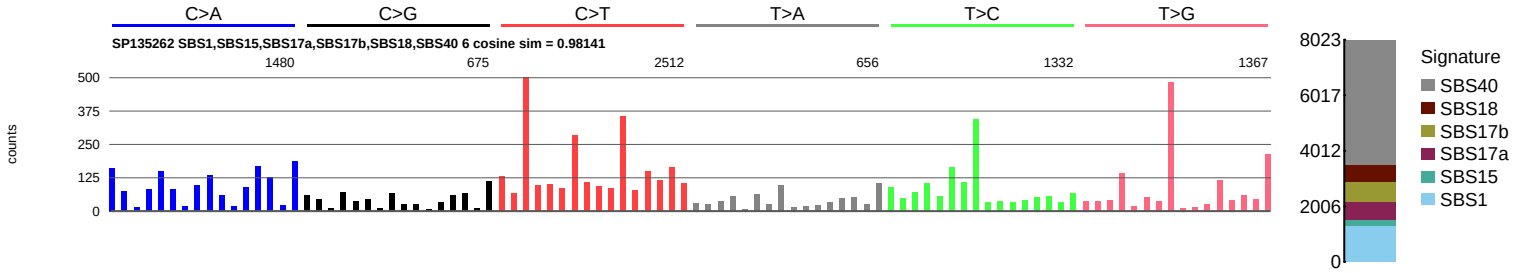
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.98141
1480
675
2512
656
1332
1367
counts
500
375
250
125
0
8023
6017
4012
2006
0
Signature
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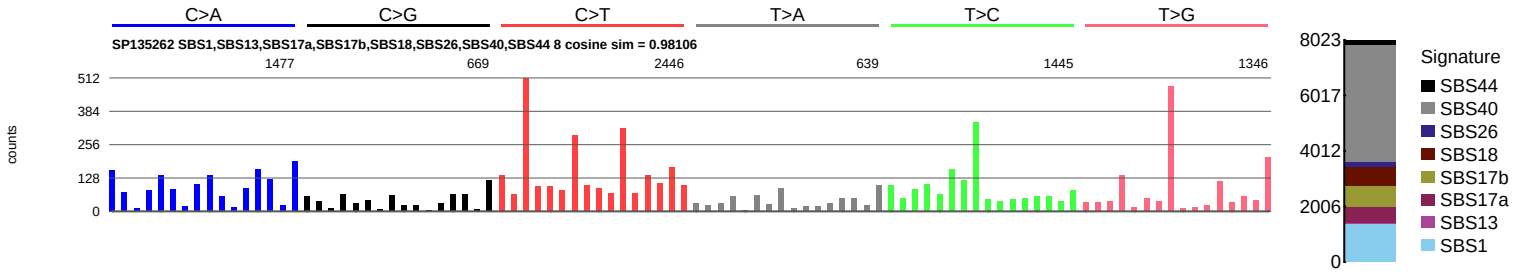
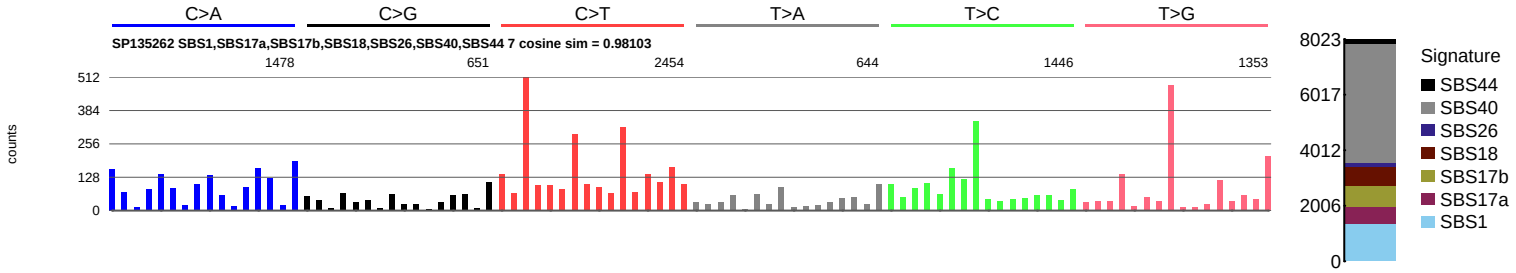
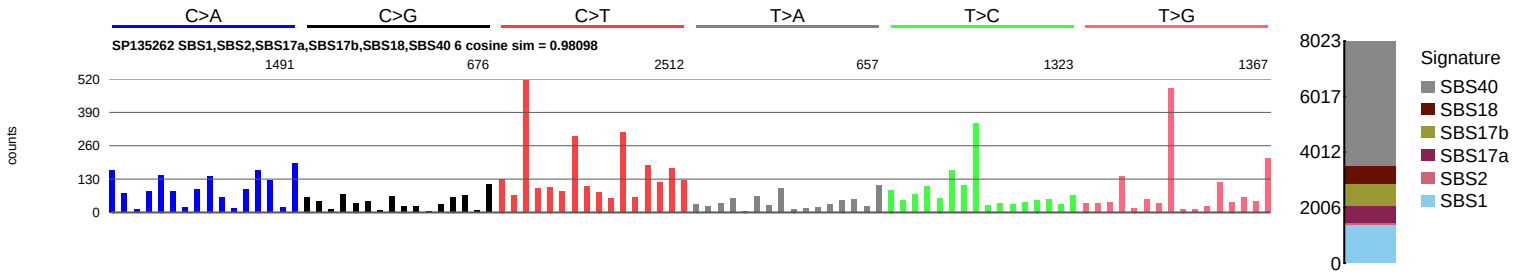
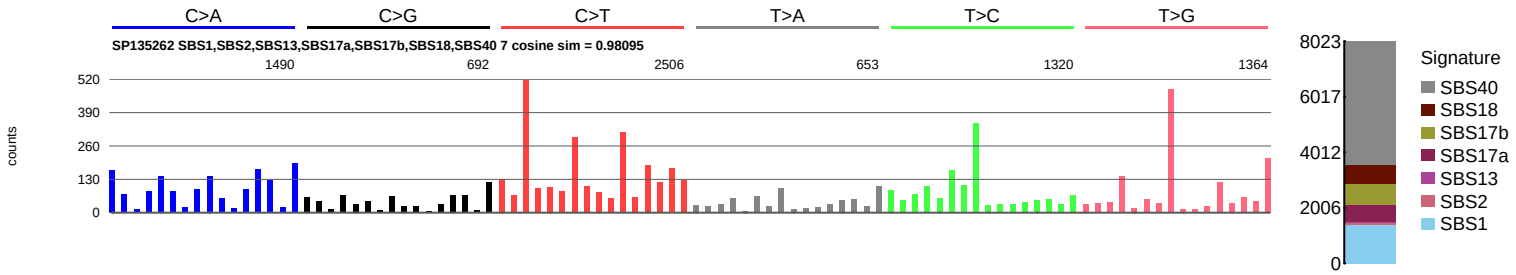
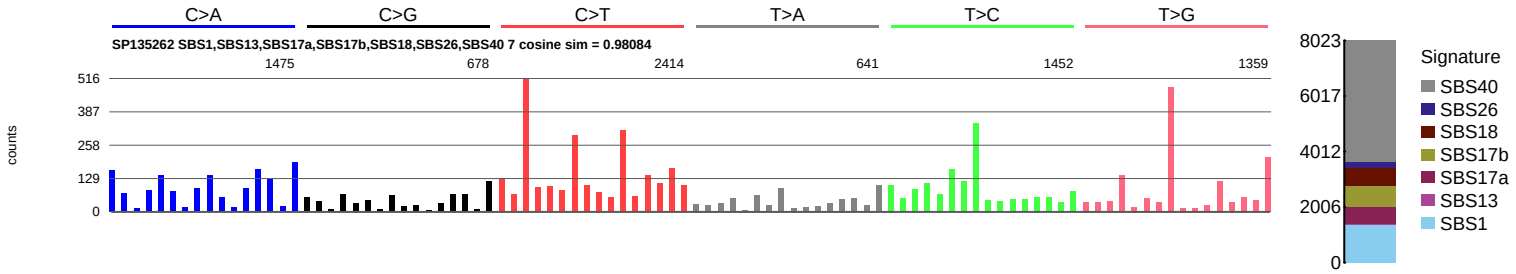
SBS15
SBS1

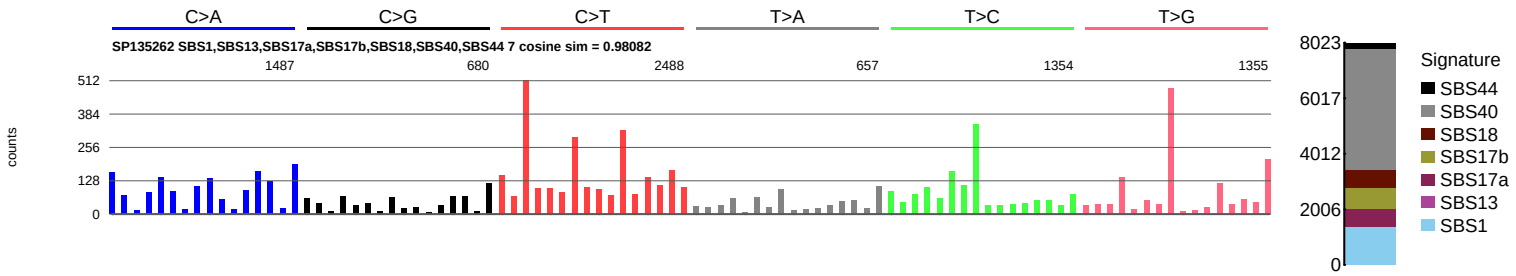

C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98106
1477
669
2446
639
1445
1346
counts
512
384
256
128
0
8023
6017
4012
2006
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS1

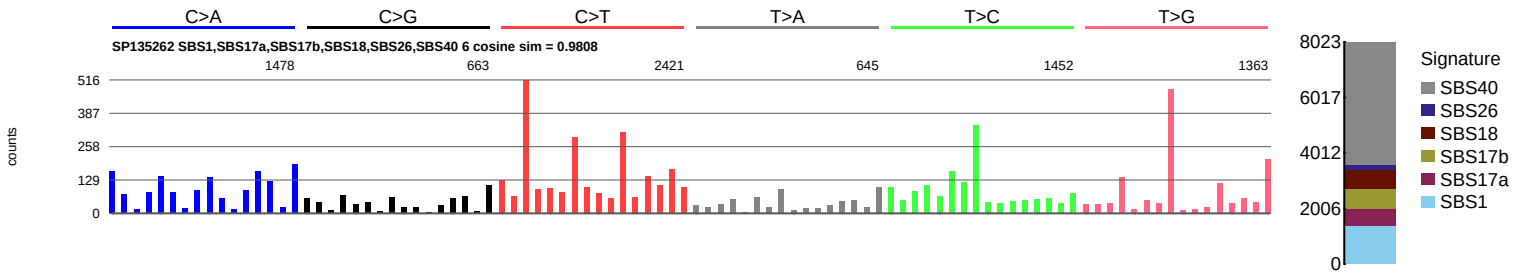

C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.98103
1478
651
2454
644
1446
1353
counts
512
384
256
128
0
8023
6017
4012
2006
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1

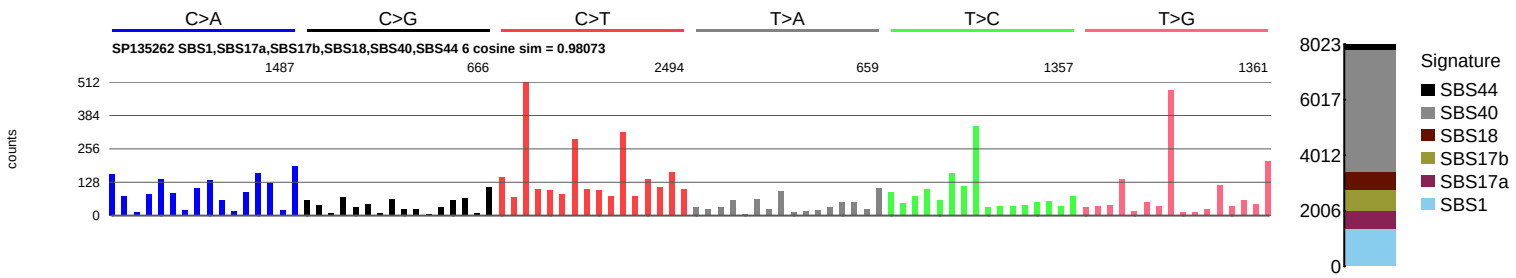

C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.98098
1491
676
2512
657
1323
1367
counts
520
390
260
130
0
8023
6017
4012
2006
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1

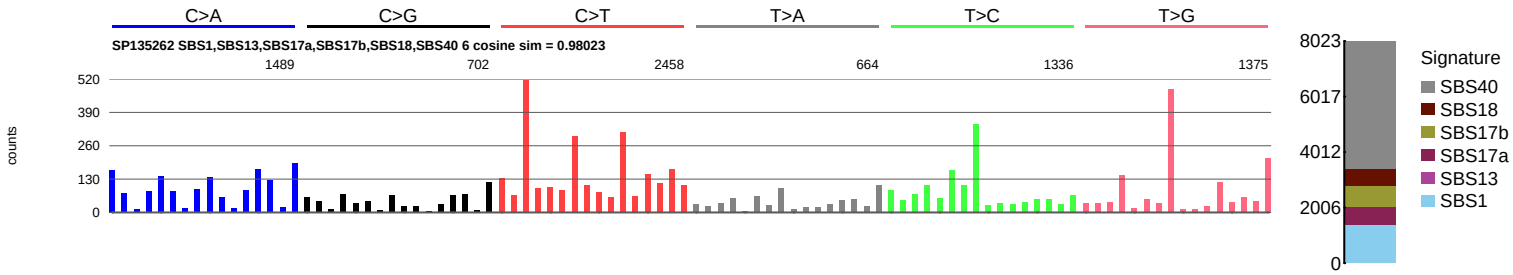

C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98095
1490
692
2506
653
1320
1364
counts
520
390
260
130
0
8023
6017
4012
2006
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

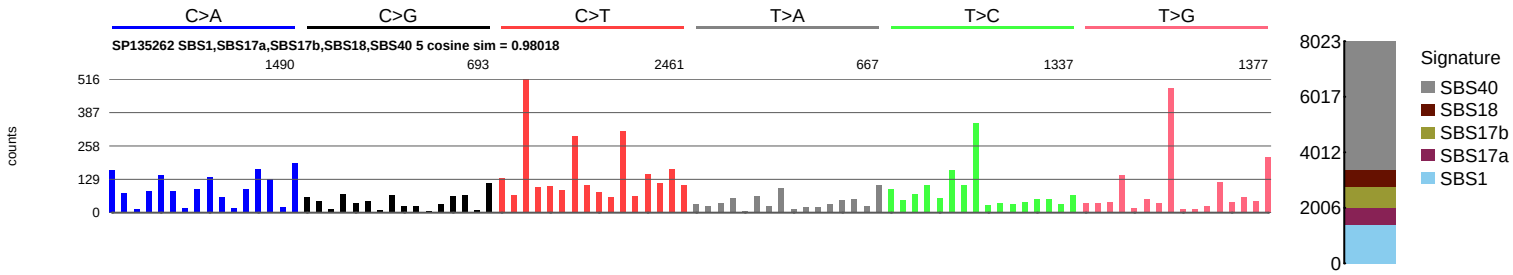

C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98084
1475
678
2414
641
1452
1359
counts
516
387
258
129
0
8023
6017
4012
2006
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS1

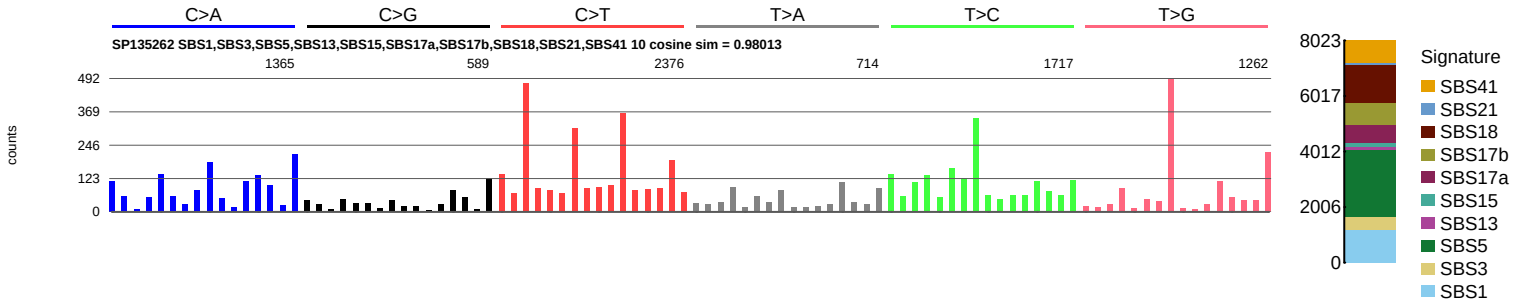

C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.98082
1487
680
2488
657
1354
1355
counts
512
384
256
128
0
8023
6017
4012
2006
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40 6 cosine sim = 0.9808
1478
663
2421
645
1452
1363
counts
516
387
258
129
0
8023
6017
4012
2006
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS17a,SBS17b,SBS18,SBS40,SBS44 6 cosine sim = 0.98073
1487
666
2494
659
1357
1361
counts
512
384
256
128
0
8023
6017
4012
2006
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.98023
1489
702
2458
664
1336
1375
counts
520
390
260
130
0
8023
6017
4012
2006
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS17a,SBS17b,SBS18,SBS40 5 cosine sim = 0.98018
1490
693
2461
667
1337
1377
counts
516
387
258
129
0
8023
6017
4012
2006
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135262 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98013
1365
589
2376
714
1717
1262
counts
492
369
246
123
0
8023
6017
4012
2006
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1